PrivaCF

A Privacy-Preserving Decentralized Recommendation Protocol

Design Document v0.3.1

> **Status:** Research specification. §10.1 lists open questions. §10.2 lists resolved prerequisites.

Notation

Symbol	Meaning	First defined
`sk`	Long-term secret key	§4.2
`null_v`	Nullifier: `Poseidon(sk, "null_v")`	§4.9.1
`epoch_id_T`	Pseudonymous identity for epoch T	§4.2
`beacon_T`	Public randomness beacon for epoch T	§4.1
`C_p(T)`	Pedersen commitment to preference vector	§4.4
`p_v`	Signed preference vector (local only)	§4.4
`r_p`	Pedersen blinding factor	§4.4
`M_v(T)`	Merkle root of behavioral history	§4.6
`π_v(T)`	Per-epoch permutation of preference vector	§4.5
`n_v(T)`	VRF-jittered gossip vector element count	§4.5
`commit_T`	Forward-secure nullifier commitment (N_fallback parallel ciphertexts)	§4.9.4
`c_T^{(i)}`	i-th ciphertext slot in `commit_T`, encrypted to committee_T^{(i)}'s threshold key	§4.9.4

Symbol	Meaning	First defined
`committee_T^{(i)}`	i-th committee for epoch T; i=0 primary, i≥1 fallback	§4.9.4
`N_fallback`	Number of parallel committees `commit_T` is encrypted to (primary + N_fallback−1 fallbacks)	§4.9.4
`W_primary`	Epochs the primary committee has to complete commit-reveal before fallback activates	§4.9.6
`W_fallback`	Epochs each fallback committee has before the next one activates	§4.9.6
`null_v_decryption`	On-chain transaction recovering null_v post-verdict	§4.9.6
`dec_nullifier`	`Poseidon(verdict_hash, null_v)`	§4.9.3
`D_v(T)`	Temporal depth of node v at epoch T	§7.2
`SUSP_SMT`	Sparse Merkle Tree of suspended nullifiers	§4.9.2
`DECRYPTION_SMT`	Sparse Merkle Tree of executed decryptions	§4.9.3
`θ_cluster`	Jaccard similarity threshold for peer selection	§5.4
`θ_behavioral`	Behavioral fingerprint similarity threshold	§6.3
`c`	Trust cap (DSybil non-overwhelming parameter)	§7.3
`n`	Admission window length in epochs	§4.3
`K_committee`	Auditor committee size	§6.4
`K_validators`	Validator set size	§4.1
`n_commit`	Epochs between public chain commits	§4.1
`δ_decay`	Per-epoch reputation decay	§6.1
`μ`	Hop-distance trust attenuation factor	§5.7
`λ`	Temporal depth decay factor	§7.2
`β`	Global/cluster trust blending factor	§3.4
`κ`	Novelty bonus scaling factor	§3.7
`r_v(X)`	Node v's local preference weight for item X (`p_v[X]`); only positive values enter trust_contribution	§3.4

Symbol	Meaning	First defined
`Δ_base`	Base trust increment per announcement	§3.4
`global_trust_total(X)`	Sum of trust_contribution over all nodes in the network that have announced X	§3.4
`cluster_trust_total(X)`	Sum of trust_contribution over nodes in the receiving node's interest cluster that have announced X	§3.4
`k_rep`	Rolling window length in epochs for reputation consistency computation	§6.1
`σ²_max`	Maximum expected score variance for an honest, fully-active node over `k_rep` epochs; normalization constant for consistency	§6.1
`f_cap`	Per-node trust cap as a fraction of `c`; no single node's accumulated trust may exceed `f_cap × c`	§7.3
`w_node_cap`	Maximum fraction of total reputation weight any single node may hold	§7.6
`w_cohort_cap`	Maximum fraction of total reputation weight any cohort may hold	§7.6
`announcement_token(v, X, T)`	Unforgeable per-announcement token: `Poseidon(sk, item_hash(X), beacon_T, "ann_token")`	§4.6
`n_peers`	Size of the PSI peer tier; grown organically via gossip referrals; requires empirical calibration	§5.7
`n_discovery`	Max parallel outgoing PSI attempts per epoch; conservative by default; calibrated against network size and bandwidth	§5.4
`n_psi_in`	Max incoming PSI requests accepted per epoch before silent drop; conservative by default	§5.4
`k_min`	Minimum PSI neighborhood size; cluster-specific behavior suspended below this threshold	§7.4
`ε`	Differential privacy budget per gossip event under Laplace mechanism; cumulative budget over T epochs is Tε	§4.5, §7.4

Abstract

Every major recommendation system learns what users like by collecting identity, history, and behavior on a server those users don't control. PrivaCF asks whether that has to be true.

The core challenge is a three-way tension: personalization requires preference data, privacy requires that data not be readable by others, and integrity requires that it reflects real human taste rather than manufactured signals from fake accounts. Prior work resolves at most two simultaneously. PrivaCF attempts all three.

Each participant holds a pseudonymous rotating identity tied to a computational admission cost that makes fake-account flooding expensive. Preferences are never transmitted in recoverable form — only shuffled, partially transmitted approximations that let similar users find each other without revealing what they actually like. Behavioral history is committed to a tamper-evident structure verified by a rotating committee of independent auditors via ZK proof, without access to the underlying data.

Suspension verdicts are permanent and survive identity rotation. A suspended node's nullifier — derived from their secret key — is inserted into a public Sparse Merkle Tree. Every future identity from the same key carries the same nullifier by construction; the membership proof fails by arithmetic, not by rule.

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1. How It Works

This section describes the system behaviorally. Cryptographic primitives are named here but defined precisely in §4; a reader encountering an unfamiliar term can defer to that section without losing the thread.

1.1 Getting Recommendations

Alice's node holds a preference vector — a signed list of how much she likes various items — kept entirely local. Periodically, her node exchanges a shuffled, partially transmitted version of this vector with a small set of peers whose tastes overlap with hers, discovered without revealing what those tastes are. PrivaCF organizes time into discrete epochs; each epoch corresponds to one block on the public chain (defined in §4.1). Each epoch, exactly n_v(T) elements are transmitted — a VRF-jittered count combining real preferences and cover items — so even the transmitted vector size reveals nothing about how many genuine preferences Alice holds.

Alice's node accumulates received vectors over time. Filtering and ranking happens entirely on her device. No recommendations are requested from any server — computation runs locally against data that has already arrived passively through the gossip protocol.

1.2 Being Part of the Network

When Alice interacts positively with an item, her node announces it to the network after a random delay, with a small amount of added noise on the rating. Negative interactions stay private. For rare items with small anonymity sets, announcement is further delayed by a VRF-derived number of epochs to prevent timing correlation.

Alice's behavioral history is summarized into a tamper-evident commitment each epoch. A rotating committee of auditors from independent clusters holds Shamir shares of her encrypted prior commitments and can verify consistency of successive commitments via ZK proof alone, without access to the underlying data. Public verdicts and anonymized reputation attestations are written to the public blockchain. Fine-grained behavioral data, continuity proofs, and handoff history are held in encrypted form under the arbitration committee's threshold custody — reconstructable only on a threshold quorum and only when an arbitration is actually invoked. Each per-interaction record (gossip push, pull response, PSI handshake, audit response) takes the form of a **co-receipt** signed by both parties and held locally; either side can later present its half to the committee if the counterparty disputes.

1.3 When Something Goes Wrong

If Alice's recommendations start degrading, her node signals this to the network. If enough independent nodes signal the same thing, a rotating committee of auditors is assembled to investigate. Audits are indistinguishable from normal protocol traffic.

If a node is found to be manipulating the network, the committee initiates a commit-reveal verdict process. Each committee member first publishes a commitment to their verdict on the public chain — locking in their decision before decryption is possible — then publishes their BLS share alongside the verdict. Once a threshold of shares is revealed, anyone can aggregate them to decrypt the node's forward-secure nullifier commitment and recover `null_v`.

The suspended node's nullifier is inserted into the suspended nullifier tree (SUSP_SMT). Creating a new identity from the same key is impossible: every future epoch ID derived from that key carries the same nullifier by construction, and the non-membership proof fails at the first handoff. Creating a new identity with a genuinely different key requires completing the full admission proof chain again, and new identities matching the behavioral fingerprint of a suspended one are flagged on arrival.

Any node can monitor the public chain for anomalous commit-reveal activity — an unexpected burst of verdict commitments with no corresponding behavioral signals triggers watchdog broadcasts and recursive oversight.

1.4 Joining the Network

Creating an identity requires completing a chain of sequential computational proofs over n epochs, interspersed with mandatory protocol interactions with existing nodes. Once all n proofs are complete and all interaction checkpoints have been passed, the identity is admitted and begins accumulating reputation. There are no partial admissions — the proof chain must be completed in full.

1.5 The Adversary Model

The threats below are defended across all transport profiles. Profile-specific adversary capabilities are summarized at the end of this section.

The opportunist creates a handful of fake accounts to boost their own content. Defended by admission cost, temporal depth, rate limits, and behavioral fingerprinting (the latter is profile-dependent — see §6.2).

The commercial promoter operates a sustained campaign of fake accounts with enough patience to build reputation before exploiting it. Defended by the non-overwhelming trust rule, smoothness detection, and within-cluster coordination detection (interest-cluster signal is transport-agnostic; behavioral-cluster signal is available under the Tor/I2P profile only).

The coordinated campaign involves many accounts pushing a narrative across many items simultaneously. The system surfaces behavioral anomaly patterns for operator review. Intent classification requires human judgment.

The epoch rotator earns a SUSPENDED verdict then attempts to re-enter under a new identity derived from the same key. Defended by the nullifier mechanism: the same key always produces the same nullifier, which is permanently in the suspended set. Re-entry from the same key is cryptographically impossible.

The dark node rotator earns a SUSPENDED verdict and goes offline before their nullifier is extracted, then attempts to re-admit from the same key. Defended by forward-secure commitment: the committee decrypts `commit_T` using the verdict as a key, without requiring node cooperation. The residual gap — nodes that go dark during the admission window before publishing any `commit_T` — is bounded by zero reputation and behavioral fingerprinting.

The rogue committee attempts mass deanonymization by extracting `null_v` from many nodes without legitimate verdicts. Defended by the commit-reveal ordering: the committee must publicly commit to verdicts before decryption is possible. Anomalous commit rates are visible on the public chain before any `null_v` is recovered, triggering watchdog signals and recursive oversight.

Transport-profile-specific adversary capabilities

Under the self-mixing Loopix profile (default):

- Defended against the **global passive adversary** observing the full wire — Loopix's per-hop Poisson mixing and constant-rate cover traffic prevent traffic-correlation attacks.
- Weaker detection of **whitewashing** — the post-suspension fingerprint match (§6.3) operates on a thin behavioral fingerprint (§6.2), reducing matching power.
- Weaker detection of **IP-coordinated botnets** — per-client IP↔epoch_id is structurally erased; only aggregate mix-layer density signals are available (§6.3).
- Weaker detection of **sub-epoch coordinated bursts** — sub-epoch timing is destroyed by mixing; the T.2 signal (§7.1a) is dead under this profile.

Under the Tor/I2P profile:

- Vulnerable to **global passive adversary** traffic-correlation attacks (Tor's known weakness). Users with this threat model must layer additional OPSEC or select the Loopix profile.
- Stronger detection of **whitewashing** — full behavioral fingerprint (§6.2) makes post-suspension identity match high-dimensional.
- Stronger detection of **IP-coordinated botnets** — only effective against lazy Sybils who do not properly rotate Tor circuits per identity; sophisticated Sybils with disciplined circuit hygiene retain anonymity comparable to the Loopix profile.
- Full **sub-epoch coordinated burst** detection (T.2, T.9 at sub-second resolution).

Nation-state adversaries, compromise of the external randomness beacon, and eclipse attacks are out of scope across all profiles.

1.6 Assumptions

A1 — Honest neighbor. Every node has at least one honest gossip peer per epoch.

A2 — Honest majority by weight. More than half of total accumulated reputation weight belongs to honest nodes.

A3 — Resource-bounded adversary. Cannot solve VDFs faster than specified delay, cannot break standard cryptographic primitives, cannot predict the drand beacon before publication, cannot find Poseidon collisions (Grassi et al., USENIX Security 2021), cannot break BLS signature unforgeability.

A4 — Time-bounded genesis. The initial bootstrap set is trusted for a finite period only. This is the standard bootstrap assumption for Byzantine-fault-tolerant systems and is treated as an axiom rather than a derived property. A network that cannot trust its genesis set generates a regress that no purely cryptographic mechanism resolves.

1.7 Security Argument

The narrative threat model in §1.5 and the assumption list in §1.6 do not, on their own, constitute a security argument. This subsection restates the protocol's intended guarantees as **game-shaped properties** with bounded adversary advantage and reductions to standard primitives or stated assumptions. Properties whose strength depends on empirically-calibrated parameters (smoothness detection, behavioral fingerprint matching) are explicitly separated from properties with cryptographic reductions. Concrete `ε` values are deferred to OQ resolution; the goal here is to fix the *shape* of each claim so that future work can fill in the constants.

Notational convention for this subsection. Symbols `ε_X` denote *adversary-advantage bounds* against the primitive or assumption named in `X` (e.g., `ε_PRF`, `ε_BLS`, `ε_IBE`, `ε_Pedersen`, `ε_DP`, `ε_perm`, `ε_Π`). They are conventional cryptographic-reduction names, not protocol parameters, and intentionally do not appear in the global notation table. `δ_DSybil` and `γ_smoothness` in P5 are empirical calibration constants reported against the Phase 4–5 experiments; their values are not pinned at the protocol layer.

P1 — Identity unlinkability. For any PPT adversary `A` observing the public chain and the wire under transport profile `Π ∈ {Loopix, Tor/I2P}`:

```
Adv_unlink(A, T, k, Π) = | Pr[A(epoch_id_T, epoch_id_{T+k}) = same_sk] − ½ |
                         ≤ ε_PRF(Poseidon) + ε_Π(traffic_correlation, T, k)
```

Reduces to (a) Poseidon PRF security on `(sk, beacon_T, null_v, "epoch_id")` and (b) the transport profile's traffic-correlation bound. Under Loopix, `ε_Π` is bounded by the per-hop Poisson mixing parameter and cover-traffic rate; under Tor/I2P it is not bounded against a global passive adversary and the property is conditionally claimed.

P2 — Preference indistinguishability in transit. For preference vectors `p^0, p^0_1` differing in a single component and `gossip_T(p)` the per-epoch gossip output:

```
Adv_pref(A, T) = | Pr[A(gossip_T(p^0)) = 0] − Pr[A(gossip_T(p^1)) = 0] |
                 ≤ ε_DP(T, ε_per_epoch) + ε_perm(π_v) + ε_Pedersen
```

Reduces to (a) per-epoch Laplace/uniform noise (§4.5) composed over `T` epochs via advanced composition (Dwork-Rothblum-Vadhan); (b) Poseidon-derived permutation secrecy of `π_v(T)`; (c) Pedersen hiding for `C_p(T)`. Lifetime DP budget `T · ε_per_epoch` is **not yet pinned** — see OQ-55 and the DP accounting note below.

P3 — Suspension persistence. Let `Admit(sk')` denote successful admission with secret key `sk'`. For any adversary holding a key `sk` whose `null_v = Poseidon(sk, "null_v")` has been inserted into `SUSP_SMT`:

```
Pr[A admits with sk' such that Poseidon(sk', "null_v") = null_v] ≤ negl(λ)
```

Reduces to Poseidon PRF collision resistance (A3) plus soundness of the SMT non-membership ZK proof in Statement 5 (§4.9.5). The only re-entry path with the same key is a Poseidon collision; the only re-entry path with a fresh key requires completing the full admission proof chain anew and incurs zero accumulated reputation.

P4 — Forward-secure nullifier extractability. Let `commit_T` be a node's forward-secure commitment for epoch `T`, encrypted to `committee_T^{(i)}`'s threshold BLS key. After a valid verdict signature `σ_T` is published:

```
Pr[anyone recovers null_v from commit_T given σ_T] = 1
Pr[anyone recovers null_v from commit_T without σ_T] ≤ ε_BLS + ε_IBE
```

Reduces to threshold BLS unforgeability (Boneh-Lynn-Shacham) + Boneh-Franklin IBE security. Multi-recipient construction (§4.9.4) preserves both bounds with `N_fallback` independent committees; stalling-minority decay is exponential in `N_fallback`. **Forward secrecy of past `commit_T`** (compromise of a *future* committee key retroactively decrypting past `commit_T` without a corresponding verdict) is currently underwritten by mandatory share deletion + slashing rather than a forward-secure cryptographic primitive. Canonical treatment in §8.2 T12.

P5 — Sybil influence bound. For an adversary controlling fraction `f` of total admission-cost units (A3, VDF-bounded):

```
E[reputation share of A] ≤ f · (1 + δ_DSybil) + f² · γ_smoothness
```

Reduces to the DSybil non-overwhelming theorem (Yu et al., IEEE S&P 2009) applied per-cluster with cap `c`, plus the FoolsGold-on-PSI-peers detection penalty applied above the smoothness threshold. The `f²` term reflects detection probability scaling with coordination cardinality — calibration-dependent and reported as an empirical bound rather than a proof.

Composition over time

Per-epoch failures of A1 (honest neighbor) and per-arbitration failures of the threshold-honesty assumption on the *currently-selected* arbitration committee compose via union bound over the protocol lifetime `T_life`:

```
Pr[lifetime failure of A1] ≤ T_life · Pr[no honest neighbor in epoch T]
Pr[lifetime failure of committee threshold] ≤ N_arbitrations · Pr[≥ K malicious
in selected committee]
```

The §4.1 redesign narrows the second bound: it is the union over **arbitrations actually invoked**, not over **all committees ever selected**. With per-epoch member rotation and slashing-enforced share deletion, no past committee retains the capability to reconstruct any `snapshot_v(T)` beyond the epoch in which it was a custodian. This **trust-localization claim** is the main security improvement of the redesign and is conditional on share deletion being enforced (§8.2 T12).

Proven vs. calibrated vs. assumed

Property	Status	Reduction or basis
P1 Identity unlinkability	proven (sk)	Poseidon PRF + transport profile bound
P2 Preference indistinguishability	proven (sk)	Pedersen hiding + permutation secrecy + DP composition (constants open)
P3 Suspension persistence	proven (sk)	Poseidon collision resistance + SMT non-membership soundness
P4 Forward-secure extraction	proven (sk)	Threshold BLS + BF-IBE; FS of past `commit_T` conditional on share deletion (§8.2 T12)
P5 Sybil influence bound	partial	DSybil cap is proven; FoolsGold-on-PSI-peers detection rate is calibrated, not proven
Smoothness detection rate	calibrated	Srivatsa et al. 2005 + empirical thresholds (OQ-9)
Behavioral fingerprint matching	calibrated	Profile-conditioned; full under Tor/I2P, thin under Loopix (§6.2)
A1 honest neighbor / A2 weight majority	assumed	Standard for gossip/BFT settings; per-epoch and lifetime-bounded by union argument
A4 honest genesis (finite period)	assumed	Standard BFT bootstrap axiom
Committee share deletion (post-rotation)	assumed	Enforced by slashing in v1; replaceable by forward-secure threshold BLS in v2 (Pixel-style)

"Proven (sk)" denotes *proof-sketchable* — the reduction is named and standard, the formal write-up is deferred to a security analysis companion document.

What this argument does **not** yet bound

- Concrete `ε_per_epoch` for P2 — depends on noise calibration in §4.5 and OQ-55.
- Concrete `δ_DSybil` and `γ_smoothness` constants for P5 — empirical, deferred to Phase 4–5 experiments.
- Forward secrecy of past `commit_T` against future-committee-key compromise — currently underwritten by slashing rather than cryptography (open thread, §8).

- Behavioral fingerprint re-identification rate under Loopix — known thin, quantification deferred.
- Network-level deanonymization under Tor/I2P against a global passive adversary — explicitly out of scope (§1.5).

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2. System Overview

PrivaCF is organized into five layers plus a single public chain, with an on-demand arbitration committee holding sensitive per-node state under threshold custody.

```
┌─────────────────────────────────────────────────┐
│  LAYER 5 · Collaborative Filtering               │
│  Computed locally · Never transmitted            │
├─────────────────────────────────────────────────┤
│  LAYER 4 · Reputation & Audit                    │
│  Behavioral commitments · Multi-auditor handoff  │
│  On-chain verdicts · Permanent health state      │
│  Commit-reveal verdict flow                      │
├─────────────────────────────────────────────────┤
│  LAYER 3 · Sybil Resistance                      │
│  Weight caps · Temporal cost · Anomaly detection │
│  Interest clusters · Behavioral clusters         │
│  Watchdog signals · Recursive oversight          │
├─────────────────────────────────────────────────┤
│  LAYER 2 · Network                               │
│  Loopix/Sphinx · Dandelion++ · PSI peer sel.     │
│  Relay submission · Niche item delay             │
├─────────────────────────────────────────────────┤
│  LAYER 1 · Identity & Privacy                    │
│  Poseidon PRF · EC-VRF · VDF · Pedersen · Perm   │
│  Merkle · Nullifier · SUSP_SMT                   │
│  ForwardCommit · DECRYPTION_SMT                  │
├─────────────────────────────────────────────────┤
│  PUBLIC CHAIN · Verdicts & Attestations          │
│  VDF-chained · BFT consensus · Deterministic     │
│  finality · Permanent verdicts · SUSP_SMT root   │
│  DECRYPTION_SMT root · commit_T per node/epoch   │
│  Light clients                                   │
```

```
├─────────────────────────────────────────────────┤
│  ARBITRATION COMMITTEE · On-Demand · Sensitive State │
│  VRF-selected · Shamir-shared · Threshold custody│
│  Continuity proofs · Fine-grained behavioral data│
│  Handoff history · Reconstructed only on quorum  │
└─────────────────────────────────────────────────┘
```

Design principle — self-report is untrusted by default. The protocol treats self-reported values as untrusted by default. All values feeding reputation scoring, audit decisions, or compound flag conditions rely on one or more of: peer attestation, ZK proof, public commitment, or committee threshold verification. No self-reported leaf types remain. The RATE_LIMIT leaf uses an announcement token set construction (§4.6) — tokens are unforgeable without `sk` and their cardinality is verifiable by the auditor without revealing which items were announced.

2.1 Node Relationships

See Appendix G for the full node relationship diagram. In brief: each node maintains peers in two tiers — an interest cluster tier (2–3 nodes confirmed by PSI) and a bridge tier (1 random DHT peer refreshed each epoch). Auditor committees and validator sets are VRF-selected each epoch with dual cluster diversity constraints; neither is predictable before the beacon is published.

On validator and committee discoverability: VRF selection means any node can verify who was selected for a given epoch after the beacon is published, but cannot predict it in advance. The cluster diversity constraints are checked deterministically against each candidate's attested cluster membership on the public chain.

Bootstrapping beyond genesis: The genesis validator set covers the period before enough nodes have accumulated sufficient temporal depth and cluster diversity. When cluster diversity constraints cannot yet be fully satisfied, the behavioral cluster constraint is relaxed during the transition period, falling back to interest cluster diversity only. The transition threshold requires empirical calibration (OQ-12).

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3. Collaborative Filtering

3.1 Goal

Given a node's interaction history and a set of gossip vectors received from peers over many epochs, produce a ranked list of items the node has not interacted with that it is likely to enjoy. The filtering and ranking computation happens entirely on the local device. The network's role is to deliver gossip vectors — it does not participate in the recommendation computation itself.

The CF algorithm is not fixed by the protocol. Item-based collaborative filtering is the default and is specified below, but any algorithm that operates on a matrix of received gossip vectors can be substituted.

3.2 Computing Recommendations

PrivaCF uses item-based collaborative filtering as its primary method. If items A and B tend to appear together in peer preference vectors, and Alice likes item A, she probably likes item B.

Each node maintains a local HNSW index built from gossip vectors received from peers over many epochs. Item similarity is computed from this index:

```
sim(item_i, item_j) = cosine_similarity(column_i(P), column_j(P))
score(item_i)       = Σ_{j ∈ interacted} sim(item_i, j) × weight(j)
```

where P is the matrix of received gossip vectors. Fully offline. Each node sends one gossip vector push per epoch and receives one per peer per epoch. Recommendation quality improves as the local index grows over many epochs.

User-based CF is applied as a within-epoch supplement. Item-based CF is preferred because it is stable across epoch rotation.

The recommendation data structure is **HNSW** — Hierarchical Navigable Small World graphs (Malkov & Yashunin, 2018). HNSW organizes items into a layered proximity graph where each node connects to its approximate nearest neighbors. Queries navigate from coarse upper layers to fine lower layers, finding approximate nearest neighbors in O(log n) time with high recall. Each PrivaCF node maintains a local HNSW index built from received gossip vectors; item similarity scores for CF are computed by querying this index. Periodic snapshots are retained to support rewind recovery (§6.6).

Peer discovery uses a separate two-step mechanism — LSH pre-filtering followed by PSI confirmation — described in §5.4.

3.3 Clusters

PrivaCF organizes nodes into two independent cluster types, both derived from public data without any central coordinator.

Interest clusters group nodes whose item interaction sets overlap significantly, as measured by Jaccard similarity via PSI. Two nodes end up in the same interest cluster because they have interacted with many of the same items — not because they were assigned to one. Interest clusters drive peer selection and CF quality.

Behavioral clusters group nodes with similar participation patterns — when they are active within an epoch, how they space out announcements, when they submit on-chain transactions. These are derived entirely from public timing data on the blockchain. Behavioral clusters are used for Sybil detection and to ensure auditor committee independence. They are never used directly for reputation — they feed the compound flag system described in §7. Availability of behavioral-cluster signal is transport-profile-conditioned: full fingerprint under the Tor/I2P profile, thin fingerprint under the Loopix profile. See §5.1 for the profile framework and §6.2 for the per-profile fingerprint definitions.

Cluster membership is not fixed. It is recomputed each epoch from current data and evolves as a node's interaction history grows.

Privacy implications: Interest cluster membership is partially inferable by an external observer who watches PSI handshake patterns over time. Behavioral cluster reconstructability is transport-profile-conditioned. Under the Tor/I2P profile, sub-epoch timing signals are visible on the wire and behavioral fingerprints are largely reconstructable from public chain data, with the on-demand arbitration model, per-n-epoch commits, relay submission, and transaction timing jitter degrading but not eliminating their precision. Under the Loopix profile, sub-epoch timing is destroyed by the mix layer — only epoch-granular presence and audit-response rate survive (see §6.2), so behavioral fingerprints are inherently thin and not reconstructable beyond those coarse signals.

3.4 Trust Weight and Local trust_total

```
trust_contribution(v, X) = max(0, r_v(X) + noise) × Δ_base × (1 + κ ×
novelty(X))
```

- `r_v(X)` is node v's local preference weight for item X (`p_v[X]`). Only positive weights are transmitted in gossip vectors; negative weights stay local. Cover items (§4.5) receive a small assigned `cover_weight` ; under the `max(0, ...)` guard they enter the sum only when their assigned weight is positive, so cover items contribute their assigned weight to `r_v(X)` and nothing more.
- `noise` is the additive announcement noise applied per §4.5: either `Laplace(0, S/ε)` under the DP deployment or a bounded uniform draw under the chopping deployment. Both are constrained by the sign-preservation requirement `|noise| < |p_v[X]|` for non-zero dimensions.
- `Δ_base` is the base trust increment per announcement. It scales the raw preference weight into the trust_total space and requires empirical calibration.

`trust_total(item)` is not stored on the blockchain. Each node maintains a local estimate updated from received announcements weighted by announcer reputation band. Divergence across nodes is acceptable — CF requires only local consistency.

```
trust_total(X) = Σ_{v : v announced X} trust_contribution(v, X)
```

Items globally popular but not popular within a node's interest cluster are softened. Each node maintains both a global and a cluster-local accumulation:

```
global_trust_total(X)  = Σ_{v ∈ all announcers of X}
trust_contribution(v, X)
cluster_trust_total(X) = Σ_{v ∈ announcers of X ∩ receiving node's cluster}
trust_contribution(v, X)

effective_trust(X) = β × global_trust_total(X) + (1−β) × cluster_trust_total(X)
item_weight(X)     = 1 / log(1 + effective_trust(X) / c)
```

`item_weight` uses a logarithmic damping analogous to IDF: items that have accumulated large trust totals contribute less marginal weight per additional interaction. The `c` in the denominator normalises the trust total against the DSybil cap, so the damping kicks in proportionally to how close the item is to the ceiling.

Both trust totals are locally computable. `trust_total` is not stored on the public chain; each node maintains its own estimate by accumulating `trust_contribution(v, X)` values from received announcements, weighting each contribution by the announcer's score band (§6.1). The cluster variant restricts the sum to announcers in the receiving node's PSI peer neighborhood (`peers_v`). Per-node divergence in these estimates is acceptable — CF requires only local consistency, not global agreement.

3.5 Dislike-Aware Scoring

Negative preference weights are never transmitted. Dislikes are applied locally as a post-processing filter only:

```
final_score(item_i) = raw_cf_score(item_i)
    − penalty × max(0, Σ_{j ∈ dislike_set} sim(item_i, j) × |p_v[j]|)
```

3.6 User-Configurable Reputation Floor

Each node sets a minimum reputation band for gossip vectors incorporated into its local HNSW index. Default: Band 2.

The reputation floor functions as a confidence threshold on peer legitimacy. Whether reputation is modeled as a continuous gradient or discrete bands (as in this protocol) only changes the granularity of that confidence: discrete bands produce two effective states per threshold — zero weight below the floor, full weight above it — while a continuous gradient would scale contributions proportionally to confidence. This connection between reputation floor and expected sybil influence per peer is made explicit in the influence model (§7.1b). Users who wish to eliminate bridge-tier sybil influence entirely may set their bridge peer weight to zero; see the note in §5.7.

3.7 Diversity and Novelty

```
novelty(item)          = clamp(1 − effective_trust(item) / c,  0, 1)
trust_contribution(v, X) = max(0, r_v(X) + noise) × Δ_base × (1 + κ ×
novelty(X))
```

`trust_contribution(v, X)` is the same quantity defined in §3.4; the novelty bonus is the `(1 + κ × novelty(X))` factor that distinguishes its role in this section.

`novelty` is clamped to [0, 1]. When `effective_trust` equals 0 the item is fully novel (novelty = 1); when it reaches `c` novelty reaches 0 and the bonus disappears. Values above `c` are possible transiently if the blended `effective_trust` exceeds the cap before the DSybil gate truncates further contributions — the clamp prevents a negative novelty from inverting the bonus into a penalty.

Novel items accumulate trust weight faster. Nodes in sparse interest clusters receive additional CF aggregation weight.

> The protocol does not prescribe a recommendation algorithm. For a full discussion of the deployment-level tensions this creates — sparsity, feedback loops, reputation interaction, epoch length, positive-only signal, niche privacy tradeoffs, and the boundary between organic popularity and coordinated pushing — see §13.

3.8 Strategy Interface and Catalog

The collaborative filtering construction in §3.1–§3.7 is the **reference recommendation strategy** shipped with the protocol. It is not the protocol. The privacy, identity, Sybil-resistance, and audit machinery in §4–§7 form a **substrate** that exposes a stable set of inputs to a strategy chosen by the node operator. Different nodes on the same network may run different strategies; the substrate guarantees Sybil resistance and privacy regardless of which strategy a node selects.

This separation is load-bearing for the project's positioning: "your node, your algorithm" is meaningful only if the substrate is general enough to host more than one algorithm.

Substrate-provided inputs to any strategy:

Input	Source	Privacy property
Trust-weighted stream of `(item, endorsement)` from peers	Gossip pulls from PSI peer neighborhood (§3.2, §5.4)	Peer endorsements visible only after PSI confirms overlap
Per-item global trust totals	Public chain aggregation (§3.4)	Public by design
Cluster membership signals	PSI peer set + behavioral fingerprint (§3.3, §6.2)	Membership inferable only by direct PSI partners
Peer trust weights	Local computation (§3.4)	Local-only

Input	Source	Privacy property
Item content / content-addressed payloads	Content-addressed fetch (out of band)	Independent of substrate
Cross-epoch continuity for follow-style trust	§4.7 handoff	Continuity without linkability

A strategy is any local computation that consumes some subset of these and produces a ranked list of candidate items. Strategies do not need substrate cooperation beyond the inputs above — they are local-only and freely swappable.

Strategy catalog (illustrative, not normative):

Strategy	Substrate inputs consumed	Locally added	Notes
Collaborative filtering (reference, §3)	All	Aggregation formula	Default; works at small scale
Content-based filtering	Item stream only; ignores peer endorsements	Local embedding model (CLIP, sentence transformers) and item-embedding gossip	Stronger privacy: no preference vector ever leaves the node
Knowledge-graph / semantic	Item stream + peer-gossiped relation edges	Local graph + spreading-activation ranker	Useful for items with rich structured metadata (music, video, papers)
LLM-based ranking	Item stream + endorsements + local history	Local LLM (e.g. Llama-class); user-authored ranking prompt	Strongest sovereignty UX: the model explains its picks in your terms
Two-tower / embedding	Item stream	Content-addressed shared item-tower artifact; locally trained user-tower	Closest analogue to modern industrial recommenders
Contextual bandit	Item stream + local reward signal	Thompson sampling or LinUCB over peer-supplied candidates	Explicit explore/exploit control

Strategy	Substrate inputs consumed	Locally added	Notes
RL with user-defined reward	Item stream + local outcomes	Local RL agent optimizing a reward function the user defines	Strongest ideological fit: you choose what is optimized
Editorial / follow-graph	Cross-epoch continuity (§4.7) only	List of trusted pseudonyms	Trivial to implement; useful as a building block
Ensemble	All of the above	User-set weights between sub-strategies	Most practical UX — combines complementary signals

Composition. Strategies may be composed: an LLM ranker may take a CF candidate list as input; an ensemble may weight a content-based and a bandit strategy; an editorial follow-list may seed a CF run. The substrate is indifferent to composition.

Substrate guarantees that bind regardless of strategy. Identity privacy (§4), preference obfuscation in transit (§4.5), tamper-evident behavioral history (§4.6), Sybil resistance (§7), and audit (§6) are properties of the substrate, not of the strategy. A node running an unusual strategy does not lose these guarantees; it only changes how it ranks the items the substrate makes visible to it.

Out of scope here. A formal strategy interface specification (concrete data types, callbacks, lifecycle hooks) is not part of this document. The catalog above describes design space, not API. See §13 for the cross-cutting deployment tensions that any strategy must address.

---

4. Identity and Privacy

Each cryptographic primitive solves one specific problem. To orient the reader before the formal definitions: a node's long-term key `sk` never leaves the device. From it, the node derives a per-epoch pseudonym (`epoch_id_T`) that is unlinkable across epochs, a permanent nullifier (`null_v`) that ties all epoch IDs together without revealing the link, and a forward-secure commitment (`commit_T`) that lets the committee extract `null_v` after a verdict without any cooperation from the node. The ZK proof system lets auditors verify properties of preference and behavior data they cannot read. The diagram below shows how these relate before each is introduced individually.

See Appendix H for the full identity/privacy relationship diagram.

4.1 Chain and Arbitration Committee

Epoch duration. PrivaCF targets an epoch duration of 2–3 hours (configurable per deployment). This window is short enough to preserve epoch-granular timing signals usable by Sybil clustering (T.1, T.5, T.9 in §7.1a) under the Loopix profile where sub-epoch timing is unavailable, while remaining long enough to amortize per-epoch handoff costs — ZK proof generation on mobile hardware (OQ-3) and committee DKG latency for the rotating audit/validator committees. Shorter epochs improve detection resolution at the cost of proof-generation throughput; longer epochs invert that tradeoff.

PrivaCF's public state lives on a single public blockchain. Per-interaction state lives in **signed co-receipts** held locally by both parties. State that must survive node rotation but cannot live on the public chain is **Shamir-shared to a rotating committee** that acts as an on-demand arbitrator rather than an active ledger keeper.

The public blockchain is the authoritative record for verdicts, anonymized reputation attestations, epoch_id registrations, `commit_T` values, score-band attestations, and the SUSP_SMT and DECRYPTION_SMT roots (see §4.9.2 and §4.9.3 for the SMT definitions). It is a VDF-chained ledger with Byzantine fault-tolerant consensus. Every node can read and verify it. Light clients verify entries via Merkle inclusion proofs against block headers.

Co-receipts. Every protocol interaction that needs after-the-fact proof — gossip push acceptance, pull response delivery, PSI handshake completion, audit response delivery — produces a receipt signed by both parties and held locally by both. Volume is low: per node per epoch, roughly one gossip push, a handful of pulls, a few PSI attempts, and the audit responses triggered for that node. Either party can later present its half of a receipt to the arbitration committee (see below) without ever publishing it to a chain. Atomic exchange of the two signatures uses a standard HTLC-style two-step swap so neither party can withhold after seeing the counterparty's signature; full construction in Appendix A.

Arbitration committee. A VRF-selected committee of size K_committee, rotating per epoch, is invoked **on demand** for three purposes: (1) verifying the per-epoch handoff package (§6.4); (2) adjudicating co-receipt disputes when a node claims a counterparty cheated; (3) holding Shamir shares of `snapshot_v(T)` — the encrypted per-node behavioral state that supports continuity proofs and Class 3 audits. The committee does **not** maintain an active ledger between invocations. Members hold encrypted shares; reconstruction requires a threshold quorum and is triggered by a specific arbitration request, not by ongoing block production. The same VRF-selection machinery used elsewhere in the protocol (§4.9.4) applies — there is no separate consensus instance.

Why this is not a chain. A full committee-held ledger would require continuous threshold consensus among members, share rotation on every membership change, and a separate availability story. The on-demand model eliminates all three. Members store their encrypted shares; the public chain records the committee's selection (via VRF output) and the handoff's score-band attestation; everything else is reconstructed only when an arbitration is actually requested. Trust assumption: threshold honesty of the **currently-selected** committee at arbitration time, not of every committee that has ever existed.

Handoff state continuity across committee rotation. When a committee rotates, the outgoing committee re-Shamir-shares each `snapshot_v(T)` it holds to the incoming committee, encrypted to the incoming members' epoch keys. This is a one-shot per-epoch handoff, not a continuous protocol. The re-share is verified by a ZK proof of correct re-encryption; failure to complete it is itself a slashable offense for the outgoing committee.

Public blockchain block structure:

```
block_T = {
    height:                 T,
    vdf_output_T:           VDF_eval(vdf_output_{T−1}, δ_block),
    prev_block_hash:        H(block_{T−1}),
    primary_entries:        [high-priority transactions for epoch T],
    beacon_T:               H(drand_T ‖ vdf_output_{T−1}),
    proposer_id:            <VRF-selected proposer epoch_id>,
    validator_sigs:         <threshold BLS aggregate signature>,
    susp_smt_root_T:        <root of SUSP_SMT>,
    decryption_smt_root_T: <root of DECRYPTION_SMT>
}
```

`primary_entries` are the high-priority transactions enumerated below (suspension verdicts, watchdog signals, epoch_id registrations, `commit_T` values, double-signing evidence, Class 3 audit results). `proposer_id` is the VRF-selected node responsible for assembling and proposing the block, per the BFT consensus protocol — analogous to the proposer in any Tendermint-style chain.

Two-level transaction pool:

```
PRIMARY POOL (packed into blocks immediately):
    committee_verdict
    verdict_commit          ← per-member verdict commitment
```

```
    verdict_reveal          ← per-member BLS share + verdict
    null_v_decryption       ← recovered null_v + DECRYPTION_SMT insertion
    double_signing_evidence
    Class 3 audit results
    epoch_transaction       ← includes commit_T and commit_T_zk_proof every
epoch
    admission_summary
    watchdog_signal

SECONDARY POOL (accumulated, merged at epoch end):
    announcement_observation
    pull_receipt
    audit_nonce_record
```

commit_T and n_commit batching: `commit_T` and its ZK proof are included in the `epoch_transaction` and submitted every epoch. `M_v(T)`, `C_p(T)`, score band, and health tier batch per `n_commit`. The rationale: `commit_T` must be tied to the current epoch's committee threshold BLS key, which rotates every epoch; batching would create key staleness. `commit_T` is opaque without a verdict signature and reveals only node presence, which is already inferrable from `epoch_id` registration. The threshold key is generated by the VRF-selected committee running a standard distributed key generation (DKG) protocol over BLS12-381 at the start of each epoch — see §4.9.4 for the construction.

DKG timing constraint: Committee DKG must complete before a node constructs and submits `commit_T`. DKG for K≈21 nodes is fast in practice. The `epoch_transaction` submission deadline is set after DKG completion.

At epoch end, all secondary pool entries are aggregated into a single `epoch_transaction`. The secondary pool entries are discarded.

Legitimacy verification: An `epoch_transaction` carries two independent authenticity guarantees. First, it is signed by the submitting node's `epoch_id` key. Second, each field is either cryptographically committed or committee-attested before the transaction is assembled. Validators verify the node's signature, the committee's threshold BLS signature on the score band, and the ZK proof certifying correct `commit_T` formation.

Block production:

```
validator_set_T = EC-VRF(sk_proposer, beacon_T ‖ "validators",
                      k = K_validators,
                      constraints = [
                          temporal_depth ≥ D_validator,
                          reputation ≥ rep_validator,
                          different_interest_clusters,
                          different_behavioral_clusters    // see §6.2; applies
under Tor/I2P profile, relaxed under Loopix profile (see §4.9.8)
                      ])

proposer_T = EC-VRF(sk_proposer, beacon_T ‖ "proposer", pool = validator_set_T)
```

The proposer assembles the block from pending transactions and broadcasts it. Validators verify and sign with BLS. Once ⌊K_validators / 3⌋ × 2 + 1 signatures are collected, the block is final. Deterministic finality. K_validators ≈ 21 is a reasonable default.

Validator incentives:

```
score_v(T) += w_validator × validator_service_indicator(T)
```

Liveness — proposer timeout: If the proposer does not produce a block within a timeout window (default: 20% of epoch duration), the next VRF-ranked validator in the set takes over.

Double-signing detection: Two BLS signatures from the same `epoch_id` on competing blocks at the same height are both on-chain and publicly verifiable. Consequence: immediate permanent SUSPENDED verdict.

Light clients: Mobile nodes store block headers only and use Merkle inclusion proofs to verify specific entries.

Per-n-epoch public chain commits: The `epoch_transaction` merge fires every `n_commit` epochs for `M_v`, `C_p`, score band, and health tier, reducing on-chain timing resolution available for behavioral fingerprinting. `commit_T` is exempt from batching because it is bound to that specific epoch's committee threshold BLS key (which rotates each epoch, per §4.9.4); batching would produce a stale key reference and break the forward-secure construction. n_commit = 2 or 3 is a reasonable starting range.

//Overall are we storing information about ALL nodes in here? score bands, health tiers, M_v sound like things that are unique to each node

4.2 Rotating Pseudonymous Identity

Problem: The same identity across epochs makes a node's activity linkable over time.

Tool: Poseidon as a keyed PRF for local per-epoch derivations, and EC-VRF (RFC 9381) for on-chain verifiable selection. Given a secret key, a nullifier, and the current beacon, the PRF produces a pseudonym for this epoch. Epoch IDs are unlinkable across epochs without knowing `sk`.

```
null_v     = Poseidon(sk, "null_v")
epoch_id_T = Poseidon(sk, beacon_T, null_v, "epoch_id")
```

Primitive split — Poseidon PRF for local derivations, EC-VRF for on-chain selection: Local per-epoch derivations (epoch ID, permutation, chop count, offset, niche delay, leaf salt) require only pseudorandomness under a secret key — a PRF, not a full VRF. Poseidon as a keyed hash satisfies this under the same assumptions used throughout the protocol, and evaluates in a few hundred Plonky3 constraints. On-chain verifiable selection (validator set, committee, relay) requires unpredictability plus publicly verifiable proof of correct evaluation — a true VRF. EC-VRF (RFC 9381) is used for these, reducing to DLEQ/DDH with production implementations in tendermint-rs (already in the stack). EC-VRF verification happens on-chain by validators, not inside ZK circuits, so circuit cost is not a concern for these uses.

Epoch ID unlinkability: `beacon_T` rotates every epoch, so `epoch_id_T` values are computationally unlinkable across epochs without `sk`, regardless of `null_v` being a shared input.

All per-epoch VRF derivations use Poseidon with explicit domain separators:

Derivation	Expression
Epoch ID	`Poseidon(sk, beacon_T, null_v, "epoch_id")`
Permutation	`Poseidon(sk, beacon_T, "perm")`
Chop count	`Poseidon(sk, beacon_T, "chop_n")`
Epoch offset	`Poseidon(sk, "epoch_offset")`
Niche announce delay	`Poseidon(sk, item_hash, beacon_T, "niche_delay")`

Derivation	Expression
Committee token	`Poseidon(committee_sk, epoch_id_v, T, "continuity_token")`
Leaf salt	`Poseidon(sk, epoch_T, "leaf_salt")`

All derivations must use distinct (domain_sep, input) pairs for any `sk`. Domain separator collision check resolved — all derivations now have explicit (domain_sep, input_structure) pairs; collision argument reduces to standard Poseidon collision resistance (OQ-4 — closed).

The leaf salt is intentionally fixed per (node, epoch): it is the same value for every leaf node v writes within epoch T, but distinct across nodes (different `sk`) and across epochs (different `epoch_T`). This shared-within-epoch property is what makes sibling hashes in the Merkle tree opaque under fixed leaf padding (§4.6) — without a per-leaf-position differentiator beyond the salt, leaf positions remain unlinkable to specific event types from an external observer who has not seen the leaf body.

4.3 Identity Admission Cost

Problem: Free instant identity creation allows unlimited fake identities.

Tool: A VDF chain. Inherently sequential — more compute does not help.

```
vdf_proof_T = VDF_eval(vdf_proof_{T-1}, δ_identity)
VDF_verify_chain([vdf_proof_{T-n}, ..., vdf_proof_T], n, δ_identity) = true
```

Admission is binary. At the epoch when admission completes, the node's first handoff must include a valid Statement 5 non-suspension proof and a valid `commit_T` ZK proof. A node whose `null_v` is already in SUSP_SMT is rejected at admission without a separate investigation. Zero CF weight and zero routing weight during the admission window.

Interaction checkpoints occur at VRF-determined epochs within the admission window. At each checkpoint: PSI handshake with a randomly selected existing node, gossip vector exchange, receipt from that exchange.

Gap tolerance: A missed VDF proof resets chain progress to zero. Accumulated temporal depth from prior completed chains is retained.

Default n = 24 (one day). High-security n = 168 (one week).

4.4 Preference Privacy

Problem: Auditors need to verify properties of a node's preference vector without reading it.

Tool: A Pedersen commitment. Binding and hiding.

```
C_p(T) = p_v · G + r_p · H
```

The preference vector `p_v` is fixed for the duration of an epoch. Updates to `p_v` are only permitted at epoch transitions. Enforcement is structural: `C_p(T)` is published at the start of the epoch with binding to a specific `p_v(T)`; Pedersen's binding property means a different `p_v'(T)` cannot open the same commitment. At the next handoff, §4.9.5 Statement 3 requires $\|p\_v(T) - p\_v(T-1)\|_1 \leq \Delta$, which constrains the magnitude of inter-epoch change but does not permit intra-epoch revision of the committed value.

4.5 Preference Obfuscation in Transit

Problem: Even without reading a preference vector, an adversary receiving it multiple times can correlate dimensions across epochs and infer preference count from vector size.

Tool: Per-epoch permutation plus variable-size transmission.

```
π_v(T)  = Poseidon(sk, beacon_T, "perm")                          // sk is
per-node, so π_v is independent across nodes
n_v(T)  = n_base + Poseidon(sk, beacon_T, "chop_n") mod n_jitter
```

Variable chopping: Exactly `n_v(T)` elements are transmitted, selected from the permuted positive preference set, padded with cover items if needed. Neither the number of real preferences nor which preferences are real is inferrable from the transmitted vector size.

Only positive preference weights are included in gossip vectors. Negative weights stay local.

Cover items:

```
cover_weight(item) = Uniform(0, cover_scale) / log(1 + trust_total(item) / c)
```

`cover_scale` is a per-deployment calibration constant bounding the maximum cover weight injected; `c` is the small constant from §3.4's `item_weight` (the DSybil cap normalizer) that prevents log explosion when `trust_total` is near zero. Both require empirical tuning (calibration entries added to OQ-list under §10).

Chopping vs. Laplace DP (mutually exclusive per deployment):

- **Chopping** (niche-friendly): transmit `n_v(T)` elements as above
- **Laplace DP** (mainstream, formal guarantee): `gossip_v(T)[i] += Laplace(0, S/ε)` with `‖p_v‖₁ = 1` enforced and L1 sensitivity S = 2. Laplace DP does **not** by itself guarantee sign preservation — the noise distribution is unbounded. Sign preservation is an additional constraint applied as a separate constraint below, which slightly weakens the pure-DP guarantee.

Sign preservation constraint:

```
|noise_i| < |p_v[i]|  for all i where p_v[i] ≠ 0
```

Enforced by per-dimension noise truncation: sample noise with σ such that the 99.7th percentile lies below `|p_v[i]|` for active dimensions, and reject-resample if a draw violates the constraint. This biases the noise away from sign-flipping while preserving most of the DP guarantee for moderate-magnitude preferences.

Niche item announcement delay:

```
announce_delay_v(item) = Poseidon(sk, item_hash, beacon_T, "niche_delay") mod
max_delay_epochs
```

This delay is a deliberate privacy mechanism: it intentionally degrades the announcement-timing behavioral signal for niche items (the very items that would otherwise be uniquely identifying). The detection cost is acknowledged — see §6.2's transport-profile-conditioned fingerprint definitions, which already account for which timing signals survive each profile.

4.6 Tamper-Evident Behavioral History

Problem: A node needs to prove to auditors that its behavior was within protocol limits without revealing its details.

Tool: A Merkle tree whose leaves are built from peer attestations collected passively as a side effect of normal protocol traffic.

Leaf structure:

```
leaf(ANNOUNCEMENT, T)  = H("ANN"  ‖ T ‖ set_of_peer_signed_observations ‖
salt_v)
leaf(PULL_RESPONSE, T) = H("PULL" ‖ T ‖ set_of_peer_signed_receipts       ‖
salt_v)
leaf(AUDIT_RESP, T)    = H("AUD"  ‖ T ‖ auditor_published_results          ‖
salt_v)
leaf(RATE_LIMIT, T)    = H("RL"   ‖ T ‖ announcement_token_set(T)          ‖
salt_v)

announcement_token(v, X, T) = Poseidon(sk, item_hash(X), beacon_T, "ann_token")

`announcement_token_set(T)` is the set of all tokens generated by node v for
items announced during epoch T. The RATE_LIMIT leaf commits to this set; the
auditor verifies only the cardinality against the rate limit ceiling, without
learning which items the tokens correspond to. Each token is unique per (node,
item, epoch) — `beacon_T` prevents reuse across epochs and `item_hash(X)`
prevents token sharing across items. A node cannot manufacture tokens for
announcements it did not make without `sk`, and cannot omit tokens for
announcements it did make without the ZK proof over the leaf set failing. The
auditor committee verifies the cardinality of `announcement_token_set(T)`
against the rate-limit ceiling during the handoff process (§6.4), via a ZK
proof over the RATE_LIMIT leaf — see also §7.4 for how rate-limit compliance
feeds the per-epoch score.

salt_v = Poseidon(sk, epoch_T, "leaf_salt")
M_v(T) = MerkleRoot(padded_leaves_v(T))
         // padded to a fixed protocol-wide maximum leaf count
```

Partial reveals are safe: Fixed padding plus per-leaf VRF salts ensure sibling hashes are opaque.

4.7 Cross-Epoch Identity Continuity

Problem: After epoch rotation, the auditor committee does not know a node's new `epoch_id` belongs to the same node.

Tool: A zero-knowledge proof of Poseidon PRF continuity, submitted to the arbitration committee only.

```
ZK { sk :
    Poseidon(sk, beacon_T,   null_v, "epoch_id") = epoch_id_T
    Poseidon(sk, beacon_T-1, null_v, "epoch_id") = epoch_id_{T-1}
    null_v = Poseidon(sk, "null_v")     // verified inside the ZK circuit; see
§4.9.5 Statement 5 for how validity and SUSP_SMT non-membership are jointly
proven without revealing null_v
}
```

`null_v` is a private witness — not a public output of this proof. Continuity proofs are never published to the public blockchain. Nodes that choose full unlinkability submit no continuity proof and sacrifice cross-epoch reputation accumulation. Sub-second on Plonky3.

Enriched rolling chain commitment:

```
rolling_chain_commitment(T) = Poseidon(
    rolling_chain_commitment(T-1),
    zk_continuity_proof(T),
    audit_interactions(T),
    SUSP_SMT_root_T
)
```

This ties each epoch's commitment to the suspension state at that epoch, preventing replay of old non-suspension proofs against a newer SMT.

Committee tokens: When a node reaches an elevated alert level, the committee issues a continuity token:

```
token_v(T)  = Poseidon(committee_sk, epoch_id_v, T, "continuity_token")
commit_v(T) = H(token_v(T))
encrypted_token = Enc(pk_v(T), token_v(T))    // pk_v(T) is node v's ephemeral
encryption public key for epoch T, derived from sk and published alongside
epoch_id_T; private decryption key never leaves the device
```

The node decrypts the token and must incorporate it into the next handoff ZK proof. A node that missed a handoff cannot produce a valid proof because it would need the token. The chain becomes a tamper-evident audit history — gaps cannot be hidden because every subsequent commitment depends on all prior ones.

4.8 PSI Cache Decay

```
psi_cache[new_epoch_id] = psi_cache[old_epoch_id] × λ_proof     (λ_proof ≈
0.95)
cache_weight(U, T)      = base_weight × λ_noproof^(T − last_verified_T)
(λ_noproof ≈ 0.7)
```

Both λ values require empirical calibration (OQ-20). The two are distinguished by the evidence available at decay time: `λ_proof` applies when a continuity proof from the peer in the new epoch confirms identity carry-over — light decay (~0.95) since the peer is still verifiably the same node. `λ_noproof` applies when continuity has not been verified for the peer — heavier decay (~0.7) reflecting the higher uncertainty that the cached PSI overlap still applies.

4.9 Nullifier, Suspension Persistence, and Dark Node Closure

4.9.1 The Nullifier

```
null_v = Poseidon(sk, "null_v")
```

`null_v` is:

- **Local** — computed entirely on device, never transmitted in normal operation
- **Private** — lives inside ZK proof circuits as a private witness only
- **Deterministic** — same `sk` always produces the same `null_v`
- **Stable** — does not change across epoch rotations; the beacon is not an input

- **Structurally intrinsic** — a valid `epoch_id_T` cannot be produced from `sk` without `null_v`, because `null_v` is an input to the epoch ID derivation. Enforced by the ZK proof at each handoff (§4.9.5 Statement 5) and non-malleability (§4.9.7): the circuit constrains `null_v` and `epoch_id_T` to share the same `sk` wire, so a forged or substituted `null_v` cannot satisfy the proof.

4.9.2 The SUSP_SMT

A Sparse Merkle Tree maintained as part of the public chain state. Each leaf position corresponds to a possible `null_v` value. A leaf is non-empty if and only if that `null_v` has been inserted following a SUSPENDED verdict. The root `SUSP_SMT_root_T` is included in every block header. The tree is append-only — leaves are never removed.

Non-membership is proven by a Merkle path from the root to the empty leaf at the position `null_v` would occupy if present. This path is computed inside the ZK circuit with `null_v` as a private input. The verifier sees only proof validity, not which position was checked.

4.9.3 The DECRYPTION_SMT

A Sparse Merkle Tree maintained alongside SUSP_SMT. Each leaf position corresponds to a possible `dec_nullifier` value:

```
dec_nullifier = Poseidon(verdict_hash, null_v)
```

A leaf is non-empty if and only if a `null_v_decryption` transaction has been finalized for that verdict. The root `DECRYPTION_SMT_root_T` is included in every block header. The tree is append-only.

Purpose: Enforces one decryption per verdict at consensus level. A second `null_v_decryption` transaction referencing the same `verdict_hash` produces the same `dec_nullifier`, which is already in the tree. Honest validators reject the block containing a duplicate. Collision resistance of `dec_nullifier` under adversarial `null_v` reduces to standard Poseidon collision resistance; `verdict_hash` is fixed on-chain before `null_v` is recovered, so chosen-input attacks are not applicable (OQ-5 — closed).

4.9.4 Forward-Secure Nullifier Commitment

Problem: `null_v` must be available for extraction after a suspension verdict even if the node is offline, without enabling covert extraction by the committee.

Tool: Boneh-Franklin Identity-Based Encryption over BLS12-381 (Boneh & Franklin, CRYPTO 2001), encrypted in parallel to `N_fallback` independent committees. Each ciphertext uses its committee's threshold BLS public key as the IBE master public key; the identity string is `"SUSPEND epoch_id_T"`; the aggregated threshold BLS signature on that string is the IBE private key for that identity; `null_v` is the plaintext; the i-th ciphertext slot `c_T^{(i)}` is one component of the composite `commit_T`. Security of each slot reduces to DBDH in the random oracle model. The implementation falls out of blst (already in the stack).

The "master key" terminology is inherited from generic IBE and is not held by any single party. Each `threshold_BLS_pk_T^{(i)}` is jointly produced by its VRF-selected committee via DKG — no member holds the corresponding master secret. Producing an IBE decryption key for the identity `"SUSPEND epoch_id_T"` against the i-th slot requires ⌋K_committee/2⌊+1 honest members of `committee_T^{(i)}` to threshold-sign that statement. Safety reduces to the honest-majority Assumption A2 applied per committee.

DST alignment requirement: The domain separation tag used for hash-to-curve in BLS signing and in IBE key derivation must be explicitly defined and the SUSPEND-IBE DST kept distinct from all other BLS signing DSTs used by the committee (block signing, gossip authentication, relay attestation). RFC 9380 governs the construction. Validators reject committee BLS signatures that use the SUSPEND-IBE DST outside the explicit verdict reveal flow, preventing a committee member from producing an IBE decryption key by signing an unrelated message. A CI-enforced collision test on the DST table is required before release.

```
// At epoch start, for each i ∈ {0, ..., N_fallback-1}:
committee_T^{(i)}        = VRF(beacon_T ∥ "audit_committee" ∥ epoch_id_v ∥ i,
...)
                           // disjoint-membership constraint: committee_T^{(i)}
∩ committee_T^{(j)} = ∅ for i ≠ j
threshold_BLS_pk_T^{(i)} = DKG_output(committee_T^{(i)})   // public, derivable
by anyone

// Node constructs N_fallback parallel ciphertexts, all encrypting the same
null_v:
commit_T = ( c_T^{(0)}, c_T^{(1)}, ..., c_T^{(N_fallback - 1)} )
    where  c_T^{(i)} = ForwardCommit(null_v, epoch_id_T,
threshold_BLS_pk_T^{(i)}; r_commit_T^{(i)})
```

Each slot `c_T^{(i)}` is decryptable if and only if the holder possesses a valid threshold BLS signature from `committee_T^{(i)}` on the statement `"SUSPEND epoch_id_T"`. The N slots are independent ciphertexts of the *same* `null_v` plaintext; Statement 5 (§4.9.5) certifies plaintext equality across slots inside the handoff ZK proof.

Decryption: Given a valid threshold BLS aggregate signature `σ_i` from `committee_T^{(i)}` on verdict `"SUSPEND epoch_id_T"`, anyone can compute:

```
null_v = ForwardCommit.Decrypt(c_T^{(i)}, σ_i)
```

Any single slot suffices to recover `null_v`. No committee member needs to be online. No escrow holder needs to act.

Step-by-step recap. (1) At epoch start, `N_fallback` committees are VRF-selected with disjoint membership; (2) each committee runs an independent DKG to produce its `threshold_BLS_pk_T^{(i)}` (all public, derivable by anyone); (3) the node computes `commit_T` as the N-tuple of forward commitments above and publishes it on chain; (4) if and only if a SUSPEND verdict is finalized, the threshold-aggregate signature σ_i on `"SUSPEND epoch_id_T"` from whichever committee completed first becomes the IBE decryption key for slot `c_T^{(i)}`; (5) anyone holding that signature plus the public `c_T^{(i)}` can compute `null_v = ForwardCommit.Decrypt(c_T^{(i)}, σ_i)`. The node need not be online after step (3); no escrow holder participates.

Why forgery is impossible: Each `c_T^{(i)}` is tied to `threshold_BLS_pk_T^{(i)}`, the public key of a specific VRF-selected committee for this node this epoch. Producing a valid aggregate signature under any one of those keys requires ⌊K_committee/2⌋ + 1 secret key shares held only by legitimately selected members of that committee. Disjoint membership across committees means no member can sign for more than one slot.

Why liveness is preserved. Stalling the suspension entirely now requires a colluding minority (more than K_committee − ⌊K_committee/2⌋ − 1 non-revealing members) in *every one* of the N_fallback committees simultaneously. Under Assumption A2, this joint probability decays exponentially in N_fallback. A single stalled committee no longer blocks the verdict — control passes to the next slot per §4.9.6.

Why confidentiality is not weakened. The security of `commit_T` is the maximum (i.e., the *weakest*) of the per-slot security levels. Because each slot independently reduces to DBDH under its own committee's honesty, and disjoint membership rules out cross-slot collusion of identical sets, the weakest slot is no weaker than the single-committee construction would have been at N=1. Multi-recipient adds liveness without trading confidentiality.

Parameter choice. `N_fallback = 3` is the suggested starting value for v1: primary plus two fallbacks tolerates two simultaneous committee-level stalls. Ciphertext size grows linearly: each `c_T^{(i)}` is a single BLS12-381 G2 element pair (≈96 bytes), so `commit_T ≈ 96 × N_fallback` bytes. Committee-side overhead grows similarly — `N_fallback × K_committee` nodes run DKG per epoch (only the primary actively audits; fallbacks hold idle keys). The audit workload itself is unchanged.

A node could attempt to evade by simply omitting `commit_T` from its handoff — but a missing `commit_T` is itself a visible chain artifact (handoff rejection per §4.9.10) and a refusal-to-sign by the committee for a legitimate SUSPEND attempt produces a non-revealing-member signal that triggers oversight (§4.9.6, §4.9.8 watchdog). Both evasion paths are detectable and penalized.

ForwardCommit security resolved as Boneh-Franklin IBE over BLS12-381 (Boneh & Franklin, CRYPTO 2001); reduces to DBDH in the random oracle model (OQ-2 — closed).

4.9.5 ZK Proofs

All four statements are evaluated together in the handoff ZK proof each epoch. (Statement 4 was retired during earlier spec consolidation; the gap in numbering is preserved to keep cross-references in implementation work stable.)

Statement 1 — Preference norm validity (Bulletproof range proof):

```
‖p_v‖₁ ≤ S
```

Statement 2 — Directional consistency (inner product argument):

```
∀ announced item X with noisy_rating > 0 : p_v[X] > 0
∀ announced item X with noisy_rating < 0 : p_v[X] ≤ 0
```

Statement 2 circuit has not been benchmarked for this construction.

Statement 3 — Temporal consistency (vector commitment difference): the ZK proof verifies that `C_p(T)` and `C_p(T-1)` open respectively to `p_v(T)` and `p_v(T-1)` (private witnesses), with the `‖p_v(T) - p_v(T-1)‖₁ ≤ Δ` bound below. Pedersen binding ensures the same commitments cannot open to different vectors, so the prover cannot substitute a fabricated `p_v(T)` that satisfies the bound but is not the committed value.

```
‖p_v(T) - p_v(T-1)‖₁ ≤ Δ
```

Statement 5 — Non-suspension and forward commitment (Poseidon evaluation + SMT non-membership + ForwardCommit certification across all N_fallback slots):

```
null_v      = Poseidon(sk, "null_v")
epoch_id_T  = Poseidon(sk, beacon_T, null_v, "epoch_id")
null_v ∉ SUSP_SMT_root_T
∀ i ∈ {0, ..., N_fallback - 1} :
    c_T^{(i)} = ForwardCommit(null_v, epoch_id_T, threshold_BLS_pk_T^{(i)};
r_commit_T^{(i)})
```

All N_fallback ForwardCommit checks share the same `null_v` wire as the first three. A node cannot include divergent plaintexts across slots — each `c_T^{(i)}` is constrained to encrypt the *same* `null_v` value within the circuit, so a SUSPEND verdict completed against any slot yields the same canonical nullifier. Extended Statement 5 circuit cost grows linearly in N_fallback (additional pairing checks per slot); mobile-hardware benchmarking required before deployment (OQ-3).

Fallback — binary ratings: If a formal influence bound is required before deployment, Config E (binary ratings) narrows the gap to the original DSybil assumptions, though a clean formal guarantee is not claimed — see OQ-10.

4.9.6 Commit-Reveal Verdict Flow

A suspension verdict follows a two-phase commit-reveal process. Committee members lock in their decision publicly before decryption becomes possible. The flow runs against the active committee slot — `committee_T^{(0)}` by default, advancing to `committee_T^{(i+1)}` if slot i stalls past its window. The canonical specification is here; §4.9.10 summarizes the full suspension flow end-to-end.

```
ACTIVE SLOT (initially i = 0; advances on stall, see "Fallback activation"
below):

COMMIT PHASE:
    Each member j of committee_T^{(i)} publishes to public chain:
        verdict_commit_j = {
            slot_index:             i,
            epoch_id_committee_j: <member epoch ID>,
            commit:                 H(BLS_share_j ‖ verdict ‖ nonce_j),
            sig:                    Sign(epoch_id_committee_j, H(commit ‖ T ‖ i))
        }

    All K_committee commits from committee_T^{(i)} must appear before reveal
phase begins.
    A verdict is invalid without all commits on-chain.
    Members cannot change their verdict after committing.

REVEAL PHASE (after all commits finalized):
    Each member j publishes:
        verdict_reveal_j = {
            slot_index:             i,
            epoch_id_committee_j: <member epoch ID>,
            bls_share_j:            <BLS secret share contribution under
threshold_BLS_pk_T^{(i)}>,
            verdict:                "SUSPEND" | "PASS",
            nonce_j:                <random nonce>,
            sig:                    Sign(epoch_id_committee_j,
                                         H(bls_share_j ‖ verdict ‖ nonce_j ‖ T ‖
i))
        }
```

```
    Validators verify:
        H(bls_share_j ‖ verdict ‖ nonce_j) = commit from verdict_commit_j ✓
        BLS share is valid under threshold_BLS_pk_T^{(i)} ✓

AGGREGATION (by anyone, permissionless):
    Once ⌊K_committee/2⌋ + 1 valid reveals from slot i are on-chain:
        σ_i    = aggregate(bls_share_j₁, ..., bls_share_jₜ)
        null_v = ForwardCommit.Decrypt(c_T^{(i)}, σ_i)

    Anyone submits null_v_decryption transaction:
        {
            verdict_hash:    H(committee_verdict),
            slot_index:      i,
            epoch_id:        <suspended node's last epoch_id>,
            null_v:          <recovered value>,
            dec_nullifier:   Poseidon(verdict_hash, null_v),
            proof:           Verify(threshold_BLS_pk_T^{(i)}, "SUSPEND
epoch_id_T", σ_i) ✓
                             AND ForwardCommit.Verify(c_T^{(i)}, null_v, σ_i)
✓
        }

    Validators verify both checks.
    Insert dec_nullifier into DECRYPTION_SMT.
    Insert null_v into SUSP_SMT.
    Update both roots in block header.
```

Critical ordering property: Committee members commit to their verdict before the aggregate signature exists and before the slot's ciphertext is decryptable. The decision is irrevocably on-chain before anyone — including the committee — can see `null_v`. This prevents `null_v` from influencing the verdict and prevents covert extraction.

Non-revealing member: A committee member who commits but refuses to reveal is in violation of protocol. Consequence: non-response penalty applied to reputation; if threshold reveals from slot i are unavailable, the slot stalls and the fallback activation rule below moves the verdict to slot i+1 without aborting the suspension.

Fallback activation. The verdict flow runs against slot i = 0 by default. If ⌊K_committee/2⌋+1 valid reveals from `committee_T^{(0)}` are not on-chain within `W_primary` epochs of the verdict being invoked, validators automatically advance the active slot to i = 1, and `committee_T^{(1)}` is required to run its own commit-reveal against `c_T^{(1)}`. Successive fallbacks activate at `W_fallback` intervals (i ← i + 1) until either some slot completes or all N_fallback slots have been exhausted. The advance is deterministic and on-chain — no human or committee action is needed to trigger it. Whichever slot completes first yields the unique `null_v`; the DECRYPTION_SMT collision check (§4.9.3) rejects any later decryption transaction from a different slot, since `dec_nullifier = Poseidon(verdict_hash, null_v)` is identical regardless of which `c_T^{(i)}` was opened. Non-revealing members of any stalled slot receive the standard L3 Suspicious classification (§4.9.8) independent of whether the fallback ultimately succeeded.

Exhausting all N_fallback slots without completion is treated as a network-wide liveness incident: the recursive oversight chain (§4.9.8) escalates to a meta-committee selected independently of any of the original N_fallback committees.

4.9.7 Non-Malleability

A node cannot substitute a different `null_v'` while keeping the same `sk`.

Fabricated null_v': The circuit checks `null_v' = Poseidon(sk, "null_v")`. If fabricated, this check fails.

Consistent but distinct null_v': Satisfying both checks with `null_v' ≠ Poseidon(sk, "null_v")` requires a Poseidon collision. Probability negligible in `λ`.

Mismatched commit_T: Committing `null_v'` in `commit_T` while satisfying the first two checks requires the circuit to hold two different values for the same wire. Impossible.

Different sk: A new `sk'` produces a new `null_v'`. This is a new identity, not a changed nullifier. Full admission cost applies.

4.9.8 Watchdog Signals and Recursive Oversight

Any node can monitor the public chain for anomalous `verdict_commit` activity:

```
watchdog_signal if:
    count(verdict_commit transactions in epoch T)
    exceeds expected_rate(T) by threshold_watchdog
    AND no corresponding behavioral signals exist at matching rate
```

Expected rate is near zero most epochs — legitimate suspensions are rare. An anomalous burst of `verdict_commit` transactions with no behavioral signal justification is immediately visible before any `null_v` is recovered.

```
watchdog_signal = {
    epoch_id:          <signaling node epoch ID>,
    epoch_T:           <T>,
    observed_commits: <count>,
    expected_rate:     <local estimate>,
    sig:               <Sign(epoch_id, H(epoch_T ‖ observed_commits ‖ T))>
}
```

Watchdog signals from multiple nodes trigger the recursive oversight chain: a meta-committee surveys the original committee; their verdict process follows the same commit-reveal ordering. Each level locks in its decision before the next level's `null_v` becomes accessible.

Why the chain converges — Chernoff security amplification: The critical structural property is that each level's committee is drawn independently via VRF with cluster diversity constraints — different interest clusters at every level, and under transport profiles that preserve sub-epoch timing (Tor/I2P, §5.1) also different behavioral clusters. Given independence, if the adversarial fraction in the committee-eligible pool is q, then:

```
P(compromise at level d) = P(Bin(K_d, q) ≥ ⌊K_d/2⌋+1)
                         ≤ exp(−K_d · D(⌊K_d/2⌋+1)/K_d ‖ q))
```

by the Chernoff bound, where D is the KL divergence. This decays **exponentially** in K_d. With committee size growing linearly across levels (K_d = K_0 + d·ΔK), P(compromise_d) itself shrinks with d, giving doubly-exponential decay in depth. Cumulative escape probability is a product of independent per-level terms each strictly less than 1, so it decays exponentially in depth. Since any polynomial instance count (budget/cost) is dominated by exponential per-instance decay, a finite depth always suffices for any target success probability threshold. This closes OQ-49 as a structural question. What remains is empirical calibration of K_0 and ΔK for the target network (Phase 5 experiment 5.42). The independence guarantee is load-bearing: if committees at different levels were drawn from the same cluster, the Chernoff amplification would not apply. The eligible-pool fraction `q` itself is bounded by Assumption A2: an adversary that pushes `q` above 1/2 in the committee-eligible pool has by definition already broken the honest-majority assumption, at which point BFT consensus, the verdict process, and watchdog signals all collapse simultaneously. q-boundedness is therefore a corollary of A2, not a separate premise.

Transport-profile dependence. The argument's strength depends on which cluster axes are available for independence. Under the Tor/I2P profile, both interest and behavioral clusters are available (per §6.2), and the dual-cluster constraint gives the strongest independence multiplier. Under the Loopix profile, behavioral-cluster signal is too thin to provide an independent axis, and the argument reduces to interest-cluster diversity alone. To recover equivalent compromise probability under the Loopix profile, `K_d` is scaled upward to compensate for the lost independence dimension — concrete per-profile `K_0` and `ΔK` calibration is Phase 5 work (experiment 5.42, parameterized by profile).

Stalling minority: A minority of members in a single committee refusing to reveal stalls only that slot's verdict — control passes to the next fallback slot per §4.9.6 without aborting the suspension. Stalling the suspension entirely requires a colluding minority in *every* one of the N_fallback committees, whose joint probability under Assumption A2 decays exponentially in N_fallback (disjoint membership across committees rules out a single coalition appearing in multiple slots). Non-revealing members in any slot receive the L3 Suspicious classification regardless of whether a fallback slot ultimately completed.

4.9.9 Suspension Persistence Across Epoch Rotation

`null_v` is derived solely from `sk` and a fixed domain separator. The beacon is not an input. Rotating to a new epoch changes `beacon_T` and therefore `epoch_id_T`, but `null_v` is the same value in every epoch for every identity sharing the same `sk`. Once inserted into SUSP_SMT, it blocks all future epoch IDs from that key.

```
Epoch T1: epoch_id_T1 = Poseidon(sk, beacon_T1, null_v, "epoch_id") → null_v ∈ SUSP_SMT → FAIL
Epoch T2: epoch_id_T2 = Poseidon(sk, beacon_T2, null_v, "epoch_id") → null_v ∈ SUSP_SMT → FAIL
Epoch T3: epoch_id_T3 = Poseidon(sk, beacon_T3, null_v, "epoch_id") → null_v ∈ SUSP_SMT → FAIL
...
```

A SUSPENDED verdict is permanent by construction — not by rule.

4.9.10 Full Suspension Flow

Visual recap of §4.9.6 — no new content; included for orientation only.

```
Class 3 audit confirms misbehavior
OR majority handoff rejection
OR double-signing detected
    │
    ▼
Active slot i ← 0  (primary committee_T^{(0)})
    │
    ▼
Commit phase (slot i):
    All K_committee members of committee_T^{(i)} publish verdict_commit_j on
public chain
    Decision locked — visible to entire network
    │
    ▼
Reveal phase (slot i):
    All K_committee members publish verdict_reveal_j on public chain
    BLS shares publicly verifiable under threshold_BLS_pk_T^{(i)}
    │
    ▼
Quorum reached within window? ──── NO ────▶ i ← i + 1   (advance to next
fallback)
    │                                         │
    │                                         ├─▶ if i < N_fallback :
restart commit phase against committee_T^{(i)}
    │                                         └─▶ else : escalate to
recursive oversight (§4.9.8)
    │ YES
    ▼
Aggregation (by anyone):
    σ_i    = aggregate BLS shares from slot i
    null_v = ForwardCommit.Decrypt(c_T^{(i)}, σ_i)
    │
    ▼
null_v_decryption transaction submitted:
```

```
        dec_nullifier = Poseidon(verdict_hash, null_v)
        Inserted into DECRYPTION_SMT
        null_v inserted into SUSP_SMT
        Both roots updated in block header
        |
        ▼
    Next epoch: node cannot produce valid Statement 5
    Handoff rejected. Epoch transition impossible.
    All future identities from same sk permanently excluded.
```

---

5. Network

5.1 Transport

PrivaCF defines a **transport interface** rather than mandating a specific transport. Each deployment selects a transport profile that satisfies the required properties below; properties marked optional vary across profiles. Instances do not federate across profiles — a deployment running the Loopix profile and one running the Tor/I2P profile are separate networks with separate Sybil pools.

Required transport properties (any conformant profile must satisfy):

- **IP↔epoch_id unlinkability.** A network-level observer cannot link a client's real IP to its rotating `epoch_id_T` identities.
- **Message integrity and authentication.** Tampering with a message in transit is detectable by the recipient.
- **Replay protection.** A captured message cannot be successfully replayed against the protocol.
- **Activity cover.** No external observer can distinguish a node's active-send periods from idle periods at finer than epoch granularity, when the node is online.

Optional transport properties (profile-dependent):

- **Per-hop timing decorrelation.** Sub-epoch send times are unobservable by external observers.
- **Fixed-size packets.** Packet sizes are uniform and content-independent, eliminating traffic-size analysis.
- **GPA (global passive adversary) resistance.** Formal anonymity holds against an adversary observing the entire wire.

Reference profile (default for MVP): self-mixing Loopix

Loopix/Sphinx mixnet (Piotrowska et al., CCS 2017) provided by PrivaCF nodes themselves — the mix layer is not an external dependency. All required and all optional transport properties are satisfied. Every unicast message — PSI handshakes, audit responses, verdict commits, gossip vector pushes — is sent as a Sphinx-onion-encrypted packet through a sequence of PrivaCF mix-role nodes, each applying a Poisson-distributed per-hop delay. No mix node learns more than the next hop. Sender, receiver, and relationship anonymity are formally analyzed under the Loopix model. See §5.1.1 for the mix-role / client-role split that makes self-mixing possible.

Loop and drop cover traffic is the mechanism that satisfies the activity-cover requirement under this profile. Every mix-capable node emits traffic at a constant Poisson rate regardless of real activity — loop covers (messages to self via the mix) and drop covers (discarded at destination). Cover traffic calibration (§5.8) sets the base Poisson rate; the formal anonymity guarantee holds under this constant-rate assumption.

Single-Use Reply Blocks (SURBs): All request-response exchanges (PSI handshakes, audit requests) include a pre-built SURB so the responder can reply anonymously without knowing the requester's mix path.

Dandelion++ (Fanti et al., SIGMETRICS 2018) is retained for the epidemic broadcast (fluff) phase of gossip propagation, where Loopix's point-to-point model does not apply. Stem phase: random linear relay ($p \approx 0.9$ to continue). Fluff phase: epidemic broadcast.

Provider role (the publicly-reachable mix-role node that buffers inbound messages for a client) sidesteps NAT traversal for home nodes — no inbound connection is required from the client. Katzenpost remains a viable reference implementation of the mix-node software, integrated into PrivaCF's node binary rather than running as a separate external network.

Alternate profile: Tor/I2P

Tor (onion routing) or I2P (garlic routing) provides the required transport properties only. IP↔epoch_id unlinkability, integrity, authentication, and replay protection are inherited from the underlying transport. **Activity cover** is satisfied by a protocol-mandated low-rate Poisson cover-packet stream emitted over the Tor/I2P transport — without this shim the profile is non-conformant. Optional properties (timing decorrelation, fixed-size packets, GPA resistance) are not provided by this profile; users with stronger threat models layer their own OPSEC (additional VPN hops, dedicated routing infrastructure).

Selection of this profile trades the Loopix profile's GPA resistance for stronger sub-epoch behavioral signals available to the Sybil-detection machinery (§6.2, §7.1a). The behavioral-cluster computation in §6.2 and the dual-cluster Chernoff argument in §4.9.8 hold their original form under this profile.

Clearnet profile

Development and testing only. Does not satisfy any of the required transport properties. Production deployments must select Loopix or Tor/I2P.

Retry on failure: Timeout-only across all profiles. After `k_retry` failures, fall back to direct broadcast.

5.1.1 Mix-role / Client-role split (self-mixing Loopix profile)

Self-mixing requires the mix layer to be implemented by PrivaCF nodes themselves. Each node may operate in two protocol-unlinkable roles on the same machine:

Mix-role identity. A long-lived router identity, IP-bound and publicly advertised in a RouterInfo-style record. Used for forwarding others' Sphinx-encrypted traffic. Only a subset of nodes opt into the mix role — uptime and bandwidth requirements rule out most mobile or intermittent clients. Mix-role identities do not rotate per epoch; they persist as long as the node continues advertising as a mix.

Client-role identity. The epoch-rotated `epoch_id_T` defined in §4.2, used for PrivaCF protocol participation (gossip, PSI, audits, handoffs). Rotates every epoch and is unlinkable across epochs without `sk`.

The two roles share a machine but are protocol-unlinkable: the mix-role identity is IP-bound and public; the client-role identity is mix-routed and pseudonymous. An observer who knows a node's mix-role IP cannot determine which client identity it operates, and vice versa.

Provider role. A further-restricted subset of mix-role nodes that are publicly reachable and offer message buffering for offline or NAT'd clients. Providers must satisfy stricter availability requirements than ordinary mix-role nodes.

Circular dependency note. Mix-layer Sybil resistance is bounded by Assumption A2 (honest majority by weight). Because mix nodes are themselves PrivaCF participants, the protocol's Sybil-resistance machinery is what keeps the adversarial fraction in the mix-eligible pool below the threshold required for Loopix anonymity. Conversely, mix anonymity is what enables much of the protocol's privacy machinery. The two are mutually load-bearing under A2; failure of either collapses the other.

Bootstrapping the initial mix set (deployment concern, not protocol). Until enough PrivaCF participants have built temporal depth to satisfy A2 in the mix-eligible pool, the circular dependency above cannot self-stabilize. This is a one-shot deployment problem rather than a protocol problem and is out of scope for the protocol spec, but worth recording:

- **Externally trusted initial mix set.** The deployment publishes a small initial mix set drawn from operators whose identity, accountability, and uptime are externally verifiable — for example, a published consortium of known infrastructure operators, university CS departments, or existing privacy-network operators (Tor, Nym, Mixmaster legacy). The set is documented and verifiable out of band; mix anonymity during bootstrap is conditional on the consortium's honesty.
- **Transition criterion.** The deployment publishes a transition criterion — an A2-equivalent measurable condition over the organically-grown mix-eligible pool — at which the externally trusted set's privileged status sunsets and selection becomes purely protocol-internal.
- **Honest bootstrap framing.** Until the transition criterion is met, the Loopix anonymity guarantee for the deployment is **conditional** on the bootstrap consortium and should be advertised that way. This is no worse than how Tor's and Nym's mix anonymity bootstraps in practice — both rely on a real set of operators whose honesty is externally evaluated — but PrivaCF should be explicit about the bootstrap dependency rather than treating A2 as self-evidently satisfied at deploy time.

The protocol itself makes no assumption about how the initial set is chosen; this paragraph is a deployment recommendation, not a normative protocol requirement.

5.2 Uniform Message Frames

//This is good even if we decide to not use Loopix
Sphinx packets are fixed-size by construction — payload length is padded to a fixed maximum at the sender before onion encryption. An observer on the wire sees fixed-size encrypted blobs and cannot distinguish message types or read contents. A gossip vector push, a Class 3 audit response, a `verdict_commit`, and a Loopix loop or drop cover are indistinguishable on the wire. Frame size estimated 4–8 KB (OQ-30).

Noise Protocol sessions are retained only for any direct connections used in development and clearnet testing; they are redundant for mix-routed production traffic where Sphinx provides per-hop encryption.

Note on audit response timing: Because Loopix's per-hop Poisson delay dominates response latency and node processing time is negligible relative to mix routing delay, audit response timing is not a usable detection signal. Audit response *rate* (fraction of challenges responded to) remains a meaningful signal and is captured in §6.2.

5.3 Node Discovery and Cluster Re-Discovery

```
NODE DISCOVERY FLOW
─────────────────────────────────────────────────────────

New node                    Network
    │                            │
    │── publish first VDF ──────►│ (public chain)
    │                            │
    │◄── VRF-selected observers──┤ submit first-observation
    │    from diff clusters      │ reports to committee
    │    observe + report        │ chain only
    │                            │
    │── complete n epochs ──────►│
    │   + interaction            │
    │   checkpoints              │
    │                            │
    │◄── admitted ───────────────┤ public chain records
    │                            │ admission decision only

PEER DISCOVERY (ongoing, each epoch)
─────────────────────────────────────────────────────────
    │
    ├── after epoch rotation: ZK continuity proof submitted?
    │       YES: committee attests continuity to existing peers
    │            update PSI cache with λ_proof decay
    │            skip re-discovery for confirmed peers
    │       NO:  proceed to candidate selection below
    │
    ├── candidate pool ←  random DHT draws (bridge tier)
    │                  +  referrals queued from prior epoch's
    │                     failed PSI, filtered by local MinHash
    │                     estimate; skip if in visited_set_T
    │                     or hop_count = 0
    │
    ├── run up to n_discovery PSI attempts in parallel
    │
    │       on PASS (similarity ≥ θ_cluster):
    │           add to PSI peer set
```

```
      │
      │       on FAIL:
      │           responder optionally attaches referral list:
      │               { epoch_id, minhash_signature, hop_count }
      │           initiator queues referrals for next epoch
      │
      └── bridge peer: refresh from random DHT each epoch

INCOMING PSI RATE LIMIT
─────────────────────────────────────────────────────────

    if incoming_psi_count(T) > n_psi_in:
        drop excess requests silently
        // no reputation penalty — absence of psi_ack is
        // indistinguishable from network delay at the initiator
```

5.4 Peer Selection — Gossip-Driven Discovery + Asymmetric PSI

Peer selection is a three-step process. Direct chain queries bootstrap discovery; gossipped referrals extend it organically across the network; PSI confirms genuine similarity.

Step 1 — LSH chain query (bootstrap)

When a node registers its `epoch_id` on the public chain, it also publishes a MinHash signature computed over its announced items. MinHash approximates Jaccard similarity: for two item sets A and B, each MinHash function returns the same value with probability `|A∩B| / |A∪B|`. Using k hash functions partitioned into b bands, two nodes sharing at least one full band are likely candidates.

A node bootstrapping its peer set queries the chain for `epoch_id`s in the same LSH band. This is a one-time or post-rotation lookup; steady-state discovery is handled by the referral mechanism below.

Step 2 — Gossip referrals

On a failed PSI (similarity below θ_cluster), the responder optionally attaches a referral list to the response:

```
referral = { epoch_id, minhash_signature, hop_count }
```

`minhash_signature` is the referred node's published chain value — no new private data is shared. The initiator estimates local Jaccard against its own MinHash before queuing the referral, discarding clearly non-similar candidates before spending a full PSI run. Loop prevention: skip any `epoch_id` already in `visited_set_T`; decrement and discard at `hop_count = 0`.

Over epochs, this creates organic convergence — similar nodes find each other transitively without central coordination and without formal cluster registration.

Step 3 — PSI confirmation (Pinkas, Rosulek, Trieu & Yanai, USENIX Security 2018)

Full asymmetric PSI is run against each candidate, using the node's complete private item set. If similarity ≥ θ_cluster the candidate joins the PSI peer set. The handshake is Loopix mix-routed (Appendix I). The responder issues a signed `psi_ack` on receipt regardless of outcome; the initiator retains it as a ZK witness for cross-cluster attempt rate.

Rate limiting and overload

`n_discovery` bounds parallel outgoing PSI attempts per epoch. It is conservative by default and should be calibrated against network size, observed bandwidth, and latency (OQ-23). A node that receives more than `n_psi_in` incoming PSI requests in an epoch drops excess silently — no reputation penalty applies, as absent `psi_ack` is indistinguishable from mix routing delay at the initiator. Both parameters should be set conservatively until empirical data from Phase 2 deployment informs tighter bounds; over-aggressive discovery degrades into a DoS vector against well-connected nodes.

Privacy properties

MinHash publication leaks only approximate similarity over already-public announced items. Referrals redistribute already-public chain data. PSI operates on the full private item set with strong cryptographic guarantees: neither party learns the other's non-overlapping items. The responder does not learn the initiator's identity (Loopix) so sharing a referral list carries no additional risk to the responder.

`θ_cluster` is community-type dependent; starting range: 0.1–0.3 (OQ-23).

5.5 Relay Submission

```
relay_T = VRF(beacon_T ‖ "relay" ‖ epoch_id_v,
              constraints = [
                  different_behavioral_cluster_from_submitter,
                  reputation ≥ median
              ])
```

Relay service contributes positively to per-epoch score via `w_relay`.

5.6 Staggered Epochs

```
offset_v = Poseidon(sk, "epoch_offset") mod epoch_duration
```

No inter-node coordination required for epoch transitions.

5.7 Two-Tier Peer Selection

```
PSI peer tier:  n_peers · Jaccard PSI confirmed · grown organically via gossip
referrals · persistent via ZK continuity
Bridge tier:    1 peer  · random DHT · refreshed each epoch
```

Cluster-specific behavior activates only when `|peers_v| ≥ k_min` (§7.4). During peer discovery, a node below k_min operates on global trust_total only until its PSI neighborhood reaches the minimum size.

Trust attenuation by hop distance (Stannat et al. 2021):

```
cf_weight(vector_from_u) = base_weight × μ^hop_distance(v, u)
```

This attenuation is one of the primary defenses against remote Sybil influence: a node many hops away can contribute at most `μ^k` of base weight regardless of reputation. μ calibration required empirically (OQ-21).

Note on bridge peer opt-out: A user whose interest cluster attracts no sybil attention and who wishes to eliminate bridge-tier sybil influence entirely may set their bridge peer weight to zero. This reduces expected sybil influence on their recommendations to zero (for the cluster tier) at the cost of losing cross-cluster discovery signal. See §7.1b for the formal decomposition. This is a per-node local setting and requires no protocol changes.

5.8 Communication Rhythm

Signal	Trigger	Rate limit	Routing
Gossip vector push	T_send = epoch_start + offset + Uniform(0, 0.3 × epoch)	1 per epoch (hard)	Loopix mix path
Item announcement (mainstream)	Positive interaction + random delay	~20/epoch	Dandelion++ stem → fluff broadcast
Item announcement (niche)	Positive interaction + VRF-derived epoch delay	Per item	Dandelion++ stem → fluff broadcast
Receipt	Epoch end (batched)	1 batch per epoch	Loopix mix path
Rewind signal	q_v(T) drop correlated with recent gossip cohort	1 per epoch per node; max 1 Class 3 trigger per N_rewind epochs	Dandelion++ stem → fluff broadcast
On-chain transaction (M_v, C_p, score)	Every n_commit epochs via relay	1 per n_commit epochs	Via relay node
commit_T + ZK proof	Every epoch via relay	1 per epoch	Via relay node
verdict_commit	Commit phase of suspension	1 per committee member per verdict	Loopix mix path
verdict_reveal	Reveal phase of suspension	1 per committee member per verdict	Loopix mix path
null_v_decryption	After threshold reveals available	Permissionless	Direct to chain

Signal	Trigger	Rate limit	Routing
watchdog_signal	Anomalous verdict_commit rate	1 per epoch per node	Dandelion++ stem → fluff broadcast
Auditor handoff	Epoch end	1 per epoch	Loopix mix path to committee
ZK continuity proof	Epoch transition (voluntary)	1 per epoch	Arbitration committee only
Loop/drop cover traffic	Constant (Poisson)	Base Poisson rate λ — calibrated per OQ-58	Loopix native (loop and drop covers)

---

6. Reputation and Audit

6.1 Per-Epoch Score

```
score_v(T) = w₁ × audit_response_rate(T)
           + w₂ × gossip_validity_rate(T)
           + w₃ × rate_limit_compliance(T)
           + w₄ × cluster_endorsement(T)
           + w₅ × (1 − rewind_signal_rate(T))
           + w_validator × validator_service_indicator(T)
           + w_relay × relay_service_indicator(T)

consistency_v(T) = clamp(1 − Var(score_v(T-k_rep), ..., score_v(T)) / σ²_max, 0, 1)
reputation_v(T)  = α × score_v(T) + (1−α) × consistency_v(T)
```

- `k_rep` is the rolling window length in epochs over which score variance is measured. A node with fewer than `k_rep` completed epochs uses however many epochs are available. Calibration target: long enough to distinguish genuine volatility from noise, short enough to be responsive to behavioral change (OQ-9).
- `σ²_max` is the maximum score variance expected from an honest, fully-active node over `k_rep` epochs under normal network conditions. It serves as a normalization constant: a node with

variance at or above `σ²_max` gets consistency = 0; a node with zero variance gets consistency = 1. `σ²_max` is set empirically from honest-node simulation (OQ-9). The clamp prevents consistency from going negative when variance exceeds `σ²_max`, which can happen for nodes with erratic but not necessarily malicious behavior.

- `α` is the blend weight between current score and historical consistency. Both `α` and the `α` in §7.3 are distinct — §7.3's `α` has been renamed `f_cap` to avoid ambiguity.

Slow reputation decay applies universally each epoch:

```
reputation_v(T) = reputation_v(T) − δ_decay
```

Weights w_1–w_5, δ_decay, and α require empirical calibration. The interactions between score components under adversarial conditions have not been formally analyzed and warrant empirical investigation (OQ-9).

Raw scores are held by the arbitration committee under threshold custody. The public chain receives only the committee-attested score band (1–4) with fuzzy boundaries, rate-limited to one change per N_band epochs.

`cluster_endorsement(v, T)` is defined as:

```
cluster_endorsement(v, T) = |{ p ∈ peers_v : p pulled from v in T }| /
|peers_v|
```

Excluded (not zeroed) for nodes with `|peers_v| = 0`: the w_4 term is omitted from the score computation and remaining weights are renormalized. This prevents penalizing nodes that have not yet grown a PSI peer neighborhood.

6.2 Behavioral Cluster Computation

Behavioral clusters are computed locally by each node from public chain timing data. The available fingerprint depends on the active transport profile (§5.1):

Tor/I2P profile — full fingerprint:

```
behavioral_fingerprint_v(T) = {
    activity_window:        which parts of the epoch the node is typically
active
    announcement_timing:    distribution of delays between interaction and
announcement
    transaction_timing:     when within the epoch the on-chain entry is
published
    audit_response_rate:    fraction of audit challenges responded to
}
```

Loopix profile — thin fingerprint:

```
behavioral_fingerprint_v(T) = {
    epoch_presence:         did the node publish in epoch T (binary or rate
over recent epochs)
    audit_response_rate:    fraction of audit challenges responded to
}
```

Sub-epoch timing components are unavailable under the Loopix profile because the mix layer destroys them by design. Epoch-granular presence and audit-response rate survive because they are counts/rates rather than timestamps. Under the Loopix profile, the content-based PSI-similarity flagging in §7.4 carries proportionally more weight, since timing-based detection is degraded.

Cluster computation is deterministic given the same chain data. Behavioral clusters have two roles: (1) input to the compound flag system (§7.8), and (2) under the Tor/I2P profile, a structural diversity axis for auditor committee selection (see below) and for validator/relay diversity constraints where invoked. They are never used for direct reputational effects — no node's per-epoch score is increased or decreased by its behavioral-cluster membership.

Auditor independence requirement is transport-conditioned. Under the Tor/I2P profile, an auditor committee must have members from different interest clusters AND different behavioral clusters (the dual-cluster constraint). Under the Loopix profile, behavioral-cluster signal is too thin to act as an independent diversity axis; the constraint reduces to interest-cluster diversity with a proportionally larger committee size (see §4.9.8 for the Chernoff implication).

6.3 Admission and First-Observation Interrogation

The admission window (n epochs): Zero CF weight, zero routing weight. VDF proof published each epoch. Interaction checkpoints at VRF-determined epochs require real network contact.

First-observation reports: When a new `epoch_id` first appears on-chain, VRF-selected existing nodes submit signed first-observation reports to the arbitration committee only.

Cross-node temporal burst aggregation: The committee aggregates `first_seen_T` and `first_seen_offset` values from first-observation reports across admission windows, computing a burst score over inter-arrival timing distributions. This detects the temporal clustering patterns described in §7.1a T.1 and T.9. The burst score enters the compound flag system at L1 only — no standalone escalation. VRF-determined checkpoint epochs are publicly verifiable after beacon publication, making checkpoint synchrony (T.9) independently verifiable from public chain data by any participant. Threshold calibration (burst window W, burst_score ceiling) is addressed in Phase 5 experiment 5.48 (OQ-59).

Provider identity in first-observation reports: First-observation reports include `provider_id`, encoding the mix-role provider node the new client connected through and providing a coarse geographic prior. The committee aggregates provider_id distributions per admission window. Geographic concentration (§7.1a T.3) enters the compound flag system at L1 only, contributing to escalation solely when combined with temporal burst and behavioral synchrony. It is never a standalone signal. Under the Tor/I2P profile this signal degrades because `provider_id` reduces to the Tor exit identity, which is shared across many clients.

Mix-network density signals (self-mixing Loopix profile). Because the mix layer is operated by PrivaCF nodes themselves (§5.1.1), the protocol can observe aggregate signals at the mix layer that are unavailable when relying on an external mixnet:

- **Mix node AS-density per epoch:** how many mix-role nodes are advertised from each AS or hosting provider. Sudden concentration in a single AS is a Sybil indicator on the mix layer.
- **Mix node admission timing correlation:** synchronized appearance of new mix-role identities, analogous to T.1 at the client layer.
- **Mix node churn rate per AS:** anomalously high churn from a specific AS suggests disposable mix infrastructure.
- **SUSPEND↔mix-disappearance correlation:** a mix-role node going offline in the same epoch a client `epoch_id` from its AS is suspended is weak but accumulable evidence of co-location.

These signals do not link to specific client `epoch_id` s — they observe the mix layer as a first-party substrate. They catch a Sybil cluster that must operate mix-capable nodes to participate at scale, complementing the per-client `provider_id` mechanism. Aggregate signals feed the compound flag system at L1 only. Under the Tor/I2P profile, mix-layer signals are not available because PrivaCF does not operate the mix layer.

Identity rotation evasion: On-chain SUSPENDED verdicts are permanent and include the behavioral fingerprint of the suspended `epoch_id` . New `epoch_id` s matching that fingerprint above threshold inherit a suspicion flag:

```
admission_flag if:
    behavioral_similarity(new_epoch_id, suspended_epoch_id) > θ_behavioral
    AND time_since_suspension < T_behavioral_window
```

Additionally, at the epoch when admission completes, the node's first handoff must include a valid Statement 5 non-suspension proof and valid `commit_T` . A node whose `null_v` is already in SUSP_SMT is rejected at admission immediately — no separate investigation required.

Suspicious restart detection:

```
suspicious_restart_flag if:
    prior_chain_abandoned_under_active_flags(T_abandon)
    AND new_epoch_id appears within T_window epochs of T_abandon
    AND behavioral_similarity(new_epoch_id, abandoned_epoch_id) > θ_behavioral
```

6.4 Multi-Auditor Encrypted State Handoff

Auditor credibility decay:

```
credibility_A(T) = credibility_A(T-1) × γ_penalty    // on confirmed false
verdict

audit_weight(result_from_A, T_result) = credibility_A(T_result)
    × e^(-λ_audit × (T_now - T_result))
```

γ*penalty and λ*audit require empirical calibration.

```
committee_T = VRF(beacon_T ‖ "audit_committee" ‖ epoch_id_v,
                  k = K_committee,
                  constraints = [
                      different_interest_clusters,
                      different_behavioral_clusters,
                      reputation ≥ median,
                      temporal_depth ≥ D_min
                  ])
```

The handoff package:

```
handoff_v(T) = {
    C_p(T),
    M_v(T),
    leaf_counts: {
        ANNOUNCEMENT:    n_ann,
        PULL_RESPONSE:   n_pull,
        AUDIT_RESPONSE:  n_aud,
        RATE_LIMIT:      n_rl    // cardinality of announcement_token_set(T)
    },
    SUSP_SMT_root_T,
    commit_T = ( c_T^{(0)}, ..., c_T^{(N_fallback-1)} ),
    threshold_BLS_pk_T^{(0..N_fallback-1)},
    ZK proof that:
        C_p(T) is a valid successor to C_p(T-1)
        M_v(T) is a valid successor to M_v(T-1)
        ||p_v(T) - p_v(T-1)||₁ ≤ Δ              (Statement 3)
        leaf_counts are consistent with M_v(T)
        token_v(T) correctly incorporated if token was issued this epoch
        null_v ∉ SUSP_SMT_root_T                (Statement 5, line 3)
        ∀ i : c_T^{(i)} = ForwardCommit(null_v, epoch_id_T,
                                         threshold_BLS_pk_T^{(i)};
r_commit_T^{(i)}) (Statement 5, line 4)
    rolling_chain_commitment: Poseidon(
        rolling_chain_commitment(T-1),
        zk_continuity_proof(T),
        audit_interactions(T),
```

```
        SUSP_SMT_root_T
    ),
    zk_continuity_proof:  <verified by committee at handoff; not published>,
    encrypted_shares: [
        Encrypt(pk_committee_i, shamir_share_i(snapshot_v(T))),
        Encrypt(pk_committee_i, shamir_share_i(peer_receipts_v(T)))
    ]
    // Per §4.1: shares are held by the on-demand arbitration committee.
    // peer_receipts_v(T) is retained as Shamir-shared encrypted state, reconstructable only when a Class 3 audit against a historical M_v(T') root is requested.
    // Eliminates per-node local receipt retention beyond one epoch (OQ-36 resolved).
}
```

Each committee member independently verifies the ZK proof. A threshold of $\lfloor K_{committee} / 2 \rfloor + 1$ must sign off. Rejection by a majority triggers the commit-reveal SUSPENDED flow (§4.9.6). Rejection by a minority triggers L3 Suspicious and a Class 3 audit.

Score band attestation: The committee publishes a threshold-BLS-signed score band to the public chain.

6.5 Audit Classes

Class 2 — passive, invisible, ~3–4 times per epoch:

```
┌───────────────────────────────────────────────────────────┐
│  CLASS 2 AUDIT FLOW                                        │
│                                                            │
│  A ──[pull request + embedded nonce]──────────────► B     │
│        indistinguishable from normal pull                  │
│                                                            │
│  A ◄──[gossip vector + H(state ‖ nonce ‖ epoch_id_B)]─── B│
│        indistinguishable from normal pull response         │
│                                                            │
│  A verifies hash against B's C_p(T) and M_v(T) on-chain   │
│  A publishes result on public chain under A's epoch_id    │
└───────────────────────────────────────────────────────────┘
```

Class 3 — committee-triggered, exceptional:

```
┌──────────────────────────────────────────────────────────┐
│  CLASS 3 AUDIT FLOW                                       │
│                                                           │
│  Q rewind signals (≥2 interest clusters, ≥D_Q depth,     │
│  correlated with same gossip cohort)                     │
│       │                                                   │
│       ▼                                                   │
│  Committee: VRF-selected (diff interest + behavioral     │
│  clusters · rep ≥ median · depth ≥ D_min)                │
│       │                                                   │
│       ▼                                                   │
│  C_i ──[challenge, formatted as pull request]──────► B   │
│                                                           │
│  B ──[Merkle proof + ZK proof (Stmts 1-3, 5)]──────► C_i │
│     via Dandelion++ stem WITH timeout retry               │
│                                                           │
│  Committee cross-checks against handoff snapshots        │
│  Contacts attestation issuers for peer-attested leaves   │
│  Threshold BLS signature on result                       │
│  Result published on public chain — permanent            │
└──────────────────────────────────────────────────────────┘
```

6.6 Rewind Signals and HNSW Snapshots

A rewind signal expresses that a node's recommendation quality has degraded and that a prior snapshot of the local HNSW index was better.

```
rewind_signal = {
    current_T:     <current epoch>,
    preferred_T:   <epoch at which quality was last acceptable>,
    cohort_epoch:  <epoch at which implicated gossip vectors entered index>,
    sig:           <Sign(epoch_id, H(current_T ‖ preferred_T ‖ cohort_epoch))>
}
```

HNSW rollback and network consequences are separate operations. The HNSW rollback is purely local. Network consequence comes from the Class 3 audit that the rewind signal triggers.

Rate limit: A node may contribute to triggering at most one Class 3 audit per N_rewind epochs.

Rewind signals as item-level velocity proxy: Rewind signals whose `cohort_epoch` timestamps cluster around a specific item's rapid `trust_total` rise are a compound signal of coordinated pushing (§7.1a T.8). No new machinery is required — the existing rewind signal mechanism surfaces this connection. When rewind signals from nodes in multiple interest clusters share the same `cohort_epoch` and that epoch coincides with anomalous `trust_total` velocity for a specific item, this enters the compound flag system as a correlated L3 signal. The novelty-kill sabotage vector (§7.3, §7.9 row 9) — a push of a niche item to suppress its novelty bonus where the item may genuinely match attacker taste — does not generate rewind signals from outside the cluster and remains O, deferred to Phase 5 experiment 5.47.

6.7 Health Tiers

Tier	Conditions	Effect
HEALTHY	All score conditions met, no flags, no adverse verdicts	Full routing weight, normal audit
DEGRADED	Score below threshold, above floor	Reduced routing weight, elevated audit
RECOVERING	Recently penalized, slow-rise trajectory	Low routing weight, elevated audit
SUSPENDED	Committee verdict on public chain via commit-reveal flow	Zero routing weight. Permanent for that epoch_id. null_v in SUSP_SMT
ABSENT	D_v < D_min, no flags	Normal routing weight
SUSPICIOUS_ABSENCE	D_v < D_min, with flags	Reduced routing weight, elevated audit
ADMITTING	Within n-epoch admission window	Zero CF weight, zero routing weight, observable
NEW	Admission complete, first N_bootstrap epochs	Minimum routing weight

7. Sybil Resistance

7.1 Attack Taxonomy

No symmetric reputation function can be Sybil-proof (Cheng & Friedman, 2005). PrivaCF's defense is not to prevent Sybil identities but to bound their influence and make them expensive.

The taxonomy below names the attacker archetypes informally; §7.1a characterizes observable behavioral properties of sybil attacks grounded in the empirical literature; §7.1b provides the influence model that connects those properties to quantifiable bounds; §7.9 formalizes the full (tactic × identity-strategy) space and states the detection contract per cell.

Attack type	What it does	Primary defense	Residual risk
Random push	Inflates trust_total with random announcements	DSybil non-overwhelming rule	CF noise
Bandwagon	Copies honest vectors then pushes target items	k_min + ε-DP impact bounding (§7.4); weight caps (§7.6)	Hard if very patient; impact bounded not detected
Segment	Targets specific interest cluster	Per-cluster reputation	Cluster-level degradation
Coordinated campaign	Multi-item narrative across clusters	Behavioral clustering, compound flags	Operator classification required
Sleeper	Builds reputation before activating	Smoothness detection, handoff chain, asymmetric penalty	Hardest to detect
Epoch rotator (same sk)	Evades SUSPENDED verdict by rotating epoch ID	Nullifier mechanism — cryptographically impossible from same sk	None for same sk
Epoch rotator (new sk)	Re-admits with fresh key after suspension	Admission cost, behavioral fingerprinting	Sophisticated adversary varying patterns
Dark node rotator	Goes dark before null_v extraction	ForwardCommit — committee decrypts without node cooperation	Admission window gap only

Attack type	What it does	Primary defense	Residual risk
Rogue committee	Mass null_v extraction without verdicts	Commit-reveal ordering — visible before any null_v recovered; watchdog signals	Threshold collusion (same as consensus assumption)

7.1a Behavioral Taxonomy of Sybil Attacks

This section characterizes observable properties that sybil attacks exhibit in practice, grounded in the empirical literature. These are not guarantees of detection — they are signals that sybil nodes tend to produce and that a detection system can exploit. Each property is stated as a tendency, not a certainty, because sophisticated adversaries can suppress individual signals at increasing cost.

The taxonomy is organized by the observable dimension rather than by attack type, since the same attack may exhibit multiple properties simultaneously and the compound of signals is more informative than any single one.

T.1 Temporal clustering on join

Sybil campaigns tend to produce bursts of identity creation within narrow time windows. This is reported consistently across OSN bot detection (Stringhini et al., ACSAC 2010; Yang et al., WWW 2014), review fraud (Mukherjee et al., WWW 2013), and DHT sybil studies (Urdaneta et al., ACM Computing Surveys 2011). The signal persists even as adversaries improve on other behavioral dimensions — Yang et al.'s successive spambot generations each showed temporal clustering despite improvements in content diversity and timing randomization of individual actions.

In PrivaCF's setting this manifests as a burst of epoch_id registrations and VDF proof submissions appearing on-chain within a narrow window, with first_seen_T values clustering tightly in first-observation reports. The committee aggregates these across admission windows as described in §6.3.

Suppression cost: low for a single campaign. An adversary who staggers identity creation across many epochs defeats this signal entirely. The cost is time, not resources — staggering admissions is free but slow, and each additional epoch of staggering is an epoch during which the identities are not yet usable.

T.2 Temporal clustering on action

Distinct from join timing. Coordinated campaigns show correlated action timing — announcements, rating submissions, protocol interactions — even when content is deliberately diversified. This is one of the most robust signals across the empirical literature: Yang et al. found it survived all three generations of spambot evolution, and Varol et al. (ICWSM 2017) found action timing among the top predictors of bot status across six bot categories.

In PrivaCF's setting this manifests as correlated announcement timing, gossip vector push timing, and on-chain transaction timing across a set of nodes, observable through behavioral cluster computation from public chain data (§6.2).

Suppression cost: moderate. Introducing timing randomization requires active coordination across the sybil set and degrades the efficiency of coordinated campaigns — nodes that randomize timing independently lose synchrony with each other, which undermines the campaign's aggregate effect.

T.3 Geographic concentration

Sybil campaigns tend to originate from a small number of geographic regions, reflecting the operational infrastructure of the attacker. This is a weak standalone signal — Stringhini et al. found it insufficient alone — but a strong compound signal that significantly improves detection precision when combined with temporal clustering and behavioral similarity.

In PrivaCF's setting geographic signal is degraded by the transport profile's IP hiding (Loopix profile destroys it entirely; Tor/I2P profile collapses it to the Tor exit). A meaningful fraction of legitimate users will also use additional VPN layers given the protocol's privacy focus. Provider identity — observable at admission through the mix-role provider node the new client connects through, under the self-mixing Loopix profile — gives a coarse regional prior that is weaker than IP but costs nothing to observe. It is recorded in first-observation reports and aggregated by the committee as described in §6.3. Under the Tor/I2P profile the provider_id signal degrades because the apparent provider is a Tor exit shared across many clients.

Suppression cost: trivial. A VPN or Tor exit node defeats IP-based geographic correlation entirely. Provider identity survives VPN use unless the adversary also routes their provider connection through an anonymity network, which is possible but adds operational friction.

T.4 Behavioral similarity within the sybil set

Sybil nodes in coordinated campaigns show higher pairwise cosine similarity in their behavior vectors than legitimate nodes in the same interest cluster. Cao et al. (IMC 2012) found that pairwise similarity distributions of sybil clusters have lower variance and higher mean than legitimate clusters — the sybil distribution is tighter around a high similarity value. This reflects the difficulty of generating genuinely idiosyncratic behavior programmatically: human taste is idiosyncratic in ways that scripted behavior is not.

In PrivaCF's setting the behavioral coherence requirement constrains attacker effectiveness: a Sybil farm that randomizes behavior enough to avoid detection also degrades the coordinated signal it is trying to inject. Impact is bounded structurally via k_min, ε-DP, and weight caps (§7.4, §7.6) rather than by detection. Anomalously similar behavioral fingerprints across nodes in the same behavioral cluster remain a signal in the compound flag system (§6.2).

Suppression cost: high and increasing. Yang et al.'s third-generation spambots introduced deliberate randomization to reduce behavioral similarity, but at the cost of reduced campaign coherence. An adversary who randomizes behavior sufficiently to evade similarity detection is also randomizing the coordinated signal they are trying to inject, which reduces attack effectiveness. There is a fundamental tension between behavioral diversity (for evasion) and signal coherence (for effect).

T.5 Anomalous trajectory smoothness

Legitimate nodes show variance in reputation and participation trajectories across epochs because real participation is noisy — connectivity varies, interest shifts, availability fluctuates. Sybil nodes maintaining a carefully controlled behavioral profile show anomalously low variance. This is distinct from temporal clustering on action: a node can have variable action timing (defeating T.2 detection) while still maintaining an anomalously smooth overall participation trajectory.

Srivatsa et al. (ICDCS 2005) identified trajectory smoothness as a signal in the TrustGuard context. In the OSN bot detection literature, Varol et al. found that temporal activity patterns — which encode smoothness indirectly — were consistently among the top predictors across bot categories.

In PrivaCF's setting this manifests as low σ^2 in the score trajectory over a rolling window, captured by smoothness detection (§7.5).

Suppression cost: moderate to high. Introducing genuine variance in participation is possible but requires the adversary to accept periods of lower participation, which reduces the campaign's throughput and may trigger the asymmetric penalty if the variance includes score drops.

T.6 Whitewashing and identity cycling

Whitewashing — abandoning a penalized identity and re-entering under a new one — is empirically the most common evasion strategy across P2P reputation systems (Hoffman et al., ACM Computing Surveys 2009). It is the dominant strategy in zero-admission-cost systems precisely because replacement is free. In high-admission-cost systems the economics change fundamentally: each new identity requires paying the full admission cost, which shifts the rational strategy away from proliferation toward identity preservation.

In PrivaCF's setting same-key whitewashing is cryptographically impossible via the nullifier mechanism. New-key whitewashing requires full admission cost and faces behavioral fingerprint matching on re-entry. The empirical dominance of whitewashing in other systems is therefore evidence that the nullifier mechanism addresses the most common real-world evasion strategy first.

Suppression cost in PrivaCF: high for same-key (cryptographically blocked), moderate for new-key (admission cost plus behavioral fingerprinting), low for new-key with deliberate fingerprint variation (admission cost only).

T.7 Solipsistic cluster structure

Sybil clusters tend to be densely internally connected and sparsely connected to the honest network, creating a detectable cut in the interaction graph. This is the basis of SybilGuard and SybilLimit's graph-cut detection. In systems with a persistent observable social graph, this is one of the strongest available signals.

In PrivaCF's setting the interaction graph is ephemeral and not directly observable, making graph-cut detection structurally unavailable — the ephemeral peer relationship model means no persistent graph exists to cut. The behavioral consequences of solipsistic cluster structure are however observable without seeing the graph: a node structurally positioned in a sybil cluster will show rare cross-cluster PSI attempts, high intra-cluster behavioral synchrony, low variance in peer set composition across epochs, and rewind signals from any bridge connections it maintains. These are behavioral proxies for graph structure rather than direct observation of it, and enter the compound flag system accordingly (§7.8).

Cross-cluster PSI attempt rate as a signal: The PSI responder issues a signed acknowledgment of receipt regardless of whether the handshake succeeds, distinct from the success receipt issued on PSI completion. The acknowledgment confirms a cross-cluster PSI attempt was received without revealing similarity score or item sets. The initiating node includes received acknowledgments as private witnesses in its handoff ZK proof, proving attempt rate without revealing who was contacted. Legitimate interest in cross-cluster discovery is genuine — any node seeking peers outside its immediate cluster has real motivation to attempt cross-cluster PSI, so the signal reflects real behavior. Nodes could spam cross-cluster PSI attempts to generate acknowledgments; this is self-limiting since anomalous handshake rates are a behavioral signal and consume responder resources, but requires an explicit rate limit (OQ-61). This mechanism requires PSI protocol change analysis before deployment.

Suppression cost: high. Maintaining genuine cross-cluster connections to defeat this signal requires genuinely interacting with the honest network, which imposes behavioral cost and creates detection surface.

T.8 Velocity of targeted item trust accumulation

Coordinated pushing campaigns produce anomalously rapid trust accumulation for targeted items concentrated in narrow time windows. Mukherjee et al. on Yelp review fraud found velocity to be the strongest single discriminator of fake review campaigns — stronger than content similarity or reviewer behavior in isolation.

In PrivaCF's setting this manifests as anomalously rapid trust_total rise for a specific item within a narrow epoch window. The trust cap c bounds the ceiling of this accumulation but does not flag the velocity. Rewind signals provide a compound proxy: a coordinated push that inflates trust_total for an item that does not match the honest network's taste will degrade recommendation quality for outside nodes, generating rewind signals pointing at the epoch when those vectors entered the index. When rewind signals from multiple interest clusters share the same cohort_epoch coinciding with anomalous trust_total velocity for a specific item, this compound signal enters L3 (§6.6).

Suppression cost: moderate for detection via rewind signals (requires the pushed item to genuinely match honest taste, which constrains attack targets), high for direct velocity detection (requires staggering the push across many epochs, reducing campaign coherence).

T.9 Interaction checkpoint synchrony

During the admission window, VRF-determined interaction checkpoint epochs differ per node by construction. Genuine synchrony in checkpoint completion timing across multiple admitting nodes is therefore improbable by chance and is a signal of coordinated admission. This is a PrivaCF-specific signal with no direct analog in the general sybil literature, arising from the VRF-based admission structure. It is subsumed into the burst score computed by the committee over first-observation timing (§6.3), since VRF-determined checkpoint epochs are publicly verifiable after beacon publication and independently checkable by any participant.

Suppression cost: low to moderate. An adversary who staggers admissions across epochs loses checkpoint synchrony but also loses the efficiency of batch identity creation.

7.1b Sybil Influence Model

No purely structural argument can bound sybil influence in PrivaCF's setting without presuppositions about adversarial behavior, and no closed-form bound is achievable without empirical data about how sybil nodes actually behave in deployment. This subsection states the influence model explicitly, derives what can be bounded structurally, and is honest about what requires empirical calibration to go further.

Baseline bound

With random peer selection and no reputation filtering, the expected fraction of sybil-influenced recommendations received by any node is exactly:

```
p(recommendation is sybil-influenced) = |S| / |N|
```

where S is the sybil set and N is the total network. This is both the worst-case upper bound with no defenses active and the exact probability under random connection. With a perfect reputation system that identifies all sybil nodes with certainty, the probability goes to zero. The actual system sits between these bounds, and the reputation system is the mechanism that drives them apart.

Cluster/bridge decomposition

The two-tier peer selection structure decomposes expected sybil influence into two independent terms:

```
E[sybil influence] =
    p(rec from cluster peer) × p(cluster peer is sybil | PSI match)
    × E[influence | sybil cluster peer]
  + p(rec from bridge peer) × p(bridge peer is sybil)
    × E[influence | sybil bridge peer]
```

The two terms have structurally different sybil exposure. Cluster peers are PSI-filtered on item overlap, so p(cluster peer is sybil | PSI match) depends on whether sybil nodes have constructed item sets that overlap with the target's niche. Bridge peers are selected from a broader pool, so p(bridge peer is sybil) approximates the base rate |S|/|N| in the worst case.

For a user whose niche attracts no sybil interest — the common case for genuine long-tail content — the cluster term approaches zero by PSI filtering alone, and only the bridge term remains. This bridge term is bounded by:

```
p(bridge influence) ≤ p(bridge rec) × |S|/|N|
```

A user who sets bridge peer weight to zero (§5.7) reduces this term to zero entirely, at the cost of losing cross-cluster discovery. For a user whose niche is actively targeted, the cluster term dominates and PSI filtering provides weaker protection, since a targeted sybil constructs its item set specifically to match. This is the targeted-PSI Sybil case (§5.7, row 21 in the detection contract).

Both terms are further attenuated by hop-distance trust attenuation (μ^hop_distance), the trust cap c, and the reputation floor each node applies to its local HNSW index (§3.6). The reputation floor functions as a confidence threshold: under discrete bands, contributions from peers below the floor are zero-weighted; under a continuous gradient, they would be proportionally attenuated. Expected influence per sybil peer is therefore structurally bounded independent of detection, though the bound is loose when the sybil peer achieves high reputation through sustained legitimate behavior before activating.

Flag compounding across epochs

For a sybil node maintaining behavioral strategy S across epochs, the probability of remaining undetected after n epochs is:

```
p(undetected after n epochs | strategy S) = ∏_i (1 − p(flag_i raised | S))^n
```

This decays toward zero exponentially in n if any flag has nonzero detection probability under strategy S. The adversary's optimization is to choose S to minimize each p(flag_i raised | S) subject to achieving sufficient influence — exactly the evasion cost argument. The behavioral taxonomy in §7.1a assigns nonzero p(flag_i raised) to sybil nodes exhibiting each property, and the suppression cost estimates characterize how adversarial investment shifts those probabilities downward.

The compound decay is meaningful rather than trivially defeatable because suppressing each flag has an independent cost. An adversary who suppresses temporal clustering (T.1) by staggering admissions, suppresses behavioral similarity (T.4) by introducing randomization, and suppresses trajectory smoothness (T.5) by introducing participation variance is paying three independent costs simultaneously, each of which reduces campaign effectiveness in addition to reducing detection probability.

The hardest case is a sleeper node (I3 in the detection contract) that generates zero flags during the accumulation phase because it behaves entirely legitimately. For this node the compounding argument does not apply during accumulation — p(flag raised) ≈ 0 for all flags by construction. The defense in this case is not detection but damage bounding: trust cap c, hop-distance attenuation, and reputation floor collectively limit what even a fully undetected high-reputation node can inject into any target's recommendations.

What requires empirical data

The flag probability terms p(flag_i raised | S) cannot be computed from first principles without presuppositions about adversarial behavior, and those presuppositions cannot be grounded without empirical data. Two approaches are available, both limited:

Empirical estimation from observed attacks. The behavioral literature (§7.1a citations) provides base rates for each signal property across documented sybil campaigns. These estimates are biased — documented attacks are not a random sample of all attacks, and sophisticated attacks that went undetected are by definition absent from the data. Additionally, many documented large-scale sybil attacks have been conducted by state-level actors who are explicitly out of scope (§1.5); under the stated assumptions (A1–A4) they would be detectable, but their behavioral profiles may not generalize to the in-scope adversary population.

Simulation over presupposed behavioral models. Phase 5 adversarial simulation (§9.2) models sybil nodes following explicit behavioral strategies and measures empirical flag raise rates. This produces calibrated estimates conditional on the presupposed strategy, with the honest limitation that real adversaries will adapt their strategies in response to protocol analysis in ways that simulation cannot fully anticipate.

Running instances of the protocol against real adversaries who are motivated to analyze and attack it is the only source of ground truth. Adversaries who have studied the protocol will attempt to either fulfill its requirements legitimately or find exploits, both of which produce different attack signatures than synthetic simulation. The interaction between literature-grounded presuppositions, simulation-calibrated estimates, and live deployment data is the right methodology — each addresses gaps the others cannot.

Honest limitations

- p(flag_i raised | S) is not known for any S without empirical calibration from live deployment.
- Documented sybil attacks are a biased sample toward detected and reported campaigns.

- Adversaries who analyze the protocol will adapt their strategies to the specific flag system, producing attack patterns that historical data cannot anticipate.
- The influence bound is loose for high-reputation sleeper nodes before activation.
- Nation-state adversaries are explicitly out of scope. Under the stated assumptions (A1–A4) they would be detectable, but those assumptions are axioms rather than derived properties.

These limitations do not invalidate the model. They bound what claims the model can honestly support: structural upper bounds on influence per tier, exponential decay in detection probability across epochs for non-sleeper strategies, and damage bounding for sleeper strategies through trust cap and attenuation. Going beyond these bounds requires empirical data from live deployment, which Phase 5 begins to approximate under controlled conditions.

7.2 Temporal Depth

```
D_v(T) = Σ_{t ∈ A_v, t ≤ T} λ^(T-t),   λ ∈ (0,1)
```

Asymmetric penalty:

```
r_v(T)   ← min(r_v(T-1), BAND_1)
r_v(T+k) ← r_v(T+k-1) + Δ_rise
```

The asymmetric penalty creates a tension: `Δ_rise` must be large enough that honest nodes recover reputation at a meaningful rate after absence, but small enough that an adversary cannot exploit recovery periods to accumulate disproportionate reputation through on-off cycling. Whether a single value satisfies both constraints simultaneously is an empirical calibration question addressed in Phase 5 experiment 5.4 (OQ-6, reclassified). λ should be calibrated per community type.

7.3 DSybil Non-Overwhelming Rule

```
trust_contribution(v, X) = max(0, r_v(X) + noise) × Δ_base × (1 + κ ×
novelty(X))

if trust_total(X) < c:   trust(v) += trust_contribution(v, X)
else:                     trust(v) += 0

trust(v) ≤ f_cap × c
```

- `r_v(X)` , `noise` , and `Δ_base` are as defined in §3.4.
- `f_cap` is the per-node trust cap as a fraction of `c` . It bounds how much of the global trust ceiling any single node can claim. This is a distinct parameter from the `α` blend weight in §6.1.

Formal bound validated on Digg with binary feedback only. Extension to continuous noisy ratings requires the per-epoch composition lemma (OQ-10).

Caveat on formal transfer. The non-overwhelming rule is borrowed from DSybil but the formal bound justifying it in that work does not transfer to PrivaCF's setting. DSybil's theorem requires a persistent social graph, trust propagation via random walks, and a sparse honest/Sybil cut. PrivaCF has none of these — peers are discovered via PSI on item overlap, trust is not propagated transitively, and the topology is intentionally ephemeral. The rule is retained as a well-motivated heuristic with empirical support from the original Digg validation, but formal justification in this setting requires empirical data from live deployment. See OQ-10.

Novelty term as passive Sybil damping. The novelty bonus serves a dual purpose. Its primary function is to accelerate trust accumulation for undersurfaced items. A secondary effect is that as `trust_total` approaches `c` , marginal adversarial contributions shrink toward zero — coordinated pushing of an already-popular item has diminishing returns by construction, and organic popularity surges are similarly self-limiting. The converse is a sabotage vector: an adversary who pushes a niche item past an early trust threshold removes its novelty bonus, suppressing the organic discovery acceleration the system provides for long-tail content. This is harder to detect than inflation attacks because it requires no ongoing coordination after the initial push, and may be difficult to distinguish from a genuine early popularity surge. Characterization of this attack and its interaction with the compound flag system is left to Phase 5 adversarial simulation.

7.4 Sybil Impact Bounding and Local PSI-Similarity Flagging

Detection difficulty is transport-profile-dependent. Under the Tor/I2P profile, sub-epoch timing signals are available and the §7.1a taxonomy operates at fine resolution. Under the Loopix profile, sub-epoch timing is destroyed by mixing, leaving content-based signals (PSI similarity, gossip vector statistics) and epoch-granular signals (admission timing, audit-response rate) as the primary detection axes. PrivaCF combines (a) structural impact bounds that limit damage independent of detection, and (b) the content-based PSI-similarity flag described later in this section, which is transport-agnostic and carries proportionally more weight under the Loopix profile.

k-anonymity for PSI neighborhoods. A node activates cluster-specific behavior (cluster_endorsement scoring, cluster-weighted trust_total) only if `|peers_v| ≥ k_min`. Below this threshold the node falls back to global trust_total only. This prevents micro-cluster targeting: a Sybil attacker cannot construct a small, highly specific cluster around a victim and accumulate identifying signal across epochs.

```
cluster_behavior_active(v, T) = (|peers_v| ≥ k_min)
```

`k_min` requires empirical calibration. A conservative starting value is k_min = 20; larger values provide stronger anonymity at the cost of slower peer discovery convergence (OQ-32).

Formal ε-DP privacy budget. Laplace noise in §4.5 is parameterized by ε. Each gossip event leaks at most ε bits of preference information. Over T epochs the cumulative budget is Tε. A Sybil attacker controlling some fraction of a node's PSI neighborhood cannot re-identify the victim from gossip output alone within this budget. See OQ-55 for the formal derivation under the SecLDP framework.

Structural caps as primary Sybil defense. With k_min enforcing minimum crowd size and ε bounding per-event leakage, any Sybils that enter a neighborhood can neither target individuals nor accumulate identifying signal efficiently. The caps in §7.6 then bound influence regardless of detection:

- `w_node_cap` / `w_cohort_cap`: no node or cohort dominates reputation weight
- `f_cap`: individual trust accumulation capped at `f_cap × c`
- Hop-distance attenuation (§5.7): remote Sybils contribute at most `μ^k` of base weight

Local PSI-similarity flagging (content-based, transport-agnostic). Each node passively computes cosine similarity of received contribution vectors against the average of its PSI peer set, flagging anomalously high pairwise similarity. This is the protocol's primary content-based Sybil signal and operates identically across transport profiles, since it depends on gossiped content rather than timing.

```
sim_flag(p, q, T) if cosine(contrib_p(T), contrib_q(T)) > θ_psi_sim
                     AND |peers_v| ≥ n_foolsgold_min
```

The flag is local, passive (computed from already-received vectors), and submitted via the handoff package — no extra messaging cost. Below `n_foolsgold_min` peers the signal is suppressed to avoid small-sample false positives.

Two-sided cross-peer flagging. Peers exchange flag-rate aggregates each epoch (not the flags themselves — only the count or rate of similarity-pairs flagged in their local view). A node whose surrounding peers all show elevated flag rates around it is itself implicated, even if its own self-reported flag rate is suspiciously low. This catches the case where a Sybil cluster suppresses its own internal flags.

On-flag PSI overlap proof escalation. When the committee receives a similarity flag, it can compel the flagged pair to run a full PSI overlap proof against each other. The proof reveals the actual intersection size *to the committee only*, not publicly. Honest nodes whose taste happens to converge benignly cleared by the proof; coordinated Sybil pairs are identified by the overlap profile. The expensive PSI overlap step is gated on the cheap cosine flag, keeping average cost low.

Transport-conditioned threshold. Under the Loopix profile, where timing signals are degraded, `θ_psi_sim` is tightened to make the content signal more sensitive. Under the Tor/I2P profile, where timing signals carry independent detection power, `θ_psi_sim` is looser to reduce false positives. Both thresholds require empirical calibration (OQ-24, profile-parameterized).

This mechanism does not solve the original attribution problem in full — a confirmed similarity pair does not by itself imply Sybil control rather than benign convergence — but escalation to a committee-mediated PSI overlap proof addresses the attribution gap by surfacing the actual evidence to a body authorized to act on it.

7.5 Smoothness Detection

```
smoothness_flag(v, T) if Var(score_v(T-k), ..., score_v(T)) < σ²_floor
```

σ²_floor requires empirical calibration (OQ-29). Calibration should explicitly include recovery trajectory patterns — anomalously smooth post-penalty recovery is a specific case of the smoothness signal and should be represented in the OQ-29 test suite.

7.6 Weight Caps and Gini Monitoring

```
G(T) = (Σ_i Σ_j |w_i - w_j|) / (2n Σ_i w_i)
```

No single node exceeds `w_node_cap` of total weight. No cohort exceeds `w_cohort_cap` of total weight. Both parameters require empirical calibration; reasonable starting ranges are w_node_cap ∈ [1%, 5%] and w_cohort_cap ∈ [10%, 20%] depending on expected network size and cluster structure (OQ-32).

7.7 Tamper Analysis

Value	Tamperable?	Detectable?	Consequence
`null_v`	Only by changing `sk`	Yes — circuit enforces `null_v = Poseidon(sk, "null_v")`; substitution fails proof	Changing `sk` produces a new identity; same `sk` always produces same `null_v`
`commit_T`	Only with matching null_v	Yes — Statement 5 circuit checks ForwardCommit formation with same null_v wire	Cannot commit a different null_v without breaking the first two circuit checks
`sk`	Changing produces a different identity	N/A	All accumulated reputation and PSI relationships lost

Value	Tamperable?	Detectable?	Consequence
`p_v`	Yes — within Statement 1 and 3 bounds	Partially — nuclear option at L5	CF manipulation within bounds
`r_p`	Loss only	Yes — Class 3 audit permanently fails	SUSPENDED verdict on public chain
`π_v(T)`	Yes	No	Node's own CF quality degrades
`q_v(T)`	Yes	Coordinated false triggering is a compound flag	False triggering costs reputation; suppression is self-harming
PSI cache	Yes	No	Node's cluster degrades. Self-harming only
Merkle leaves (before commitment)	Yes	Yes — handoff ZK proof commits to counts	Consistent ZK proof while lying requires breaking the ZK system
Merkle leaves (after commitment)	Computationally infeasible	N/A	Root is on public chain and VDF-chained
Peer attestations	Cannot forge	Yes if attempted	Fails signature verification during audit
Handoff ZK proof	Cannot forge	Yes if attempted	Committee rejects. SUSPENDED on public chain
On-chain verdict	Cannot alter	N/A	Permanent. Epoch rotation does not erase it
SUSP_SMT leaf	Cannot alter without forking the chain	N/A	Append-only, VDF-chained
DECRYPTION_SMT leaf	Cannot alter without forking the chain	N/A	Append-only, VDF-chained

Value	Tamperable?	Detectable?	Consequence
Arbitration committee entry	Cannot alter without threshold compromise	Detectable via public chain Merkle root	Requires compromising threshold committee
verdict_commit	Cannot alter after publication	Yes — on-chain, signed, VDF-chained	Non-alterable commitment to verdict
verdict_reveal	Cannot mismatch with commit	Yes — validators verify H(share ‖ verdict ‖ nonce) = commit	Invalid reveal rejected by validators

7.8 Compound Flag System and Alert Levels

Signals present	Alert level	Automated action	Human review?
High announcement rate alone	L1 Watch	None	No
Low announcement diversity alone	L1 Watch	None	No
High rate + low diversity	L2 Elevated	Increase audit frequency	No
L2 + cohort temporal clustering (same behavioral cluster)	L3 Suspicious	Elevate + flag	Recommended
L3 + PSI neighborhood below k_min threshold	L4 Probable Sybil	Suspend cluster behavior + Class 3	Yes
L3 + cross-cluster behavioral similarity	L4 Probable Coordinated	Reduce routing weight + Class 3	Yes
L4 + Class 3 fail OR handoff rejection (majority)	L5 Confirmed	Commit-reveal SUSPENDED flow on public chain	No
L4 + Class 3 pass	L2 Elevated	Maintain elevated audit, clear L4	No
Smoothness flag alone	L1 Watch	None	No

Signals present	Alert level	Automated action	Human review?
Smoothness flag + low temporal depth	L2 Elevated	Increase audit frequency	No
First-observation reports inconsistent with VDF start	L2 Elevated	Extend admission observation	No
Temporal burst score above threshold (§6.3)	L1 Watch	None — contributes to compound flag only	No
Geographic concentration flag (§6.3)	L1 Watch	None — contributes to compound flag only when combined	No
Temporal burst + geographic concentration + behavioral synchrony	L2 Elevated	Increase audit frequency	No
New epoch_id null_v already in SUSP_SMT	Rejected at admission	Statement 5 proof fails — no handoff possible	No — cryptographic
New epoch_id behavioral fingerprint matches suspended	L2 Elevated	Elevated audit from admission	No
Suspicious restart detection	L3 Suspicious	Elevated audit + flag	Recommended
Q rewind signals (≥2 interest clusters, correlated cohort)	L3 Suspicious	Trigger Class 3	No
Rewind signals correlated with specific item trust_total velocity	L3 Suspicious	Trigger Class 3	No
Coordinated rewind signals from same behavioral cluster	L3 Suspicious	Elevate + flag	Recommended
Handoff rejection (minority of committee)	L3 Suspicious	Trigger Class 3	No
Handoff rejection (majority of committee)	L5 Confirmed	Commit-reveal SUSPENDED flow	No

Signals present	Alert level	Automated action	Human review?
Validator double-signing detected	L5 Confirmed	Commit-reveal SUSPENDED flow, permanent	No
Anomalous verdict_commit rate, no behavioral signal justification	L3 Suspicious	Watchdog signal broadcast + oversight chain	Recommended
Stalling committee (commit phase started, reveals missing)	L3 Suspicious for non-revealing members	Non-response penalty + oversight chain	Recommended
L5 Confirmed + persistent manipulation signals despite Class 3 pass	L5 Nuclear	Committee demands full rolling chain commitment traversal	Committee review required

---

7.9 Detection Contract

§7.1 enumerates attackers and §7.8 enumerates detection actions, but the mapping between them has so far been informal. This subsection makes the mapping explicit. For each meaningful intersection of *manipulation tactic* and *identity/persistence strategy*, the contract states what triggers detection, which mechanism handles it, the strength of the guarantee, and what slips through. Unresolved cells are named so reviewers can falsify them.

7.9.1 Axes

Manipulation tactic (T) — what the adversary injects:

ID	Tactic	Description
T1	Random injection	Random items, random ratings — no specific target
T2	Targeted push	Inflate `trust_total` of a specific item
T3	Coordinated narrative	Multi-item campaign across one or more interest clusters
T4	Novelty kill	Push a niche item past the trust threshold to suppress its novelty bonus (§7.3 sabotage vector)

ID	Tactic	Description
T5	Semantic poisoning	Genuine engagement patterns, manipulated semantic intent

Suppression in PrivaCF reduces to T2 against competitors of the target — dislikes are local-only and not transmittable, so direct nuke attacks are not in the tactic space.

Identity / persistence strategy (I) — how the adversary persists or evades:

ID	Strategy	Description
I1	Single identity, immediate	One identity, payload deployed immediately after admission
I2	Single identity, sustained	One identity active across many epochs, payload throughout
I3	Single identity, sleeper	One identity builds reputation passively, then deploys payload
I4	Sybil swarm, fresh keys	Many identities each with a distinct `sk`, used briefly
I5	Same-key rotator (post-suspension)	Identity suspended; new `epoch_id` derived from same `sk`
I6	New-key rotator, similar fingerprint	Identity suspended; new `sk`, behavioral pattern preserved
I7	New-key rotator, varied fingerprint	Identity suspended; new `sk`, behavioral pattern deliberately altered
I8	Dark node rotator	Identity goes offline before nullifier extraction, then re-admits
I9	Rogue committee (audit-layer attack)	Compromised committee threshold attempts mass deanonymization or covert decryption

Guarantee level (G) — strength of the detection promise:

Level	Meaning
C	**Cryptographic.** Detection (or block) follows by arithmetic from an honest-validator check; failure requires breaking a primitive listed in A3.
B	**Behavioral-probabilistic.** Detection rate is governed by a calibrated threshold; false positive and false negative rates are characterized empirically. The named OQ governs calibration.

Level	Meaning
H	**Heuristic.** A signal is surfaced into the compound flag system; final adjudication requires operator review or correlated evidence across multiple flags.
O	**Out of scope.** Explicitly not promised. Listed so reviewers can confirm the gap is intentional, not overlooked.

7.9.2 Contract Table

Each row names one meaningful (T, I) intersection. Cells not listed collapse to a row that handles them — see §7.9.3.

#	(T, I)	Adversary archetype	Triggering signal	Mechanism (§ref)	G	Residual / Calibration
1	T2 × I1	Naive opportunist	Bursty announcements, low diversity	§7.8 L1→L2 (rate + diversity)	B	OQ-32
2	T1 × I4	Random Sybil flooder	Admission-rate spike; Gini drift; influence bounded by caps	§6.3 first-observation; §7.4; §7.6	B	OQ-32
3	T2 × I2	Patient pusher (sustained, single identity)	DSybil cap reached on target item; smoothness flag if score variance is artificially low	§7.3 cap `c`; §7.5 smoothness	B (smoothness); damage structurally bounded by cap `c` regardless of detection	OQ-15, OQ-29
4	T2 × I3	Sleeper activator (single identity)	Score variance + sudden trajectory change after dormancy	§7.5 smoothness; §7.2 asymmetric penalty	B	OQ-29, OQ-6 (`Δ_rise`); hardest single-identity case
5	T2 × I2 (mimic)	Bandwagon mimic	Influence bounded by k_min + ε-DP + weight caps; no direct detection	§7.4 impact bounding; §7.6	B	OQ-32
6	T3 × I4	Coordinated campaign, single cluster	Behavioral cluster overlap; influence bounded by caps	§6.2 behavioral cluster signal; §7.4; §7.6	B	OQ-28

#	(T, I)	Adversary archetype	Triggering signal	Mechanism (§ref)	G	Residual / Calibration
7	T3 × I4-distrib	Coordinated campaign, cross-cluster	Cross-cluster behavioral similarity (timing, audit response distribution)	§7.8 L4 Probable Coordinated	H	OQ-28; operator review required
8	T3 × I3 swarm	Cross-cluster sleeper swarm	Smoothness flag across multiple identities + cross-cluster behavioral similarity	§7.5 + §6.2 + §7.8	H	OQ-28, OQ-29; hardest multi-identity case
9	T4 × any	Novelty-kill saboteur	Item crosses trust threshold with abnormally fast accumulation + low contributor diversity	None currently; collapses into "organic surge or coordinated push"	**O** (today) → H (future)	§7.3 caveat; deferred to Phase 5 adversarial simulation
10	T5 × any	Semantic poisoner	Indistinguishable from genuine signal at the protocol layer	None — by construction	O	Acknowledged residual (§8.1 "Semantic poisoning")
11	any × I5	Same-key rotator	`null_v ∈ SUSP_SMT` — Statement 5 fails	§4.9.5 Statement 5 + §4.9.2 SUSP_SMT	C	None — fails by arithmetic
12	any × I6	New-key rotator, similar fingerprint	`behavioral_similarity > θ_behavioral` within `T_behavioral_window`	§6.3 admission_flag; §7.8 L2 Elevated	B	OQ-34, OQ-35
13	any × I7	New-key rotator, varied fingerprint	None at identity layer; falls back to whatever T-level defense applies	§4.3 admission cost (rate-limits but doesn't block); T-row applies	B (T-only)	Acknowledged residual (§8.1 "Sophisticated epoch rotators")
14	any × I8 post-publish	Dark node, post-`commit_T`	ForwardCommit decryption recovers `null_v` from on-chain `commit_T`	§4.9.4 ForwardCommit; §4.9.6 commit-reveal	C	None — committee can decrypt without node cooperation

#	(T, I)	Adversary archetype	Triggering signal	Mechanism (§ref)	G	Residual / Calibration
15	any × I8 admission window	Dark node, pre-`commit_T`	Zero reputation + behavioral fingerprint of admission attempt	§4.3 admission cost; §6.3 first-observation	B (bounded)	OQ-16; bounded but not eliminated; admission-window-only gap
16	I9 on-chain	Rogue committee, on-chain extraction	Anomalous `verdict_commit` rate vs. behavioral signals	§4.9.8 watchdog + recursive oversight	B	OQ-48; bounded by Chernoff (§4.9.8) under A2
17	I9 off-chain	Rogue committee, covert decryption	None until a `null_v_decryption` transaction is submitted; off-chain σ aggregation leaves no trace	§8.1 "Threshold collusion" carve-out	O (below A2)	OQ-17, OQ-53; bounded by Byzantine majority assumption
18	orthogonal	Validator double-signer	Two BLS signatures on competing blocks at same height	§4.1 double-signing detection	C	None — immediate permanent SUSPENDED
19	I-orthogonal	Reputation laundering (proactive rotation)	Re-admission cost + behavioral fingerprint match if pattern preserved; if varied, falls to row 13	§4.3 + §6.3	B (with similar fingerprint) / O (with varied)	OQ-57 (cost amortization across cycles)
20	I-orthogonal	Cluster splitting via inconsistent PSI	Self-harming (degrades node's own cluster); detectable as PSI cache divergence across observers	§7.7 (PSI cache row)	B (self-harming)	Self-harming only; no operator action needed
21	T2 × I-targeted PSI	PSI-targeted Sybil	Sybil constructs item set to enter victim's interest cluster	§5.4 PSI; §7.4 k_min + ε-DP impact bounding	B (partial)	§8.2 T5 — formal bound not established

7.9.3 Collapse rules

The full Cartesian product of T (5) × I (9) is 45 cells. The table above lists 21 distinguishable rows. The rest collapse:

- **Row 11 absorbs (any T) × I5** — Statement 5 fails before any payload can be deployed.
- **Row 14 absorbs (any T) × I8 post-publish** — `null_v` is recoverable from the prior epoch's `commit_T` regardless of what payload was attempted.
- **Row 10 absorbs T5 × (any I)** — semantic poisoning is undetectable at the protocol layer regardless of identity strategy.
- **(T1) × (I1, I2, I3)** collapse into row 1 (the signal that triggers detection — high rate, low diversity — is identical for random and targeted single-identity push at this scale).

7.9.4 Where the contract is weakest

Three regions of the matrix are where the protocol's promise is genuinely thin and worth flagging to reviewers:

1. **Row 9 (T4 — novelty kill).** No mechanism currently distinguishes a coordinated novelty-suppression push from an organic early-popularity surge. The §7.3 sabotage vector is named but unaddressed. Currently O; should become H once Phase 5 simulation produces a separator.
2. **Row 13 (I7 — new-key rotator with varied fingerprint).** Once the adversary changes both `sk` and behavioral pattern, the only remaining defense is admission cost, which rate-limits but does not detect. Acknowledged in §8.1.
3. **Row 17 (I9 off-chain).** A colluding committee threshold can recover `null_v` covertly without on-chain trace. Bounded by A2 (the same assumption consensus depends on), but no protocol-layer detection exists below that ceiling. This is the weakest link in the entire identity-privacy story.

The contract is strongest at rows 11, 14, 18 (cryptographic guarantees by arithmetic) and at rows 1–6, 12, 15 (behavioral-probabilistic with named calibration OQs). Rows 7, 8, 16 are heuristic and depend on operator review or compound-signal correlation.

7.9.5 Implications for Phase 5 simulation

Each B-level and H-level row in the table corresponds to at least one Phase 5 experiment in §9.2 (with the exception of self-harming rows like row 20, where TPR/FPR characterization is not meaningful — the attacker's own CF quality degrades and operator action is unnecessary). The detection contract above can be used as a checklist: for each non-self-harming row marked B or H, the corresponding experiment must produce a TPR/FPR characterization with an explicit calibration recommendation. Rows marked O are not eliminated by Phase 5 — they are documented residuals.

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8. Known Limitations

8.1 What Is Not Protected

- **Within-cluster preference privacy.** Interest cluster peers accumulate observations over time. Accepted tradeoff.
- **Long-term differential privacy.** T epochs of ε-DP gives Tε total loss. Not claimed.
- **Nation-state adversaries.** Loopix provides formal sender/receiver/relationship anonymity under the Poisson traffic model, but mix node compromise and global traffic analysis remain out of scope.
- **Semantic poisoning.** Structurally unresolvable at the protocol layer — an epistemic problem, not a security one.
- **Announcement token unforgeability assumes sk integrity.** The RATE_LIMIT leaf construction proves announcement count without revealing items, but relies on the node's `sk` not being compromised. A node with a stolen key could have tokens manufactured on its behalf; this is subsumed by the general key compromise threat and is not specific to this construction.
- **Dark node rotators — admission window only.** The dark node gap is closed for nodes that have published at least one `epoch_transaction`. The residual gap is nodes that go dark during the admission window before any `commit_T` is published. Bounded by zero reputation, full admission cost, and behavioral fingerprinting.
- **Sophisticated epoch rotators with key change.** An adversary generating a fresh key evades the nullifier mechanism entirely. Admission cost and non-overwhelming trust are the remaining defenses.
- **Poseidon maturity.** Less cryptanalytic history than EC constructions. Resolved by primitive split: local derivations use Poseidon as a keyed PRF; pseudorandomness under standard Poseidon assumptions suffices (OQ-1 — closed).
- **ForwardCommit instantiation maturity.** BLS-based ForwardCommit resolved as Boneh-Franklin IBE over BLS12-381; reduces to DBDH in the random oracle model (OQ-2 — closed).
- **Behavioral fingerprint as persistent identifier.** Derived from public on-chain data and available to any observer. Degraded but not eliminated by per-n-epoch commits, relay submission, and transaction timing jitter.

- **commit_T as per-epoch presence signal.** Published every epoch, exempt from n_commit batching. Opaque without verdict signature but confirms node liveness every epoch.
- **Approximate interest cluster graph reconstruction.** Loopix relationship anonymity substantially degrades this but does not eliminate it — PSI traffic patterns over many epochs may still leak cluster topology to a patient adversary observing mix nodes.
- **Poisson cover traffic bandwidth.** Every node emits traffic at a constant Poisson rate regardless of real activity. This is non-trivial on metered or mobile connections (OQ-58).
- **Arbitration committee threshold trust.** Compromise of a threshold quorum of the **currently-selected** arbitration committee (§4.1) can reconstruct any `snapshot_v(T)` whose shares that committee holds — fine-grained behavioral data and peer-receipt sets for the epoch. Bounded by per-epoch committee rotation and Shamir handoff to the incoming committee; not bounded by the lifetime trust of every prior committee, as a full encrypted ledger would have been.
- **Score verification trustlessness.** Ordinary nodes cannot recompute raw scores from public chain data alone. Verification is committee-attested. Accepted cost of the committee-attested score model.
- **p_v targeted inflation.** An adversary can inflate preference weights for target items within the bounds of Statements 1 and 3.
- **Threshold collusion for null_v extraction.** A colluding supermajority of a committee can aggregate σ off-chain and decrypt `commit_T` without going through the commit-reveal flow, recovering `null_v` without an on-chain trace. Damage is limited to targeted epoch ID linkage — no suspension, no preference exposure, no protocol consequence unless an on-chain transaction is subsequently submitted (at which point the action becomes visible). Acting on the recovered `null_v` in any protocol-meaningful way requires submitting a `null_v_decryption` transaction, which is publicly visible. ForwardCommit provides security against covert deanonymization for all adversaries who cannot both compromise a threshold of committee members AND suppress watchdog signals across the honest majority simultaneously. This requires compromising the same threshold that breaks consensus — the privacy assumption is bounded by the Byzantine majority assumption already in the threat model.

8.2 Unresolved Design Tensions

Privacy tradeoffs

- **T2 — Niche signal vs. privacy protection.** Chopping vs. Laplace. Mutually exclusive per deployment.
- **T9 — n_commit batching vs. audit freshness.** Larger n_commit improves privacy but widens

the undetected drift window.

- **T10 — commit_T per-epoch submission vs. behavioral fingerprinting.** commit_T is exempt from n_commit batching and provides a per-epoch liveness signal. Mitigated by commit_T's opacity — it reveals presence only, not behavior.

Security tradeoffs

- **T1 — Δ_rise calibration vs. on-off attack defense.** If no single Δ_rise satisfies both path-responsiveness and on-off attack resistance, governance must choose a point on the tradeoff curve. Characterized empirically in Phase 5 experiment 5.4.
- **T5 — PSI-targeted Sybils.** Formal bound not established under patient targeted adversary who constructs an item set specifically to enter a victim's interest cluster.
- **T12 — Forward secrecy of past `commit_T`.** A future arbitration committee whose threshold BLS shares are compromised could, in principle, decrypt **past** `commit_T` values without a corresponding verdict — recovering `null_v` for prior epochs in which that key still applied. v1 underwrites the gap with mandatory share deletion at epoch end, enforced by slashing if a past `commit_T` is later decrypted without a matching `verdict_commit`. This is cheap but unverifiable in the positive case (you cannot prove a key was deleted). A proper cryptographic fix is Pixel-style forward-secure threshold BLS composed with a CHK forward-secure PKE wrapper, deferred to v2 pending production-grade FS-BLS support in `blst`. Referenced by §1.7 P4.

Liveness tradeoffs

- **T7 — Validator incentives.** Addressed by service score bonus and shirking penalty. Long-term sufficiency without tokens requires analysis.
- **T8 — Shamir re-share availability during committee rotation.** The one-shot re-share from outgoing to incoming arbitration committee (§4.1) must complete inside the rotation window. Failure modes: outgoing committee partially online, re-share ZK proofs failing, incoming committee key publication slow. Mitigated by slashing the outgoing committee for non-completion, but the residual availability gap during the rotation window itself requires calibration.
- **T11 — Commit-reveal latency vs. suspension throughput.** The commit-reveal process adds latency to suspension finality. High volumes of simultaneous suspensions may create block space contention.

Calibration questions

- **T3 — Validator set size vs. collusion resistance.**

- **T4 — Admission window length vs. UX.** No middle states by design.
- **T6 — Behavioral cluster false positives.** Real users in the same timezone with similar routines may cluster together. Mitigated by compound flag system rather than direct reputational effects.

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9. Implementation Plan

9.1 Minimal Viable PrivaCF

Reference transport profile. The MVP targets the self-mixing Loopix profile (§5.1) as the default. The Tor/I2P alternate profile is fully specified but not implemented in Phase 1–2; reference-profile interoperability with the alternate profile is not required (per §5.1, instances do not federate across profiles).

Core hypothesis: Decentralized CF with rotating pseudonymous identities, self-mixing Loopix/Sphinx transport (mix layer provided by PrivaCF mix-role nodes themselves, §5.1.1), Dandelion++ broadcast routing, Jaccard PSI peer selection, Merkle-committed behavioral history with peer attestations, multi-auditor encrypted handoff, a single public chain with arbitration-committee custody of sensitive state, nullifier-based suspension persistence, forward-secure commitment for dark node closure, commit-reveal verdict observability, and light Sybil resistance can surface content that a popularity baseline misses, without a trusted server.

Stack: Python or Rust · NumPy/ndarray · hnswlib · EMP-toolkit (Pinkas PSI) · tendermint-rs (BFT consensus) · blst (BLS signatures) · Poseidon crate (arkworks-rs) · Simulated network for Experiments 1–3 · self-mixing Loopix/Sphinx mix-node implementation (Katzenpost code reused as the in-tree mix-node component rather than as an external network) + Dandelion++ for Experiment 4.

Datasets:

Dataset	Purpose
LastFM HetRec 2011	Long-tail music; sparse; DP-CF benchmark
MovieLens 1M/25M	Dense mainstream; baseline contrast
RateYourMusic exports	Extreme niche; genuinely sparse
User platform exports	Real niche signal; preference initialization

Experiments:

E1 — Does it recommend at all? Precision@K, NDCG, HR, long-tail discovery rate per segment (head / long-tail). Gate: beats popularity baseline on long-tail discovery.

E2 — Content discovery under noise per segment. Chopping vs. Laplace across head / long-tail. Gate: long-tail precision ≥ popularity baseline.

E3 — Sybil damage by attack type and SSP scenario. All RobuRec types × Dense/Distributed/Sparse SSP. Gate: damage measurable and bounded.

E4 — PSI peer selection and identity rotation. Gate: PSI improves precision@K; VRF degrades by < 20%.

9.2 PoC Phases

Phase 1 — Identity, Network, and Blockchain Core

Poseidon PRF epoch IDs and permutation; null_v local derivation; staggered epoch offsets; variable chopping with VRF-jittered n_v(T); niche item announcement delay; identity VDF chain with interaction checkpoints; Pedersen commitment; Merkle tree with peer-attested leaves; HNSW snapshot storage for rewind; Shamir distribution to auditor committee; **self-mixing Loopix/Sphinx transport** — mix-role node implementation (Sphinx packet processing, Poisson per-hop delay scheduler, RouterInfo advertisement), client-role / mix-role split per §5.1.1, provider-based NAT traversal; SURB-based request-response; Dandelion++ for broadcast fluff phase with timeout retry; Sphinx fixed-size packet padding; Noise Protocol sessions for clearnet/dev only; Loopix loop and drop cover traffic at base Poisson rate; communication rhythm; entry protocol; Class 2 passive audit; first-observation report collection and arbitration committee submission including provider_id field; burst score aggregation over first-observation timing; mix-network density aggregate signals (§6.3); ADMITTING health tier.

ForwardCommit construction and per-epoch commit_T publication; Statement 5 extended circuit (four checks + SMT path) in Plonky3; DECRYPTION_SMT initialization; verdict_commit and verdict_reveal transaction types; null_v_decryption transaction type and permissionless aggregation; watchdog_signal transaction type; DKG protocol for per-epoch committee threshold BLS key.

Public blockchain: block structure and VDF chaining; SUSP_SMT initialization; VRF validator and proposer selection with dual cluster constraints; BLS threshold signatures for block finality; Tendermint-style proposer timeout fallback; double-signing detection; genesis validator set and transition protocol; light client block header storage with Merkle inclusion proofs; per-n-epoch commit batching.

Arbitration committee: threshold-held encrypted ledger; committee handoff protocol between rotating committees; Shamir share rotation; continuity proof storage; first-observation report storage; fine-grained behavioral data storage.

Relay nodes: VRF selection; submission batching; reputation model integration.

PSI acknowledgment message type: signed acknowledgment of cross-cluster PSI receipt, independent of handshake success; rate limit implementation (OQ-61 pending full analysis).

Exit: N nodes cycling through staggered epochs over the self-mixing Loopix/Sphinx network (mix layer provided by a subset of those same nodes operating mix-role identities) with public state on the public chain and sensitive state held under arbitration committee threshold custody. Block finality verified. Double-signing detection verified. Light client sync verified. Class 2 audit flow end-to-end verified. Relay submission verified. Statement 5 proof generation and verification end-to-end verified. commit_T published and verified each epoch. Commit-reveal verdict flow end-to-end verified including permissionless aggregation. DECRYPTION_SMT insertion verified. Watchdog signal broadcast verified. Dark node extraction verified — node goes offline after publish, committee decrypts from commit_T. Burst score computation verified against synthetic admission timing data. Provider_id field included in first-observation reports. Mix-role/client-role unlinkability verified under adversarial probing.

Phase 2 — Reputation and Sybil Resistance

Per-epoch score including validator/relay service components; slow reputation decay; consistency and smoothness detection including recovery trajectory patterns in test suite; rewind signal trigger with cohort correlation; item-level trust_total velocity correlation with rewind cohort_epoch; HNSW snapshot rollback; multi-auditor encrypted handoff; Class 3 audit trigger; justified disclosure; compound flag system including coordinated rewind signal detection and item-velocity compound signal; behavioral cluster computation; inter-cluster reputation; trust attenuation; commit-reveal SUSPENDED verdicts with null_v extraction and SUSP_SMT insertion; behavioral fingerprint matching; suspicious restart detection; committee-attested score band publication; watchdog signal compound flag entry; stalling minority detection and non-response penalty; recursive oversight chain with hard depth limit.

Exit: Multi-auditor handoff correctly detects inconsistent state. On-chain verdicts survive epoch rotation. null_v correctly inserted into SUSP_SMT on suspension. Re-admission from same sk correctly rejected by Statement 5. Behavioral fingerprint matching flags suspicious new admissions. Watchdog signal correctly triggers on anomalous commit rate. Stalling member correctly flagged. Oversight chain resolves correctly under honest majority. Rogue committee simulation correctly detected and blocked before null_v recovery. Item-velocity compound signal correctly triggers on synthetic coordinated push.

Phase 3 — Cryptographic Layer

ZK consistency proofs (Statements 1–3, 5); sign preservation constraint enforcement; ZK continuity proofs (arbitration committee); handoff ZK proof including PSI acknowledgment witnesses; rewind signal aggregation; mobile benchmarks for all ZK components; Poseidon PRF security analysis (closed — OQ-1); domain separator collision check (closed — OQ-4); ForwardCommit security analysis (closed — OQ-2); extended Statement 5 mobile benchmark; dec_nullifier collision resistance analysis (closed — OQ-5); BLS-based ForwardCommit instantiation formal review; permutation reconstruction security benchmark; Pinkas PSI performance on mobile.

Phase 4 — CF and Noise Calibration

Full signed preference model; asymmetric PSI cache decay; two-tier peer selection; variable chopping with empirical n calibration; cover items post-n_cover; trust attenuation in CF weight; dislike-aware scoring; noise system comparison per segment (head / long-tail); adjacent-epoch weight validation; behavioral fingerprint calibration; n_commit calibration; niche announcement delay calibration.

Phase 5 — Adversarial Simulation

5.1 Recommendation poisoning — all RobuRec types × all SSP scenarios
5.2 Sybil flooding at genesis — join-rate spike detection
5.3 Patient adversary — smoothness detection calibration
5.4 On-off attack — Δ*rise tension validation*
5.5 Diversity bonus exploitation
5.6 Interest cluster seeding — gradual infiltration
5.7 Behavioral cluster false positives — co-located legitimate users
5.8 Coordinated announcement vs. viral discovery — TPR/FPR
5.9 Semantic poisoning within chopping obfuscation
5.10 Eclipse attack exposure
5.11 Noise system comparison under attack
5.12 k_min calibration — PSI neighborhood size vs. anonymity/quality tradeoff
5.13 Multi-auditor handoff false rejection rate
5.14 Auditor collusion resistance — K_committee calibration
5.15 Validator collusion resistance — K_validators calibration
5.16 VDF gap tolerance from mobile connectivity data
5.17 Adjacent-epoch CF weight calibration
5.18 Cover item threshold n_cover calibration
5.19 Jaccard PSI threshold calibration
5.20 First-observation aggregation calibration

5.21 Admission window length n calibration

*5.22 Behavioral fingerprint matching θ*behavioral and T_behavioral_window calibration

5.23 Epoch rotation evasion — sophisticated adversary varying behavioral patterns

5.24 Blockchain liveness under validator dropout

5.25 Block VDF proposer timeout calibration

5.26 Genesis transition timing — when is the validator pool viable?

5.27 Variable chopping privacy gain vs. CF quality — n_base and n_jitter per sparsity tier (head / long-tail)

5.28 Relay submission batching — timing obfuscation vs. submission latency

5.29 n_commit calibration — behavioral fingerprint degradation vs. audit freshness

5.30 Reputation decay δ*decay — legitimate absence tolerance vs. maintenance incentive*

5.31 Arbitration committee availability under member dropout and rotation

5.32 Rewind signal adversarial triggering — N_rewind calibration

5.33 p_v targeted inflation within Statement 1/3 bounds — damage quantification

5.34 Bootstrap behavioral cluster relaxation — transition timing calibration

5.35 HNSW snapshot storage overhead vs. rewind recovery quality

*5.36 Niche item announcement delay — τ*niche and max_delay_epochs calibration

5.37 Dark node re-admission damage — admission window gap

5.38 SUSP_SMT growth rate and non-membership proof latency as function of suspended node count

5.39 Statement 5 circuit performance on mobile hardware across SUSP_SMT depth

5.40 Commit-reveal latency vs. epoch duration — suspension finality timing

5.41 Watchdog signal false positive rate — threshold_watchdog calibration

5.42 Oversight chain depth limit — damage during resolution window

5.43 DKG timing constraint — epoch_transaction submission deadline calibration

5.44 Stalling minority attack — damage before oversight chain resolves

5.45 commit_T presence signal — behavioral fingerprinting impact vs. n_commit

5.46 Reputation laundering — sustained proactive identity rotation; cost amortization across cycles; behavioral fingerprint stability under deliberate variation (OQ-57; §7.9 row 19)

5.47 Novelty-kill saboteur — coordinated push of niche items past trust threshold to suppress novelty bonus; separation from organic early-popularity surges (§7.3 sabotage vector; §7.9 row 9)

5.48 Coordinated admission burst threshold — separating organic community growth from sybil admission campaigns; calibration of burst window W and burst_score ceiling (OQ-59)

5.49 Per-transport-profile Chernoff K_d calibration — committee size needed under interest-cluster-only diversity (Loopix profile) vs. dual-cluster diversity (Tor/I2P profile), per §4.9.8

5.50 Per-transport-profile PSI-similarity threshold calibration — θ_psi_sim under Loopix (tight) vs. Tor/I2P (loose), per §7.4

5.51 Mix-network density signal calibration (self-mixing Loopix profile only) — AS-density anomaly threshold, mix-churn baseline, SUSPEND↔mix-disappearance correlation strength

Tor/I2P alternate profile. The alternate transport profile (§5.1) is fully specified but not in the Phase 1–2 implementation scope. A separate deployment track may instantiate it post-Phase 5 once per-profile calibration data is available; profile selection is a deployment-time decision.

9.3 Evaluation Metrics

Recommendation quality: Precision@K, NDCG, HR, long-tail discovery rate per popularity segment (head: top-X% of `trust_total` distribution; long-tail: remainder — boundary defined per deployment following Park & Tuzhilin, 2008), privacy-utility curve per segment, coverage, Prediction Shift. HNSW rewind recovery quality after poisoning event. The two segments have distinct signal properties: head items have dense co-occurrence signal and DP noise is relatively cheap; long-tail items are sparse, where chopping dominates and Laplace is destructive.

Privacy: Cross-epoch linkability; behavioral fingerprint re-identification rate under the arbitration-committee model; interest cluster graph reconstruction rate from PSI traffic patterns; score trajectory entropy; permutation reconstruction time; ε-DP per gossip event (mainstream only); cover item anonymity set per segment; niche item announcement timing anonymity set size.

Sybil resistance: PS and HR per attack type and SSP scenario; p_v inflation damage within Statement bounds; patient adversary accumulation rate; smoothness detection false positive rate; announcement anomaly TPR/FPR; coordinated rewind signal detection TPR/FPR; item-velocity compound signal TPR/FPR; epoch rotation evasion detection rate; dark node re-admission damage; burst score false positive rate vs. organic growth events.

Nullifier mechanism: False rejection rate for honest nodes under Statement 5; SUSP_SMT non-membership proof latency on mobile as tree depth grows; Statement 5 circuit benchmark on mobile.

Audit and blockchain: False audit claim rate; handoff rejection false positive rate; auditor collusion detection rate; ZK proof times on mobile; block finality latency; validator dropout tolerance; double-signing detection latency; light client sync overhead; arbitration committee availability under rotation.

Network health: Health tier distribution over time; false positive rate per scenario; recovery time per health tier; Gini coefficient trajectory; gossip convergence time; VDF admission cost per hardware tier; public blockchain storage per light client; arbitration committee storage per committee member.

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10. Open Questions and Status

10.1 Open Questions

ID	Question	Field	Framework	Tier	Effort	Status
OQ-3	What is the maximum acceptable Statement 5 proof generation time on target mobile hardware across SUSP_SMT depths?	Cryptography	empirical	3	1	open
OQ-3b	What is the maximum acceptable Statement 2 proof generation time on target mobile hardware at d=128 and d=256?	Cryptography	empirical	3	1	open
OQ-9	What weight calibration (w_1–w_5, δ_decay, α) correctly balances reputation score components under adversarial conditions — can any pair of components be driven in opposing directions, and which dominates?	Reputation	empirical	3	2	open
OQ-10	What is the empirically observed sybil influence on recommendation quality in live deployments, and how do flag raise rates vary across documented sybil behavioral strategies?	Sybil resistance	empirical	2	5	**partially resolved** — structural bounds established in §7.1b; closed-form bound requires empirical data from live deployment; paths A (stochastic block model) and B (MeritRank ratio) remain as routes to sharpen structural bounds; calibrated flag probability terms deferred to Phase 5 and live deployment

ID	Question	Field	Framework	Tier	Effort	Status
OQ-11	Is there a ZK circuit construction for behavioral cluster membership that supports justified disclosure without revealing the underlying fingerprint?	Cryptography	theoretical → empirical	2	4	resolved (desktop) — two viable paths: (1) BBS+ attestation: arbitration committee issues a signed cluster attestation each epoch; node uses BBS+ selective disclosure to prove cluster = C without revealing epoch_id or fingerprint values; reuses BLS keys already in the stack (§4.9); simpler to implement first. (2) ZK k-means over public centroids (Plonky3): because centroids are public, squared-distance comparison reduces to an inner product of private F with public coefficient vectors plus a range proof on the result — no quadratic terms; estimated ~2–3k constraints at d=100, k=10; proof time under 100ms on desktop with Plonky3. The remaining design work for path (2) is tying the private fingerprint witness to epoch_id inside the circuit (likely a Merkle path or hash preimage), which adds cost but is not a feasibility blocker. Mobile is a compatibility target, not a PoC requirement; both paths are deferred on mobile pending benchmarks.

ID	Question	Field	Framework	Tier	Effort	Status
OQ-12	What is the minimum genesis set size for bootstrap viability, and at what point can the behavioral cluster diversity constraint be enforced without relaxation?	Network	empirical	3	3	partial — the genesis set floor is now bounded by interest-cluster committee viability (not dual-cluster coverage), since the recommendation layer remains useful during the behavioral-cluster-relaxed bootstrap phase via content-based cold-start (§13). The enforcement threshold (when behavioral fingerprints have sufficient temporal depth and cluster density) remains open; see OQ-44 for the tightening schedule.
OQ-13	Are validator incentives sufficient to prevent long-term shirking without token rewards?	Reputation	theoretical | empirical	2	3	open
OQ-14	What is the correct staleness window for SUSP_SMT_root references in Statement 5 — how old can the root be before the non-membership proof is no longer meaningful?	Cryptography	theoretical	1	2	**closed** — the staleness window is zero by construction. The handoff package (§6.4) fixes exactly one `SUSP_SMT_root_T` per epoch as a named field; it is also chained into the rolling_chain_commitment. There is no prover choice of root — the epoch-T root is the only valid input. Any null_v inserted in epoch T-1 is present in every epoch-T root, because the commit-reveal flow spans multiple epochs and cannot finalize a suspension mid-epoch without prior evidence epochs. The question of staleness does not arise: the protocol admits exactly one valid root per epoch, and that root is always post-suspension for any node suspended in a prior epoch.

ID	Question	Field	Framework	Tier	Effort	Status
OQ-15	Does trust_total converge to a stable distribution under sustained adversarial flooding, or does it oscillate?	Sybil resistance	theoretical	1	3	open
OQ-16	What is the expected damage from dark node re-admission during the admission window gap, as a function of n and c?	Sybil resistance	theoretical → empirical	2	3	open
OQ-17	Can off-chain committee collusion executing a covert ForwardCommit decryption be detected, and what is the damage bound below the Byzantine majority threshold?	Cryptography	theoretical	2	3	open
OQ-18	What is the correct VDF gap tolerance policy — at what point should a partially completed chain be invalidated vs. allowed to resume?	Identity	empirical	3	2	open

ID	Question	Field	Framework	Tier	Effort	Status
OQ-19	What is the minimum cover item count n_cover that provides meaningful anonymity set protection without degrading CF quality?	Privacy	empirical	3	2	**partially resolved** — structural bound derived analytically. Under a capacity-bounded colluding-receiver adversary controlling k_adv peers, item X with K honest likers achieves (δ, ε*conc)-failure-probability anonymity when:* `n_cover(K) ≥ C_effective/(N−K) × [m_fixed/δ_anon − K_eff + √((N−K)×ln(1/δ_conc)/2)]` *where m_fixed = k_adv×(T_epochs−τ*niche), K_eff = Σ min(1, n_v(T)/d_v) over honest likers, δ = δ*anon*+δconc+δ*loopix, and δ*loopix = m_fixed×ε*loopix. Requires Loopix sender-anonymity guarantee (*εloopix-approximate uniformity). Numerical evaluation blocked on OQ-58 (ε_loopix uncharacterized). CF quality degradation calibration remains empirical (Phase 3). §4.5 confirmed: permutation is over positive-preference set (line 499: "selected from the permuted positive preference set") — d_v denominator in p_real = min(1, n_v(T)/d_v) is correct.
OQ-20	What are the correct PSI cache decay values λ*proof and λ*noproof across community sparsity profiles?	Network	empirical	3	2	open
OQ-21	What value of μ correctly attenuates hop-distance trust without suppressing legitimate long-range CF signal?	CF	empirical	3	2	open

ID	Question	Field	Framework	Tier	Effort	Status
OQ-22	How long does permutation reconstruction take under realistic gossip observation rates, and does it meaningfully threaten preference privacy?	Privacy	empirical	3	2	open
OQ-23	What Jaccard PSI threshold θ_cluster optimally balances peer quality against cluster size across different community types?	Network	empirical	3	2	open
OQ-24	What value of k_min correctly balances PSI neighborhood anonymity against recommendation quality and peer discovery convergence speed across community density profiles?	Sybil resistance	empirical	4	3	open
OQ-62	Can Byzantine-robust aggregation (coordinate-wise median or trimmed mean) be applied to committee aggregation steps in PrivaCF without requiring nodes to open committed values beyond what the existing audit mechanism already reveals?	Sybil resistance	theoretical	3	3	open

ID	Question	Field	Framework	Tier	Effort	Status
OQ-25	Does the 0.5× adjacent-epoch CF weight default correctly discount stale gossip vectors, or does it over-penalize nodes with slow interaction rates?	CF	empirical	3	2	open
OQ-26	What auditor committee size K_committee minimizes collusion probability while remaining viable under cluster diversity constraints?	Reputation	empirical	4	3	open
OQ-27	What validator set size K_validators correctly balances BFT safety margin against participation overhead?	Network	empirical	3	2	open
OQ-28	What behavioral cluster similarity threshold minimizes false positives from co-located legitimate users while catching coordinated Sybils?	Sybil resistance	empirical	4	3	open
OQ-29	What smoothness variance floor σ^2_floor separates legitimate consistent participation from adversarial reputation smoothing, including recovery trajectory patterns?	Sybil resistance	empirical	4	3	open

ID	Question	Field	Framework	Tier	Effort	Status
OQ-30	What is the correct maximum frame size across all protocol message types, including the PSI acknowledgment message type?	Network	empirical	1	1	open
OQ-31	What interaction checkpoint schedule within the admission window best balances re-identification risk against Sybil detection signal?	Identity	empirical	3	2	open
OQ-32	How many first-observation reports are needed for a meaningful coordinated admission signal?	Reputation	empirical	3	2	open
OQ-33	What is the maximum acceptable handoff ZK proof generation time on target mobile hardware?	Cryptography	empirical	1	1	open
OQ-34	What behavioral fingerprint similarity threshold θ_behavioral correctly flags suspicious new admissions without over-penalizing legitimate users?	Sybil resistance	empirical	4	3	open

ID	Question	Field	Framework	Tier	Effort	Status
OQ-35	What is the correct T_behavioral_window — how long should a suspended fingerprint continue to flag new admissions?	Sybil resistance	empirical	4	2	open
OQ-36	Is one epoch sufficient for peer attestation retention, or do audit windows require longer retention?	Reputation	empirical	1	1	**resolved** — one epoch is sufficient for Class 2 audits (current-epoch pull response receipts only). Insufficient for Class 3 audits, which may challenge any epoch T' for which M_v(T') is committed on-chain. Resolution: receipts (set_of_peer_signed_receipts composing PULL_RESPONSE leaves) are included in the encrypted_shares payload of the auditor handoff, retained by the arbitration committee under threshold custody for the lifetime of the node's audit chain. Class 3 audits query the arbitration committee directly for historical receipts, eliminating the need for per-node local retention beyond one epoch. See §6.4 handoff package update.
OQ-37	What n_commit value optimally degrades behavioral fingerprinting precision without creating an audit freshness gap?	Privacy	empirical	3	2	open

ID	Question	Field	Framework	Tier	Effort	Status
OQ-38	What reputation decay rate δ_decay tolerates legitimate absence without allowing dormant adversarial nodes to maintain inflated reputation?	Reputation	empirical	3	2	open
OQ-39	What arbitration committee availability protocol survives member dropout and rotation without creating handoff gaps?	Reputation	empirical	3	3	open
OQ-40	What relay node batching window provides meaningful submission timing obfuscation without unacceptable submission latency?	Network	empirical	3	2	open
OQ-41	What n_jitter range provides sufficient vector size obfuscation without degrading CF quality in sparse communities?	Privacy	empirical	3	2	open
OQ-42	What N_rewind rate limit prevents adversarial Class 3 triggering while allowing legitimate quality degradation signals through?	Reputation	empirical	3	2	open

ID	Question	Field	Framework	Tier	Effort	Status
OQ-43	What is the expected p_v inflation damage within Statement 1 and 3 bounds, and at what point does it meaningfully degrade recommendation quality?	Sybil resistance	empirical	4	3	open
OQ-44	What is the correct bootstrap behavioral cluster relaxation schedule — how quickly should the diversity constraint tighten as the network matures?	Network	empirical	3	2	open
OQ-45	How many HNSW snapshots should be retained, and at what depth does rewind recovery quality degrade below usefulness?	CF	empirical	3	2	open

ID	Question	Field	Framework	Tier	Effort	Status
OQ-46	What announcement delay parameters τ_niche and max_delay_epochs correctly balance niche item re-identification risk against discovery signal propagation?	Privacy	empirical	3	2	**partially resolved** — tradeoff structure derived analytically from OQ-19 bound. τ*niche reduces the adversary's effective observation window, replacing T_epochs with (T_epochs*−τniche) in m_fixed. The tradeoff between n_cover and τ*niche is linear with constant substitution rate Λ = C_effective×k_adv/((N−K)×δ*anon): `n_cover = Λ×(T_epochs−τ_niche)`, `dn_cover/dτ_niche = −Λ`. Any (n_cover, τ*niche) pair on the line `n_cover + Λ×τ_niche = Λ×T_epochs` achieves the target failure probability — the choice between high cover/low delay and low cover/high delay is a UX decision, not a security one. Numerical calibration of Λ requires k_adv estimate (Phase 5) and OQ-58 resolution (ε*loopix). Calibration against community sparsity profiles remains empirical.
OQ-47	How does SUSP_SMT non-membership proof latency scale with suspended node count on target mobile hardware?	Cryptography	empirical	3	2	open
OQ-48	What watchdog threshold correctly separates legitimate suspension bursts from rogue committee activity?	Reputation	empirical	3	2	open

ID	Question	Field	Framework	Tier	Effort	Status
OQ-49	What oversight chain depth limit and escalation schedule bounds damage during the resolution window without creating liveness risks?	Reputation	empirical	4	3	closed — structural question resolved by Chernoff bound; empirical calibration (K_0, ΔK) remains in Phase 5
OQ-50	What DKG completion deadline correctly constrains the epoch_transaction submission window without creating liveness pressure on slow committee members?	Network	empirical	3	2	open
OQ-51	What is the behavioral fingerprinting impact of per-epoch commit_T submission relative to n_commit batching?	Privacy	empirical	3	2	open
OQ-52	What is the expected damage from a stalling minority attack before the oversight chain resolves?	Reputation	empirical	4	3	open
OQ-53	Can off-chain committee collusion be detected in arrears from public chain data, and what damage is possible below the Byzantine majority threshold?	Cryptography	theoretical	2	3	open

ID	Question	Field	Framework	Tier	Effort	Status
OQ-54	Which noise mechanism (chopping vs. Laplace) and at what parameters correctly trades privacy for CF quality across head and long-tail segments?	CF	empirical	3	2	open

ID	Question	Field	Framework	Tier	Effort	Status
OQ-55	Can the SecLDP trust model from the unified DP-DL MF framework (Cyffers et al., arXiv:2510.17480, 2025) be adapted to derive formal GDP bounds on PrivaCF's gossip exchange, parameterized by n_v(T), the Laplace noise scale, and the permutation key?	Privacy	theoretical	2	3	**partially resolved** — SecLDP framework maps cleanly onto PrivaCF's gossip mechanism. The permutation key sk_perm is the hidden secret S in the SecLDP trust model; n_v(T) elements of the positive-preference set (size d_v) are the partially observable output. Lemma 14 (Cyffers et al.) removes the permutation-attributable component from the adversary's view matrix, giving effective sensitivity sens_eff = sens × (n_v(T)/d_v) — privacy amplification by subsampling at rate p = n_v(T)/d_v. Under Gaussian mechanism: Theorem 8 gives 1/σ*eff-GDP where σ*eff = ν/sens_eff = ν×d_v/(sens×n_v(T)). Under Laplace mechanism (PrivaCF's deployed mode): privacy amplification by subsampling yields effective ε*eff = log(1 + p×(e^ε−1)) ≈ p×ε for small ε, but formal equivalence between Lemma 14's matrix argument and the Laplace subsampling amplification theorem (Balle et al., ICML 2020) has not been verified. Remaining blocking work: (a) verify that Lemma 14 applies to Laplace mechanism, or derive a parallel statement; (b) map Statement 1 noise injection to Theorem 3/4 of Cyffers et al. to confirm the sens*Π(C;B) computation; (c) characterize interaction between ε_loopix and the GDP bound (OQ-58). See §10.1 notes on OQ-55 assumptions.

ID	Question	Field	Framework	Tier	Effort	Status
OQ-56	Does PrivaCF's item-based CF on accumulated gossip vectors achieve comparable recommendation quality to gossip-based matrix factorization (Hegedűs et al., ECML PKDD 2019) under matched sparsity conditions on MovieLens and RateYourMusic?	CF	empirical	3	2	open
OQ-57	Can sustained reputation laundering — proactive identity rotation across many cycles — amortize admission cost in a way that defeats the rate-limiting effect of the VDF chain, and does behavioral fingerprinting hold against a launderer who deliberately varies patterns across cycles?	Sybil resistance	theoretical | empirical	2	3	open

ID	Question	Field	Framework	Tier	Effort	Status
OQ-58	What base Poisson rate λ for Loopix loop and drop cover traffic provides sufficient anonymity against a mix-node-observing adversary without imposing unacceptable bandwidth overhead on metered or mobile deployments, and what is the correct λ per config (A–E)?	Network	empirical	3	2	open
OQ-59	What threshold W and burst_score ceiling separates coordinated admission bursts from organic community growth at realistic network sizes?	Sybil resistance	empirical	3	2	open — Phase 5 experiment 5.48
OQ-61	What rate limit on cross-cluster PSI acknowledgment attempts prevents spam while preserving the signal, and what are the privacy implications of the acknowledgment message type?	Network	theoretical → empirical	2	3	open — requires PSI protocol change analysis before Phase 5 scoping

Notes on selected questions

OQ-10 (Sybil influence bound — reframed): The question has been reframed from "does a stochastic block model argument yield a budget-scaled influence bound" to "what is the empirically observed sybil influence in live deployments." The reason is that §7.1b establishes that no closed-form bound is achievable without presuppositions about adversarial behavior, and those presuppositions cannot be grounded without empirical data. The structural bounds (baseline, cluster/bridge decomposition, flag compounding decay, damage bounding) are established and do not require further theoretical work. What remains is empirical calibration of the flag probability terms p(flag_i raised | S), which cannot be computed from first principles.

Two formal paths remain open as routes to sharpen the structural bounds:

Path A — stochastic block model: PSI-selected interest clusters and VRF-selected behavioral clusters together provide independent random sampling from a pool with bounded adversarial fraction. The Chernoff argument that closed OQ-49 is the structural template. The open work is adapting it to a continuous influence quantity rather than a binary compromise event — requiring a sensitivity argument on the item-based CF scoring function under bounded adversarial weight injection.

Path B — MeritRank ratio bound: Nasrulin et al. (IEEE BRAINS 2022) prove $\lim_{|S|\to\infty} w^{+}(\sigma S) / w^{-}(\sigma S) \leq c$ under random-walk trust propagation. PrivaCF's trust cap c directly caps per-item adversarial weight injection by construction. Adapting the ratio bound to PrivaCF's PSI-based ephemeral topology — no transitive trust, peer sets refreshed each epoch — requires a new argument but the target statement and proof structure are available.

Both paths sharpen bounds but do not eliminate the presupposition dependency. Live deployment data is the only source of ground truth, and the sleeper attack escapes any purely structural argument since it satisfies cluster membership criteria honestly.

OQ-49 (Oversight chain depth limit — closed): See §10.2 and §4.9.8. Empirical calibration of K_0 and ΔK remains, addressed in Phase 5 experiment 5.42.

OQ-53 (Off-chain committee collusion): The Chernoff argument that closed OQ-49 bounds committee compromise for protocol-visible selection events. Off-chain collusion is the residual attack it does not cover — a colluding supermajority can aggregate σ and decrypt `commit_T` without going through the commit-reveal flow, leaving no on-chain trace unless they subsequently submit a `null_v_decryption` transaction.

OQ-55 (SecLDP-GDP — explicit assumptions): The partial resolution rests on four assumptions that must be stated explicitly before the bound can be used in a security argument:

4. *Epoch independence.* Beacon rotation produces a fresh permutation key each epoch. Cross-epoch information leakage is bounded by Loopix sender anonymity; the adversary's view from epoch T is independent of epoch T' given ε_loopix. Without this, the subsampling amplification calculation degrades because the adversary accumulates views.
5. *Cover item approximate uniformity.* Cover items are drawn approximately uniformly from the local cluster. This is a conservative direction: if cover items are clustered around the node's true interests, p = n_v(T)/d_v overestimates the adversary's information per epoch, making the bound tighter than reality (safe). The cover weight formula in §4.5 is trust-decay-weighted, not uniform, so this assumption is approximate; the error is bounded by the trust-weight variance over cover items.
6. *Capacity-bounded adversary.* The adversary controls k_adv fixed peers (not a fraction of traffic). This prevents circularity in the n_cover bound (OQ-19/OQ-46). Under a rate-bounded model (α × traffic), m_fixed grows with n_cover and the bound becomes vacuous.
7. εloopix-approximate Loopix uniformity._ Loopix provides ε*loopix-approximate sender anonymity: for any two senders u, u', the adversary's posterior over sender identity given the observed traffic pattern is within multiplicative factor e*^{εloopix} of uniform. This is the standard Loopix anonymity model. OQ-58 must characterize ε_loopix per config (A–E) before numerical bounds from OQ-19/OQ-46/OQ-55 can be instantiated.

OQ-55 remaining blocking work (theoretical, pre-experiment):

Step 1 — Lemma 14 / Laplace equivalence. The partial resolution uses Lemma 14 (Cyffers et al.) to remove permutation-attributable components from the adversary's view matrix, then invokes Theorem 8 for Gaussian mechanism. PrivaCF's deployed mode uses Laplace noise. The question is whether the matrix factorization argument of Lemma 14 is compatible with Balle et al. (ICML 2020) subsampling amplification for Laplace: if the subsampling structure induced by the permutation matches the "Poisson subsampling" model of Balle et al., then ε_eff = log(1 + p×(e^ε−1)) where p = n_v(T)/d_v is formally justified and OQ-55 closes under Laplace. If the permutation induces "sampling without replacement" instead, the Balle et al. bound is slightly different (tighter) and an explicit theorem reference is needed. This check requires reading §3 of Balle et al. against the Lemma 14 matrix structure — no new math, just confirming which subsampling model applies.

*Step 2 — sens*Π(C;B) computation._ Cyffers et al. Theorem 3/4 defines sens*Π(C;B) = max*{G,G': neighbors} ||C(G−G')||*{B†B} where G is the gossip graph and C, B are the mixing and observation matrices. For PrivaCF's gossip exchange: C is the n_v(T)×d_v subsampling matrix (one row per transmitted element, column per preference dimension), B is the identity (adversary directly observes transmitted elements with Laplace noise). The computation reduces to bounding the column norm of C* *under the permutation constraint — each column has exactly one nonzero entry (the element lands in exactly one position), so ||C(G−G')||{B†B} = ||G−G'||2 restricted to the transmitted coordinates. The sensitivity is then sens = max change in preference weight per dimension = 2 (L1-normalized vector, bounded by §4.5). Writing this out explicitly closes Step 2 and gives a concrete σ*eff formula ready for numerical instantiation once OQ-58 resolves ε_loopix.

OQ-58 analytical lower bound (pre-experiment): OQ-58 is listed as empirical (calibrating λ per config), but a conservative analytical bound on ε*loopix is derivable from Piotrowska et al. (CCS 2017) before any experiment. The Loopix anonymity analysis gives sender anonymity as a function of λ (Poisson cover rate), mix topology (number of mixes, stratification), and adversary observation fraction. Specifically, the probability that the adversary correctly identifies a sender given traffic observation is bounded by a function of λ and the number of cover messages per real message. Inverting this gives a conservative* εloopix(λ) per config. This analytical bound would unblock numerical evaluation of OQ-19, OQ-46, and OQ-55 without waiting for Phase 5 experiments — the experiments would then refine the bound rather than provide it from scratch.

10.2 Resolved Prerequisites

All blocking prerequisites have been resolved or reclassified as non-blocking empirical calibration.

Closed or reclassified as non-blocking

ID	Resolution
OQ-C1	`trust_total` convergence under adversarial flooding — closed by inspection. Bounded above by `c` by construction; the update rule halts contributions once the cap is reached regardless of flooding rate. The residual question (can adversaries push `trust_total` to `c` before honest contributions arrive) is a restatement of the Sybil influence problem addressed by OQ-10, not a separate convergence concern.
OQ-C2	Single-decryption enforcement — closed by the DECRYPTION_SMT design. A second `null_v_decryption` transaction for the same `verdict_hash` produces the same `dec_nullifier`, already in the tree; honest validators reject the containing block.

ID	Resolution
OQ-1	Poseidon PRF + EC-VRF security — resolved by primitive split. Local per-epoch derivations (epoch ID, permutation, chop count, offset, niche delay, leaf salt) use Poseidon as a keyed PRF; pseudorandomness under standard Poseidon assumptions suffices and no verifiability property is required. On-chain verifiable selection (validator, committee, relay) uses EC-VRF (RFC 9381), reducing to DLEQ/DDH with production implementations in tendermint-rs. See §4.2.
OQ-2	ForwardCommit security — resolved as Boneh-Franklin IBE over BLS12-381 (Boneh & Franklin, CRYPTO 2001). Security reduces to DBDH in the random oracle model. DST alignment between BLS signing and IBE key derivation must be explicitly specified. Collusion carve-out bounded by Byzantine majority assumption. See §4.9.4.
OQ-4	Domain separator collision resistance — all derivations using `sk` now have a unique explicit (domain_sep, input_structure) pair after adding `"epoch_id"` and `"niche_delay"` separators. Collision argument reduces to standard Poseidon collision resistance. See §4.2.
OQ-5	dec_nullifier collision resistance — reduces to standard Poseidon collision resistance. `verdict_hash` is fixed on-chain before `null_v` is recovered, so chosen-input attacks are not applicable.
OQ-6	Δ_rise calibration — reclassified as empirical. The tension between path-responsiveness and on-off attack defense is real but is a calibration question, not a blocking one. Addressed in Phase 5 experiment 5.4 and tracked as §8.2 T1. See §7.2.
OQ-7	Oversight chain liveness — terminates by induction under honest majority (assumption A2). At each recursive level, a VRF-selected meta-committee with escalating size runs the same commit-reveal flow; under A2, each level terminates in finite expected time by the standard BFT argument (Castro & Liskov, 1999). The hard depth limit ensures termination even under degraded honesty conditions by capping the recursion before the honest majority fraction can fall below the BFT threshold. What remains open is the empirical question of how much damage accumulates during the resolution window — addressed by OQ-49.
OQ-8	Off-chain committee collusion — damage limited to targeted off-chain epoch ID linkage with no protocol consequence unless an on-chain transaction is submitted (at which point it becomes visible). Fully bounded by Byzantine majority assumption. See §8.1.
OQ-3	Statement 5 mobile benchmark — reclassified as non-blocking. Not a PoC blocker; required before any mobile deployment. See §10.1.
OQ-3b	Statement 2 mobile benchmark — reclassified as non-blocking. Same status as OQ-3. See §10.1.

ID	Resolution
OQ-49	Oversight chain depth limit — closed by Chernoff bound argument. Given that each level's committee is drawn independently via VRF with dual cluster diversity constraints, P(compromise at level d) = P(Bin(K_d, q) ≥ ⌊K_d/2⌋+1), which decays exponentially in K_d by the Chernoff bound. Cumulative escape probability is a product of independent per-level terms, each strictly less than 1, giving exponential decay in depth. With K_d growing linearly (K_d = K_0 + d·ΔK), per-level compromise probability itself shrinks with d, yielding doubly-exponential decay overall. Since any polynomial instance count (budget/cost) is dominated by exponential per-instance decay, a finite depth always suffices for any target success probability. The remaining calibration questions — concrete values of K_0 and ΔK for the target network — are addressed in Phase 5 experiment 5.42. See §4.9.8.

10.3 Proposed Experiments

Experiments are ordered by dependency tier. Tier 1 requires no infrastructure and no theoretical prerequisites — pen-and-paper or straightforward measurement work only. Tier 2 requires theoretical groundwork before experiment design is meaningful. Tier 3 requires Phase 1 infrastructure. Tier 4 requires Phase 2 or later. Only open questions appear here; closed questions (OQ-4, OQ-5, etc.) have been resolved and are not listed.

Tier 1 — No prerequisites

OQ-14 — SUSP_SMT staleness window. **Closed** — see §10.1 status. No experiment required.

OQ-15 — trust_total convergence under adversarial flooding. Model trust_total dynamics under sustained adversarial flooding as a bounded accumulator with a cap at `C`. Show analytically whether honest contributions can be crowded out before the cap is reached under different adversarial budget assumptions. Expected output: a characterization of the crowding-out condition as a function of adversarial budget and arrival rate. Effort: moderate.

OQ-30 — Frame size. Enumerate all protocol message types from Appendix A, including the PSI acknowledgment message type. Measure maximum serialized size for each under worst-case inputs. Set frame size to the maximum plus a fixed overhead margin. Expected output: a concrete frame size recommendation. Effort: minimal.

OQ-33 — Handoff ZK proof generation time. Build the Statement 5 circuit in Plonky3. Run on target mobile hardware (mid-range Android, iPhone SE class). Measure wall-clock proof generation time. Gate: under 2 seconds. Expected output: a benchmark table across hardware tiers. Effort: low once circuit exists.

OQ-36 — Peer attestation retention. Derive analytically the maximum audit query window across all audit classes. Confirm whether one epoch of attestation retention covers all cases or whether longer retention is required. Expected output: a retention period recommendation with justification. Effort: minimal.

Tier 2 — Theoretical groundwork required first

OQ-10 — Sybil influence empirical baseline. Collect flag raise rate data from Phase 5 simulation runs across all B-level and H-level detection contract rows. For each row, produce a TPR/FPR characterization conditioned on the simulated adversarial strategy. Use this to ground the p(flag_i raised | S) terms in §7.1b's flag compounding model. Paths A and B (stochastic block model and MeritRank ratio) may be pursued in parallel to sharpen structural bounds but are not prerequisites for empirical calibration. Expected output: a table of empirical flag probability estimates per strategy per detection mechanism, with confidence intervals and explicit statement of which adversarial behavioral presuppositions generated each estimate. Effort: very high.

OQ-17 — Off-chain committee collusion. Analyze the maximum damage a colluding committee supermajority can cause by executing a covert off-chain ForwardCommit decryption. The ceiling is the Byzantine majority assumption; the question is whether meaningful damage is possible below that ceiling and whether it is detectable in arrears from public chain data. Expected output: a damage bound and a detectability argument. Effort: moderate.

OQ-53 — Off-chain collusion detectability. Related to OQ-17 but focused on the detection side: given only the public chain, can an observer reconstruct evidence of a covert decryption that did not go through the commit-reveal flow? Expected output: either a detection protocol or a proof that covert decryption is undetectable below threshold collusion. Effort: moderate.

OQ-61 — PSI acknowledgment rate limit and privacy analysis. Analyze the privacy implications of the signed cross-cluster PSI acknowledgment message type. Determine what rate limit prevents acknowledgment spam while preserving the cross-cluster attempt rate signal. Expected output: a rate limit recommendation, a privacy analysis of the acknowledgment message under Loopix, and a determination of whether the mechanism requires a full PSI protocol change or can be implemented as an extension. Effort: moderate.

Tier 3 — Requires Phase 1 infrastructure

OQ-3 — Statement 5 mobile benchmark. Instrument the full Statement 5 circuit in Plonky3. Benchmark on target mobile hardware across a range of SUSP_SMT depths. Gate: proof generation under per-epoch budget on mid-range hardware. Expected output: a benchmark table and a recommended maximum SUSP_SMT depth. Effort: low once circuit exists.

OQ-3b — Statement 2 mobile benchmark. Instrument the inner product argument for directional consistency at $d=128$ and $d=256$ on Plonky3/WASM. Benchmark on target mobile hardware. Gate: proof generation under per-epoch budget alongside Statement 5. Expected output: a benchmark table and a recommended maximum vector dimension. Effort: low once circuit exists.

OQ-18 — VDF gap tolerance. Run the admission VDF chain on simulated mobile connectivity profiles. Measure chain failure rates as a function of gap tolerance policy. Expected output: a gap tolerance recommendation calibrated to realistic mobile usage. Effort: low.

OQ-19 — Cover item threshold. Vary n_cover and measure anonymity set size against CF quality degradation on LastFM and RateYourMusic. Expected output: a recommended n_cover per sparsity tier (head / long-tail). Effort: low.

OQ-20 — PSI cache decay calibration. Simulate epoch rotation with varying λ*proof and λ*noproof values. Measure peer set stability against re-identification risk. Expected output: recommended λ values per community sparsity profile. Effort: moderate.

OQ-21 — μ calibration. Vary μ and measure CF quality at different hop distances on MovieLens and LastFM. Expected output: a recommended μ value and sensitivity curve. Effort: low.

OQ-22 — Permutation reconstruction time. Simulate an adversary observing gossip vectors across multiple epochs and attempting to reconstruct the underlying permutation. Measure reconstruction success rate as a function of observation count and n_jitter range. Expected output: a minimum observation count required for meaningful reconstruction. Effort: moderate.

OQ-25 — Adjacent-epoch CF weight. Compare recommendation quality with the 0.5× adjacent-epoch discount applied vs. equal weighting, across dense and sparse datasets. Expected output: a recommended weight and sensitivity analysis. Effort: low.

OQ-37 — n_commit calibration. Simulate behavioral fingerprinting attack against public chain timing data at varying n_commit values. Measure re-identification rate against audit freshness window. Expected output: a recommended n_commit value. Effort: moderate.

OQ-47 — SUSP_SMT proof latency scaling. Measure SMT non-membership proof generation time on target mobile hardware as tree depth grows to projected maximum. Expected output: a latency curve and a recommended maximum tree depth. Effort: low.

OQ-59 — Burst score threshold calibration. Run synthetic admission timing data through the committee burst score aggregation at varying network sizes. Simulate both organic community growth events and coordinated admission campaigns. Measure false positive rate (organic growth flagged) and false negative rate (campaigns not flagged) as a function of burst window W and burst_score ceiling. Expected output: recommended W and ceiling values per network size tier. Effort: moderate.

Tier 4 — Requires Phase 2 or later

OQ-24 — k_min calibration. Simulate PSI neighborhood growth across community density profiles. Measure recommendation quality degradation and peer discovery convergence time as a function of k_min. Expected output: recommended k_min per community density tier. Effort: moderate.

OQ-26 — Auditor committee size K_committee. Simulate committee collusion attempts at varying K_committee values under cluster diversity constraints. Measure collusion probability as a function of adversarial budget and cluster structure. Expected output: a recommended K_committee with a collusion probability bound. Effort: moderate.

OQ-28 — Behavioral cluster false positive rate. Simulate co-located legitimate users and measure behavioral cluster false positive rate under the compound flag system. Expected output: a characterization of false positive conditions and a recommended θ_behavioral. Effort: moderate.

OQ-29 — Smoothness floor calibration. Run smoothness detection against legitimate consistent nodes and adversarial sleeper nodes across varying σ^2*floor values. Test suite must explicitly include recovery trajectory patterns as a distinct test case. Measure TPR/FPR. Expected output: a recommended* σ^2floor per community type with recovery-trajectory-specific characterization. Effort: moderate.

OQ-34 — Behavioral fingerprint matching calibration. Simulate sophisticated epoch rotators varying behavioral patterns across re-admission attempts. Measure detection rate as a function of θ_behavioral and T_behavioral_window. Expected output: recommended parameter values with a characterization of the evasion boundary. Effort: high.

OQ-43 — p_v inflation damage quantification. Simulate targeted p_v inflation within Statement 1 and 3 bounds on MovieLens and RateYourMusic. Measure recommendation quality degradation as a function of inflation magnitude and target item popularity segment (head / long-tail). Expected output: a damage characterization per segment. Effort: moderate.

OQ-9 — Reputation weight calibration. Simulate adversarial nodes attempting to maximize individual score components while degrading others. Sweep w_1–w_5, δ_decay, and α across ranges. Measure which component dominates under each adversarial strategy and identify weight combinations that resist manipulation. Expected output: recommended default weight values with adversarial resistance characterization. Effort: moderate.

OQ-54 — Noise system calibration per segment. On MovieLens and RateYourMusic, split items into head and long-tail by popularity (Park & Tuzhilin boundary). Apply chopping and Laplace DP noise at varying parameters for each segment. Measure Precision@K, NDCG, long-tail discovery rate, and privacy-utility curve per segment. Expected output: recommended noise mechanism and parameters per segment. Effort: moderate. Note: if content-based bootstrapping (§13) is available for long-tail items, the effective signal density for those items improves, shifting where chopping and Laplace diverge — the experiment should be run both with and without content bootstrapping.

OQ-49 — Oversight chain depth limit. **Structural question closed** by Chernoff bound argument (see §4.9.8, §10.2). Empirical calibration of K_0 and ΔK remains: simulate recursive oversight chains under varying adversarial committee compositions to produce a recommended depth limit and escalation schedule. Effort: high.

OQ-52 — Stalling minority damage. Simulate a stalling minority at varying committee sizes. Measure damage accumulation before the oversight chain resolves. Expected output: a damage bound as a function of minority size and epoch duration. Effort: moderate.

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11. Comparative Analysis

11.1 Identity and Privacy Properties

System	No central server	Pseudonymous identity	Admission cost	Behavioral privacy	Deployment
EigenTrust	Partial (DHT)	No	None	None	Research
TrustGuard	Yes	No	None	None	Research

System	No central server	Pseudonymous identity	Admission cost	Behavioral privacy	Deployment
SybilGuard	Yes	No	None	None	Research
DSybil	Partial	No	None	None	Research
McSherry & Mironov (2009)	No (central required for aggregation)	No	None	ε-DP formal bound on aggregation phase	Research (Netflix-scale)
GOSSPLE	Yes	Weak (proxy, long-term linkable)	None	Bloom filter digests only; no formal bound	Research (PlanetLab)
Hegedűs et al. (2020)	Yes	No	None	None (model params in plaintext)	Research
Web3Recommend + MeritRank (2023)	Yes	Pseudonym (persistent edges, linkable)	None	None	PoC (MusicDAO)
Unified DP-DL MF (2025)	Yes	No	None	GDP formal (LDP / PNDP / SecLDP)	Research
PrivaCF (Loopix profile)	Yes	Yes (Poseidon PRF + EC-VRF)	VDF chain + checkpoints	Chopping/Laplace + ZK + niche delay + GPA-resistant transport	PoC
PrivaCF (Tor/I2P profile)	Yes	Yes (Poseidon PRF + EC-VRF)	VDF chain + checkpoints	Chopping/Laplace + ZK + niche delay (no GPA resistance at transport)	Spec only

The two PrivaCF rows reflect the transport profile choice in §5.1. Profile-independent properties (identity rotation, admission cost, content-level chopping/Laplace, ZK preference privacy) are identical across rows; the differentiator is wire-level anonymity against a global passive adversary.

11.2 Sybil Resistance and Audit Properties

System	Sybil resistance	On-chain verdicts	Suspension persistence	Dark node closure	Observable extraction	Recommendation output
EigenTrust	None	No	N/A	N/A	N/A	No
TrustGuard	Oscillation detection	No	N/A	N/A	N/A	No
SybilGuard	Graph-cut	No	N/A	N/A	N/A	No
DSybil	Non-overwhelming rule (heuristic in PrivaCF's setting)	No	N/A	N/A	N/A	Partial
McSherry & Mironov (2009)	None	No	N/A	N/A	N/A	Yes (Netflix-scale)
GOSSPLE	None (certs assumed externally)	No	N/A	N/A	N/A	Yes (empirical)
Hegedűs et al. (2020)	None	No	N/A	N/A	N/A	Yes (gossip MF, empirical quality)
Web3Recommend + MeritRank (2023)	Formal ratio bound (random-walk setting)	No	None	N/A	N/A	Yes (MusicDAO PoC)
Unified DP-DL MF (2025)	None	No	N/A	N/A	N/A	No

System	Sybil resistance	On-chain verdicts	Suspension persistence	Dark node closure	Observable extraction	Recommendation output
PrivaCF (Loopix profile)	Non-overwhelming trust cap + temporal depth + k_min/ε-DP impact bounding + content-based PSI-similarity flagging (§7.4) + epoch-granular behavioral signals + mix-layer density signals. Sub-epoch timing signals unavailable. Empirical calibration deferred to Phase 5 (OQ-10).	Yes (permanent)	Hard (nullifier + SUSP_SMT)	Yes — ForwardCommit	Yes — commit-reveal, watchdog	Yes

System	Sybil resistance	On-chain verdicts	Suspension persistence	Dark node closure	Observable extraction	Recommendation output
PrivaCF (Tor/I2P profile)	Non-overwhelming trust cap + temporal depth + k_min/ε-DP impact bounding + full behavioral clustering (§6.2 4-tuple fingerprint) + sub-epoch coordination detection + PSI-similarity flagging. Mix-layer signals unavailable. Empirical calibration deferred to Phase 5 (OQ-10).	Yes (permanent)	Hard (nullifier + SUSP_SMT)	Yes — ForwardCommit	Yes — commit-reveal, watchdog	Yes

11.3 Narrative Analysis

Against the P2P reputation baselines (EigenTrust, TrustGuard, SybilGuard, DSybil): each addresses one or two of the three core tensions — personalization, privacy, integrity — but none address all three. None provide pseudonymous identities, on-chain verdicts, or suspension persistence. DSybil provides a partial non-overwhelming trust bound but no privacy guarantees and no recommendation output. PrivaCF's contributions over these baselines are clear: rotating pseudonymous identity with cryptographic unlinkability; admission cost via sequential VDF; preference privacy via Pedersen commitments and gossip vector obfuscation; behavioral privacy via Merkle-committed history with ZK consistency proofs; permanent on-chain suspension verdicts surviving epoch rotation via the nullifier mechanism; dark node closure via forward-secure commitment; and observable extraction via commit-reveal ordering with public watchdog signals.

Graph-cut detection as used in SybilGuard and SybilLimit is structurally unavailable in PrivaCF due to the ephemeral peer relationship model — no persistent graph exists to cut. The behavioral consequences of solipsistic cluster structure are observable without explicit graph visibility through the compound flag system (§7.1a T.7, §7.8), at the cost of a behavioral-probabilistic rather than graph-cut deterministic guarantee. The tradeoff is intentional.

Against McSherry & Mironov (2009): this is the strongest comparison on privacy formalism. Their system achieves formal ε-DP for the Netflix Prize algorithms but requires a central server for the aggregation phase — the core architectural trade-off PrivaCF rejects. PrivaCF cannot currently make a comparable formal privacy claim. OQ-55 investigates whether the GDP framework from Cyffers et al. (2025) can close this gap for PrivaCF's gossip exchange without reintroducing central aggregation.

Against GOSSPLE: structurally closest to PrivaCF's gossip peer discovery. GOSSPLE uses Bloom filter digests over proxy-based two-hop paths to find interest-similar peers without revealing identity, validated empirically on PlanetLab. The anonymity is weak — proxy paths are linkable over time and no Sybil resistance exists. PrivaCF replaces the proxy with PSI (stronger non-revelation), adds identity rotation, admission cost, and the full audit stack. GOSSPLE's convergence results (~14–20 gossip cycles to stable peer sets) provide a useful lower-bound expectation for PrivaCF's PSI-based discovery phase.

Against Hegedűs et al. (2020): their gossip-based matrix factorization on fully distributed data — no central server, model parameters exchanged peer-to-peer — achieves comparable quality to federated learning. This directly validates that decentralized gossip is a viable substrate for CF quality. PrivaCF's gossip vector approach is architecturally similar but trades MF-level accuracy for stronger preference privacy (gossip vectors vs. raw latent factors). OQ-56 formalizes this comparison: whether PrivaCF's item-based CF on accumulated vectors matches Hegedűs-style MF under matched sparsity, giving PrivaCF its first empirical quality benchmark.

Against Web3Recommend + MeritRank (2023): the closest architectural competitor. Fully decentralized, gossip-based, Sybil-resistant, with a PoC deployment on a real music platform. MeritRank's formal ratio bound (`lim|S|→∞ w⁺/w⁻ ≤ c`) is a tighter formal statement than PrivaCF currently claims. The critical difference: Web3Recommend operates under pseudonyms with persistent graph edges — any observer can reconstruct the full trust graph and link identities across time. PrivaCF closes this with epoch rotation, arbitration-committee custody of sensitive state, and the nullifier mechanism. Web3Recommend also has no suspension persistence, no dark node closure, and no formal admission cost. The MeritRank ratio bound is being adapted as Path B for OQ-10.

Against Unified DP-DL MF (2025): provides formal GDP bounds for decentralized learning under three trust models. Their SecLDP model — privacy conditional on a hidden secret — maps directly onto PrivaCF's permutation-secret gossip exchange (the permutation key is the hidden secret; $n_v(T)$ elements are the partially observable output). This is the technical basis for OQ-55. PrivaCF's Mafalda-SGD-equivalent would be the gossip vector chopping + Laplace noise mechanism; deriving the corresponding GDP bound would give PrivaCF its first formal privacy guarantee without architectural changes.

Summary of residual gaps: PrivaCF is the only system in this comparison with unlinkable rotating identity, suspension persistence, dark node closure, and decentralized operation simultaneously. The gaps are: no formal privacy bound (OQ-55 targets this), no formally proven Sybil influence bound (OQ-10 reframed as empirical — structural bounds established, flag probability terms require live deployment data), and no empirical recommendation quality benchmark (OQ-56). The residual architectural gap is threshold collusion for off-chain nullifier extraction, bounded by the same Byzantine majority assumption the consensus layer already depends on.

Transport-profile design choice. PrivaCF defines two transport profiles (§5.1) rather than mandating one — Loopix profile prioritizes wire-level anonymity, Tor/I2P profile prioritizes Sybil-detection signal richness. The choice reflects an explicit acknowledgement that anonymity-set strength and behavioral observability are antagonistic at the wire layer; instead of picking one absolute ranking, the spec lets each deployment choose. No other system in this comparison exposes this choice as a deployment-time decision: most mandate clearnet (and therefore implicitly Tor-equivalent observability), and McSherry & Mironov mandate central aggregation. The profile-based framing is novel as a comparative axis.

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12. Related Work

Reference	Relevance
Douceur (IPTPS 2002)	Sybil attack definition and impossibility
Cheng & Friedman (P2PEcon 2005)	Sybil-proofness impossibility for symmetric functions
Kamvar, Schlosser & Garcia-Molina (WWW 2003)	EigenTrust
Srivatsa, Kambhampati & Liu (ICDCS 2005)	TrustGuard — oscillation detection; asymmetric penalty

Reference	Relevance
Yu, Kaminsky, Gibbons & Flaxman (SIGCOMM 2006)	SybilGuard
Stannat, Ileri, Gijswijt & Pouwelse (AAMAS 2021)	Sufficient conditions for personalized-seed Sybil resistance
Yu, Kaminsky, Gibbons & Flaxman (IEEE S&P 2009)	DSybil non-overwhelming trust
Viswanath, Becker, Gummadi & Mislove (IEEE S&P 2008)	SybilLimit
Tran, Min, Li & Subramanian (NDSS 2009)	SybilInfer
Damiani et al. (2002)	Reputation-based P2P trust
Fanti et al. (SIGMETRICS 2018)	Dandelion++
Pinkas, Rosulek, Trieu & Yanai (USENIX Security 2018)	Unbalanced PSI
Kolesnikov et al. (CCS 2016)	KKRT circuit PSI (balanced)
Malkov & Yashunin (2018)	HNSW
Indyk & Motwani (STOC 1998)	LSH
McSherry & Mironov (KDD 2009)	Differentially private CF
Parsarad & Wagner (2025)	DP harm to sparse user recommendations
Dwork, Rothblum & Vadhan (2010)	Basic DP composition theorem
Werthenbach & Pouwelse (arXiv:2306.15044, 2023)	SSP attack taxonomy
Nasrulin et al. (IEEE BRAINS 2022)	MeritRank

Reference	Relevance
Sun, Han & Liu (2008)	Asymmetric penalty mechanism
Fung et al. (RAID 2020)	FoolsGold — Sybil detection via contribution vector similarity; not adopted (incompatible with privacy stack, see §7.4)
Perrin (2018)	Noise Protocol Framework
Jøsang & Ismail (2002)	Beta Reputation System
Maram et al. (CCS 2021)	CanDID — distributed credential storage
Boneh et al. (2018)	VDF constructions
Wesolowski (2018)	Efficient VDF verification
Castro & Liskov (OSDI 1999)	PBFT
Buchman (2016)	Tendermint BFT consensus
Boneh, Drijvers & Neven (2018)	BLS multi-signatures
Arshad et al. (IEEE Access 2022)	REPUTABLE
Anonymous (J. Sens. Actuator Netw. 2023)	Blockchain + MPC decentralized reputation
Shi, Zhang, Yin, Chi & Liu (Results in Engineering 2025)	DSRep
Möser et al. (2018)	Monero transaction graph analysis; decoy selection distribution
Sasson et al. (IEEE S&P 2014)	Zcash — nullifier pattern for spent-note detection
WhiteHat (2019)	Semaphore — ZK group membership with nullifiers
Boneh, Bünz & Fisch (CRYPTO 2019)	Batched accumulator witnesses; SMT non-membership
Grassi, Khovratovich et al. (USENIX Security 2021)	Poseidon hash function
Camenisch & Lysyanskaya (Eurocrypt 2001)	Traceable anonymous credentials

Reference	Relevance
Chaney, Stewart & Engelhardt (2018)	Recommendation feedback loops and filter bubble dynamics
Adomavicius & Tuzhilin (2005)	CF assumptions survey
Hu, Koren & Volinsky (2008)	Implicit feedback limitations
Leskovec, Adamic & Huberman (2007)	Information diffusion patterns, organic vs. coordinated spread
Blanchard et al. (2017)	Byzantine-resilient aggregation (Krum)
Sundaram & Hadjicostis (2011)	Resilient distributed averaging
Holland, Laskey & Leinhardt (1983)	Stochastic block models — suggested framework for OQ-10 Path A
Boneh, Lynn, Shacham (ASIACRYPT 2001)	BLS signatures — threshold BLS used in commit-reveal flow
Pedersen (CRYPTO 1991)	Verifiable secret sharing — basis for commit-reveal scheme
Siddarth, Ivliev, Siri & Berman (Frontiers in Blockchain, 2020)	"Who Watches the Watchmen?" — survey of subjective approaches to Sybil-resistance; frames the "who verifies the verifier" regress that the recursive oversight chain addresses
Park & Tuzhilin (RecSys 2008)	Formal head/long-tail popularity split in CF evaluation; basis for head/long-tail segmentation in §9.3 and OQ-54
Abdollahpouri, Burke & Mobasher (FLAIRS 2019)	Popularity bias and re-ranking in recommender systems; head/long-tail framing
Kermarrec, Van Roy, Ganesh & Voulgaris (MIDDLEWARE 2010)	GOSSPLE — gossip-based anonymous social network using Bloom filter digests and proxy routing for interest-similar peer discovery; convergence results (~14–20 cycles) bound PrivaCF's PSI discovery expectations
Hegedűs, Danner & Jelasity (ECML PKDD 2019)	Gossip-based matrix factorization on fully distributed data; empirically comparable to federated learning; basis for OQ-56 CF quality benchmark

Reference	Relevance
Trautwein, Ishmaev & Pouwelse (arXiv:2307.01411, 2023)	Web3Recommend — decentralized social recommendation with MeritRank Sybil resistance; formal ratio bound `lim|S|→∞ w⁺/w⁻ ≤ c`; basis for OQ-10 Path B; persistent graph linkability is the key gap vs. PrivaCF
Cyffers et al. (arXiv:2510.17480, 2025)	Unified GDP bounds for decentralized learning via matrix factorization; SecLDP trust model maps onto PrivaCF's permutation-secret gossip exchange; basis for OQ-55
Stringhini et al. (ACSAC 2010)	OSN bot detection — temporal clustering on join and action as persistent signals; basis for T.1 and T.2 in §7.1a
Yang et al. (WWW 2014)	Spambot evolution across three generations — temporal clustering survives content diversification; behavioral similarity degrades with randomization; basis for T.1, T.2, T.4
Mukherjee et al. (WWW 2013)	Yelp review fraud — velocity as strongest discriminator of fake campaigns; basis for T.8
Urdaneta et al. (ACM Computing Surveys 2011)	DHT sybil studies — temporal clustering on join; basis for T.1
Varol et al. (ICWSM 2017)	Bot detection across six categories — action timing among top predictors; trajectory smoothness as indirect predictor; basis for T.2 and T.5
Cao et al. (IMC 2012)	Pairwise similarity distributions of sybil clusters — lower variance, higher mean than legitimate clusters; basis for T.4
Hoffman et al. (ACM Computing Surveys 2009)	Whitewashing as dominant evasion strategy in P2P reputation systems; basis for T.6

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13. Recommendation Layer: Open Problems for Deployment

The CF algorithm is intentionally not fixed by the protocol. Different communities have different content types, interaction patterns, sparsity profiles, and privacy/utility tradeoffs. A single recommendation algorithm would serve none of them well. What follows is a speculative enumeration of tensions any deployment will need to address when designing or selecting a recommendation layer. No solutions are proposed — these are community-dependent design problems, not protocol-level ones.

Sparsity and cold start. Item-based CF requires sufficient co-occurrence signal to produce reliable similarity estimates. For genuinely niche items the co-occurrence matrix may be too sparse for cosine similarity to be meaningful regardless of privacy mechanisms. The novelty bonus accelerates accumulation but does not substitute for signal density. Deployments targeting extreme long-tail content should evaluate hybrid fallbacks (content-based priors, cluster-level popularity estimates) that do not require centralizing metadata the protocol deliberately avoids.

Feedback loop and filter bubble risk. Recommendations influence interactions, which influence preference vectors, which influence future recommendations. This loop is not modeled in the current CF sketch. Under some conditions it could produce runaway local optima or filter bubbles that degrade recommendation diversity over time. Randomness injection at the recommendation layer is a speculative mitigation; calibration is community and content-type dependent. See Chaney, Stewart & Engelhardt (2018).

Reputation system interaction. Niche users structurally interact with fewer items and maintain smaller peer sets than mainstream users. Participation-rate-weighted reputation could systematically push niche users toward lower reputation bands, causing their announcements to carry less weight in others' `trust_total` estimates — suppressing exactly the long-tail signal the CF layer is designed to amplify. The niche-specific protocol handling (announcement delays, cluster weight adjustments) partially addresses this but the interaction with reputation bands is not formally analyzed.

Epoch length and content type. A single epoch length is simultaneously too long for fast-moving content and potentially too short for slowly-evolving taste domains. Preference vector freezing at epoch boundaries quantizes updates to the epoch cadence. Epoch length should be treated as a community-tunable parameter calibrated to the dominant content type.

Positive-only signal and neutral interactions. Dislikes are retained locally and applied as a post-processing filter, which is a meaningful improvement over standard implicit feedback systems that conflate interaction with preference. The residual ambiguity — items interacted with neutrally — is narrower than the standard implicit feedback problem but remains, and is more significant in sparse domains where false positive signal is costlier.

Niche announcement privacy-utility tradeoff. Small anonymity sets for niche items create a tension specific to long-tail communities: VRF-derived announcement delays reduce re-identification risk but also delay signal propagation that benefits niche discovery. The right balance is community-dependent and should be evaluated empirically against the anonymity set sizes characteristic of that community's item distribution.

Content-based hybrid as a future direction. For content types with publicly available metadata (films, music, books), each node can independently fetch item features from external sources (IMDB, MusicBrainz, Spotify, etc.) without centralizing anything — the metadata is already public. For user-generated content (videos, posts), local AI-based embedding (transcript via Whisper, text or visual similarity via on-device models) produces item similarity signals without sharing embeddings. In both cases the content layer is either public or computed locally; only the interaction signal is shared via the gossip protocol. This directly addresses the new-item cold-start gap for these domains: a new item with zero interactions can be bootstrapped via content similarity to existing items that have accumulated collaborative signal. Sequential recommendation signals (using the order of a user's own interaction history, not just the unordered bag) are similarly local — a user's own sequence is known on-device and can feed a local sequential model without any additional sharing. None of this is protocol-level; it belongs to the recommendation layer each deployment builds on top of PrivaCF.

Distinguishing coordinated pushing from organic popularity surges. Naive coordinated pushing is detectable through behavioral synchrony signatures (same epoch, uniform timing, uniform rating distributions). Sophisticated mimicry of organic diffusion patterns — seeding from multiple behavioral clusters with staggered timing — is harder to distinguish from genuine popularity surges using the observables currently available. This is a soft boundary that empirical testing should characterize rather than a problem the protocol claims to solve.

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14. Beyond Recommendations: Decentralized Learning Profile

The substrate (§4–§7) is not specific to recommendation. The same machinery — pseudonymous rotating identity with cross-epoch continuity, PSI peer neighborhoods, trust-weighted contribution aggregation, FoolsGold-on-PSI-peers Sybil suppression, and the Loopix transport — describes a general **privacy-preserving, Sybil-resistant contribution aggregation protocol**. Recommendation is one instantiation. Decentralized learning is another.

This section stakes the claim without specifying the protocol. A full DL profile is out of scope for v0.3.1; the goal here is to make the substrate's scope explicit and to record what would be required to host a DL workload.

14.1 Why DL fits the substrate

In decentralized learning (Lian et al., NeurIPS 2017; Hegedűs, Danner & Jelasity, ECML PKDD 2019; Bittensor) peers exchange model updates directly without a central aggregator. Convergence depends on a mixing matrix that propagates each peer's gradient through the network, and on Sybil-resistance over the contribution pool. Both are problems the substrate already solves for the recommendation case:

Recommendation construct	Decentralized learning analogue
Preference vector `p_v`	Local gradient or weight delta `g_v`
PSI on item-interaction sets	PSI / similarity on gradient direction or local data distribution
`trust_weight(v)` over peers	Per-peer aggregation weight
Gossip vector push	Gradient push to PSI peer neighborhood
`announcement_token` rate limit	Per-round update budget per pseudonymous client
FoolsGold-on-PSI-peers (§7.4)	Direct Sybil defense — FoolsGold was originally a federated-learning defense
Handoff state (§6.4)	Optimizer state / momentum carry-over across epoch rotations
Cross-epoch continuity (§4.7)	Stable per-client contribution history without persistent linkability
Content-addressed item payloads	Content-addressed model artifacts (architectures, base weights, training scripts)

FoolsGold's presence in §7.4 is the clearest tell: the protocol already runs a federated-learning Sybil defense over PSI peer neighborhoods. The construction transfers directly when the contribution is a gradient rather than a preference endorsement.

14.2 What the substrate would need to add

A DL profile is not a free upgrade. At minimum:

- **Secure aggregation.** Recommendation tolerates peers seeing each other's contributions through PSI — preferences are coarse-grained and obfuscated (§4.5). Gradients leak training-data reconstruction via gradient-inversion attacks (Geiping et al., NeurIPS 2020), so a DL profile must layer Bonawitz-style masked aggregation or homomorphic accumulation over the gossip exchange. The Loopix transport and the multi-recipient commit_T construction (§4.9) compose cleanly with masked aggregation but neither substitutes for it.
- **Distribution-aware peer selection.** Recommendation peer selection uses PSI on item-interaction sets — overlap predicts taste similarity. DL convergence under non-IID data requires peers whose distributions are heterogeneous enough to propagate signal across the network. The PSI primitive applies; the selection criterion inverts.

- **Model-artifact distribution.** Items in the recommendation case are content-addressed and pulled out of band. Model artifacts (architectures, base weights, training scripts) follow the same pattern but at substantially larger size. No protocol change is required, but a deployment must provide content-addressed storage sized for model artifacts.
- **Convergence accounting.** Recommendation quality is judged at any time by the local node. DL workloads converge over rounds; the substrate must expose round-aligned epochs (the existing epoch boundary suffices) and a way to publish "model at round T" as a content-addressed artifact for late-joining clients to bootstrap from.
- **Reward / contribution accounting (if incentivized).** Bittensor-style economic incentives over decentralized training require a contribution-quality signal. The per-epoch score (§6.1) is the natural anchor — adapted from "endorsement uptake" to "gradient quality / convergence contribution" — but the formula is a deployment choice.

14.3 Privacy and Sybil properties under a DL profile

All substrate guarantees survive a DL profile unchanged:

- IP↔epoch_id unlinkability (§5.1) holds — gradients are transmitted over the same transport.
- Cross-epoch contribution unlinkability without `sk` (§4.2) holds.
- DSybil + FoolsGold + temporal depth (§7) directly defend the aggregation step.
- The compound flag system (§7.8) reuses unchanged.

The DL-specific risk added beyond the recommendation case is **gradient inversion** — a per-message rather than per-protocol risk — which the secure-aggregation requirement above is exactly designed to address.

14.4 Status

Stated, not specified. A v0.4 or later revision may define a normative DL profile if the recommendation deployment validates the substrate. The framing here is to make explicit that the substrate's scope is broader than recommendation, and that the project's "post-big-tech" positioning extends naturally to model training — the domain where centralized compute monopolies are most entrenched and decentralized alternatives are most needed.

Appendix A — Full Message Schemas

All messages transmitted inside uniform fixed-size encrypted frames via Noise Protocol sessions.

```
// GOSSIP VECTOR PUSH
{
  "type":          "gossip_push",
  "epoch_id":      "<Poseidon(sk, beacon_T, null_v, 'epoch_id')-derived epoch
ID>",
  "vector":        "<permuted float array, exactly n_v(T) elements>",
  "noise_system": "chopping | laplace",
  "sig":           "<Sign(epoch_id, H(vector ‖ T))>"
}

// PULL REQUEST + CLASS 2 AUDIT CHALLENGE
{
  "type":         "pull_request",
  "epoch_id":     "<requester epoch ID>",
  "audit_nonce": "<random 256-bit nonce>",
  "sig":          "<Sign(epoch_id, H(nonce ‖ T))>"
}

// PULL RESPONSE + CLASS 2 AUDIT RESPONSE
{
  "type":         "pull_response",
  "epoch_id":     "<responder epoch ID>",
  "vector":       "<permuted float array, exactly n_v(T) elements>",
  "audit_hash":  "<H(state ‖ nonce ‖ epoch_id)>",
  "receipt_sig": "<Sign(responder_epoch_id, H(vector ‖ T))>",
  "sig":          "<Sign(epoch_id, H(vector ‖ audit_hash ‖ T))>"
}

// CLASS 3 AUDIT CHALLENGE
{
  "type":          "pull_request",
  "epoch_id":      "<committee member epoch ID>",
  "audit_nonce":  "<Poseidon(beacon ‖ 'c3_nonce' ‖ committee_epoch_id)>",
  "audit_class":  "3",
```

```
  "challenged_T": "<suspect epoch>",
  "sig":          "<Sign(epoch_id, H(nonce ‖ challenged_T ‖ T))>"
}

// CLASS 3 AUDIT RESPONSE
{
  "type":         "audit_response",
  "epoch_id":     "<target epoch ID>",
  "merkle_proof": "<branch paths for challenged leaf categories>",
  "zk_proof":     "<Plonky3 proof for Statements 1-3 and 5>",
  "sig":          "<Sign(epoch_id, H(merkle_proof ‖ zk_proof ‖ T))>"
}

// AUDITOR HANDOFF
{
  "type":                       "auditor_handoff",
  "epoch_id":                   "<node epoch ID>",
  "new_commitment":             "<C_p(T)>",
  "new_merkle_root":            "<M_v(T)>",
  "leaf_counts": {
      "ANNOUNCEMENT":           "<n>",
      "PULL_RESPONSE":          "<n>",
      "AUDIT_RESPONSE":         "<n>",
      "RATE_LIMIT":             "<n>"
  },
  "susp_smt_root_ref":          "<SUSP_SMT_root_T>",
  "commit_T":                   "<ForwardCommit(null_v, epoch_id_T,
threshold_BLS_pk_T; r)>",
  "threshold_bls_pk_T":         "<public key of this epoch's committee>",
  "successor_zk_proof":         "<proof: valid successor + leaf count consistency
+ Statement 5 (4 checks) + PSI acknowledgment witnesses>",
  "rolling_chain_commitment": "<Poseidon(prior_chain ‖ current_proof ‖
audit_interactions ‖ SUSP_SMT_root_T)>",
  "zk_continuity_proof":        "<arbitration committee only — not in public
handoff>",
  "encrypted_shares":           ["<Encrypt(pk_auditor_i, shamir_share_i)>",
"..."],
  "sig":                        "<Sign(epoch_id, H(all fields ‖ T))>"
```

```
}

// ON-CHAIN EPOCH TRANSACTION
{
  "type":                "epoch_transaction",
  "epoch_id":            "<node epoch ID>",
  "commit_T":            "<ForwardCommit — every epoch, exempt from n_commit>",
  "commit_T_zk_proof": "<Statement 5 ZK proof — every epoch>",
  "threshold_bls_pk_T":"<committee threshold BLS public key for this epoch>",
  // Fields below submitted only every n_commit epochs:
  "score_band":          "<committee-attested band 1-4>",
  "health_tier":         "<self-reported — committee verdicts override>",
  "commitment":          "<C_p(T)>",
  "merkle_root":         "<M_v(T)>",
  "vdf_proof":           "<identity chain proof for this epoch>",
  "band_attestation":  "<threshold BLS sig from committee on score_band>",
  "susp_smt_root_ref": "<SUSP_SMT_root_T>",
  "inter_cluster_rep": "<float>",
  "sig":                 "<Sign(epoch_id, H(all fields ‖ T))>"
}

// ON-CHAIN VERDICT COMMIT
{
  "type":                    "verdict_commit",
  "committee_member_id":   "<member epoch ID>",
  "target_epoch_id":       "<audited node epoch ID>",
  "verdict_epoch":         "<T>",
  "commit":                  "<H(bls_share_i ‖ verdict ‖ nonce_i)>",
  "sig":                     "<Sign(committee_member_id, H(commit ‖ T))>"
}

// ON-CHAIN VERDICT REVEAL
{
  "type":                    "verdict_reveal",
  "committee_member_id":   "<member epoch ID>",
  "target_epoch_id":       "<audited node epoch ID>",
  "verdict_epoch":         "<T>",
  "bls_share_i":           "<BLS secret share contribution>",
```

```
  "verdict":                "SUSPENDED | PASS",
  "nonce_i":                "<random nonce>",
  "sig":                    "<Sign(committee_member_id, H(bls_share_i ‖ verdict
‖ nonce_i ‖ T))>"
}

// ON-CHAIN NULL_V DECRYPTION
{
  "type":             "null_v_decryption",
  "verdict_hash":     "<H(committee_verdict_transaction)>",
  "epoch_id":         "<suspended node's last epoch_id>",
  "null_v":           "<recovered nullifier value>",
  "dec_nullifier":    "<Poseidon(verdict_hash, null_v)>",
  "sigma":            "<aggregated threshold BLS signature>",
  "proof":            "<Verify(threshold_BLS_pk_T, 'SUSPEND epoch_id_T', sigma)
= true
                       AND ForwardCommit.Verify(commit_T, null_v, sigma) =
true>"
}

// ON-CHAIN COMMITTEE VERDICT
{
  "type":                    "committee_verdict",
  "committee_ids":           ["<member epoch IDs>"],
  "target_epoch_id":         "<audited node epoch ID>",
  "verdict_epoch":           "<T>",
  "verdict":                 "SUSPENDED | DEGRADED | PASS",
  "reason":                  "class3_fail | handoff_reject | double_sign |
vdf_fail",
  "behavioral_fingerprint": "<fingerprint hash — SUSPENDED only>",
  "bls_aggregate_sig":       "<threshold BLS aggregate signature>"
}

// ON-CHAIN WATCHDOG SIGNAL
{
  "type":               "watchdog_signal",
  "epoch_id":           "<signaling node epoch ID>",
  "epoch_T":            "<T>",
```

```
  "observed_commits":  "<count>",
  "expected_rate":     "<local estimate>",
  "sig":               "<Sign(epoch_id, H(epoch_T ‖ observed_commits ‖ T))>"
}

// ON-CHAIN AUDIT RESULT (Class 2)
{
  "type":           "audit_result",
  "auditor_id":     "<auditing node epoch ID>",
  "target_id":      "<audited node epoch ID>",
  "audit_epoch":    "<T>",
  "result":         "pass | fail | no_response",
  "auditor_sig":    "<Sign(auditor_id, H(all fields ‖ T))>"
}

// ON-CHAIN ADMISSION SUMMARY
{
  "type":           "admission_summary",
  "epoch_id":       "<admitting node epoch ID>",
  "admitted":       "true | false",
  "admitted_T":     "<epoch>",
  "committee_sig": "<threshold BLS sig>"
}

// ARBITRATION COMMITTEE — FIRST OBSERVATION REPORT
{
  "type":               "first_observation",
  "observer_id":        "<existing node epoch ID>",
  "new_epoch_id":       "<admitting node epoch ID>",
  "first_seen_T":       "<epoch>",
  "first_seen_offset": "<seconds within epoch>",
  "provider_id":        "<mix-role provider ID under Loopix profile, Tor exit identity under Tor/I2P profile — coarse geographic prior, see §5.1>",
  "channel":            "chain | gossip | PSI",
  "observer_sig":       "<Sign(observer_id, H(report))>"
}

// ARBITRATION COMMITTEE — ZK CONTINUITY PROOF
```

```
{
  "type":           "zk_continuity",
  "epoch_id_T":     "<current epoch ID>",
  "epoch_id_prev":  "<prior epoch ID>",
  "zk_proof":       "<Plonky3 Poseidon PRF continuity proof>",
  "committee_sig":  "<threshold BLS sig acknowledging receipt>"
}

// REWIND SIGNAL
{
  "type":           "rewind_signal",
  "epoch_id":       "<signaling node epoch ID>",
  "current_T":      "<T>",
  "preferred_T":    "<epoch at which quality was last acceptable>",
  "cohort_epoch":   "<epoch at which implicated gossip vectors entered index>",
  "sig":            "<Sign(epoch_id, H(current_T ‖ preferred_T ‖
cohort_epoch))>"
}

// PSI ACKNOWLEDGMENT (cross-cluster attempt receipt)
{
  "type":         "psi_ack",
  "responder_id":"<responder epoch ID>",
  "epoch_T":      "<T>",
  "ack_nonce":    "<random 256-bit nonce from initiator's PSI request>",
  "sig":          "<Sign(responder_id, H(ack_nonce ‖ T))>"
}

// COVER TRAFFIC
{
  "type":    "cover",
  "payload": "<random bytes padded to frame size>"
}
```

Appendix B — What Each Node Holds

Private — never leaves the device:

Item	What it is
`sk`	Long-term signing key
`null_v`	Nullifier: `Poseidon(sk, "null_v")`. Never transmitted except as circuit witness
`r_commit_T`	Blinding factor for commit_T this epoch
`p_v`	Full signed preference vector. Updated only at epoch transitions
`r_p`	Pedersen blinding factor. Losing it locks out Class 3 audit responses for that identity
`π_v(T)`	Per-epoch permutation
Interaction history	Raw log. Aggregated into Merkle leaves; raw data stays local
`q_v(T)`	Local acceptance rate
HNSW snapshots	Periodic snapshots retained for rewind recovery
PSI cache	Per-peer Jaccard similarity scores and ZK continuity weights
PSI acknowledgments	Received cross-cluster PSI acknowledgments, retained as ZK witnesses for handoff proof
IP address	Hidden from all other nodes by Loopix/Sphinx mixnet transport

On the public blockchain:

Item	What it reveals
`epoch_id_T`	Pseudonym for this epoch. Derived as `Poseidon(sk, beacon_T, null_v, "epoch_id")`. Unlinkable to prior epochs without `sk`
VDF proof chain	At least n epochs of sequential computation with mandatory interaction checkpoints
`C_p(T)`	Pedersen commitment. Binding but reveals nothing about contents

Item	What it reveals
`M_v(T)`	Merkle root of behavioral history. Fixed-size padding prevents activity level inference
Score band (1–4)	Coarse reputation quartile — committee-attested
Item announcements	Item hash + noisy rating, positive interactions only. Niche items subject to VRF-derived delay
Committee verdicts	SUSPENDED and other health determinations. Permanent
Behavioral fingerprint	Included only in SUSPENDED verdicts
`SUSP_SMT_root_T`	Root of suspended nullifiers SMT
`commit_T`	Forward-secure commitment to null_v. Published every epoch. Opaque without verdict signature
`commit_T_zk_proof`	ZK proof certifying commit_T is correctly formed
`DECRYPTION_SMT_root_T`	Root of executed decryptions SMT
`verdict_commit` transactions	Per-committee-member verdict commitments
`verdict_reveal` transactions	Per-committee-member BLS shares and verdicts

Held by the arbitration committee — under threshold custody, accessible only on quorum:

Item	What it contains
ZK continuity proofs	Links `epoch_id_T` to `epoch_id_{T-1}`
Fine-grained behavioral data	Per-epoch timing distributions, audit response rates
Handoff history	Encrypted state chain for each node under audit
First-observation reports	Observer timing data and provider_id for coordinated admission detection
Raw score	Full computed score per epoch

Off-chain entirely:

Item	Where it lives
PSI relationships	Local device only
Gossip vector exchanges	Direct peer-to-peer
Receipt collection	Local device, retained one epoch for auditor queries

---

Appendix C — Reputation Decision Tree

```
SIGNAL RECEIVED
|
├── Protocol violation?
|   ├── Critical (VDF fail, Class 3 non-response, handoff rejected by majority,
|   |             double-signing detected, Statement 5 proof invalid,
|   |             commit_T ZK proof invalid)
|   |   └── → Commit-reveal SUSPENDED flow (see §4.9.6 for canonical
specification)
|   └── Minor (rate limit hit, Class 2 mismatch, single auditor handoff
dispute)
|       └── → reduce relevant score component

ANOMALOUS VERDICT_COMMIT RATE
  Any node detects rate exceeds expected bound without behavioral signal
justification
      → broadcast watchdog_signal
  Multiple watchdog_signals this epoch
      → L3 Suspicious for implicated committee
      → meta-committee assembled via VRF
      → same commit-reveal ordering applies recursively

EPOCH END
  commit_T + Statement 5 ZK proof submitted every epoch (exempt from n_commit)
  Remaining fields submitted every n_commit epochs
  Committee verifies handoff: majority accept → normal · majority reject →
SUSPENDED flow
  Score held under arbitration committee custody; band attestation → public
chain
```

```
ADMISSION — nullifier check
  Node completes n-epoch VDF admission chain
  First handoff must include valid Statement 5 proof + valid commit_T
      null_v ∉ SUSP_SMT: admission proceeds normally
      null_v ∈ SUSP_SMT: admission rejected immediately — cryptographic

IDENTITY ROTATION EVASION
  Same sk: cryptographically impossible
  New sk: fresh null_v', full admission cost required
      Behavioral fingerprint matching → L2 Elevated from admission
  Dark node (same sk): closed by ForwardCommit
  Dark node (admission window, no commit_T published):
      Residual gap — bounded by zero reputation + fingerprinting
```

Appendix D — Node Lifecycle

```
SETUP (one-time)
  Generate sk
  Compute null_v = Poseidon(sk, "null_v")
  Compute epoch offset: offset_v = Poseidon(sk, "epoch_offset") mod
epoch_duration
  Begin admission window (n epochs)
      Each epoch: compute and publish identity VDF proof on-chain
      At VRF-determined checkpoint epochs:
          PSI handshake with random existing node (Loopix mix-routed, SURB
reply)
          Gossip vector exchange + collect receipt
      First-observation reports submitted to arbitration committee only
(including provider_id)
      Public chain records admission decision only on completion
  Epoch n: binary — full admission if chain complete and all checkpoints passed
  First handoff after admission must include valid Statement 5 proof + valid
commit_T
      If null_v ∈ SUSP_SMT: admission rejected immediately
```

```
EACH EPOCH (offset_v stagger)

  EPOCH START
  ├── Rotate epoch_id_T = Poseidon(sk, beacon_T, null_v, "epoch_id")
  ├── Rotate π_v(T) = Poseidon(sk, beacon_T, "perm")
  ├── Derive n_v(T) = n_base + Poseidon(sk, beacon_T, "chop_n") mod n_jitter
  ├── Renew C_p(T)  [p_v updates only permitted here]
  ├── Fetch current SUSP_SMT_root_T from public chain
  ├── Await DKG completion → obtain threshold_BLS_pk_T
  ├── Construct commit_T = BF-IBE.Encrypt(null_v, "SUSPEND epoch_id_T",
threshold_BLS_pk_T; r_commit_T)
  ├── Generate Statement 5 ZK proof (4 checks + SMT path)
  ├── Optional ZK continuity proof (arbitration committee only)
  ├── Refresh bridge peer (DHT + Loopix provider)
  ├── Update PSI cache (λ_proof or λ_noproof)
  └── Store HNSW snapshot if snapshot interval reached

  DURING EPOCH
  ├── Push gossip vector at T_send (first 30% of epoch, randomized)
  ├── Pull from peers — embed Class 2 audit nonce in every pull request
  ├── Respond to pulls — embed audit hash + receipt signature in every response
  ├── Respond to cross-cluster PSI attempts — issue signed psi_ack regardless
of handshake outcome
  ├── Collect received psi_ack messages as ZK witnesses for handoff proof
  ├── Retain issued receipts for one epoch (for auditor queries)
  ├── Announce positive interactions (delay + noise on rating)
  ├── Collect peer attestations passively as receipts arrive
  ├── Update local HNSW from received vectors
  ├── Compute recommendations offline
  ├── Track q_v(T) — rewind signal if acceptance rate drops
  ├── Monitor public chain for anomalous verdict_commit rate
  │    → broadcast watchdog_signal if rate exceeds expected
  └── Maintain constant cover traffic

  EPOCH END (spread across randomized window)
  ├── Assemble Merkle leaves (fixed protocol-wide maximum padding)
  ├── Compute M_v(T)
```

```
  ├── Submit commit_T + Statement 5 ZK proof via relay → public chain (every
epoch)
  ├── Generate auditor handoff:
  │   successor ZK proof + leaf counts + Statement 5 + commit_T + encrypted
shares
  │   + PSI acknowledgment witnesses (cross-cluster attempt rate proof)
  │   ZK continuity proof submitted to arbitration committee only
  ├── Distribute handoff to committee via Dandelion++ stem
  ├── Await committee threshold signature
  ├── If commit epoch (T mod n_commit = 0):
  │   Submit M_v, C_p, score band, health tier via relay → public chain
  ├── If Class 3 audit pending: submit Merkle + ZK proof (Stmts 1–3, 5) via
stem with retry
  │   If rewind confirmed: roll back HNSW to preferred_T snapshot
  │                          discard gossip vectors from implicated cohort
  ├── Compute score_v(T) — held under arbitration committee custody
  ├── Consistency + smoothness check
  ├── Send batch receipts (Dandelion++ stem)
  └── Check public chain for committee verdicts against any peers

 BLOCKCHAIN (validator nodes only)
  ├── Proposer: collect pending transactions, assemble block
  ├── Proposer: verify all commit_T ZK proofs in epoch_transactions
  ├── Proposer: verify verdict_reveal matches against verdict_commit
  ├── Proposer: verify null_v_decryption proofs; update DECRYPTION_SMT and
SUSP_SMT
  ├── Proposer: compute block VDF_eval(vdf_output_{T−1}, δ_block)
  ├── Proposer: update SUSP_SMT_root_T and DECRYPTION_SMT_root_T if updated
this epoch
  ├── Proposer: broadcast candidate block to validator set
  ├── Validators: verify block contents, VDF output, committee BLS sigs,
  │              root consistency, commit_T ZK proofs, verdict_reveal
consistency
  ├── Block final when ⌊K/3⌋×2+1 signatures collected
  ├── Final block broadcast to network
  └── Light clients: sync block header + verify relevant entries via Merkle
proof
```

```
  COMMITTEE (burst score aggregation)
  └── At end of each admission window:
      Aggregate first_seen_T and first_seen_offset across first-observation
reports
      Aggregate provider_id distributions
      Compute burst score over inter-arrival timing distributions
      Compute geographic concentration signal
      Both enter compound flag system at L1 only
```

Appendix E — Implementation Readiness

```
TIER 1 — Build now
  Poseidon PRF                        arkworks-rs Poseidon crate; domain
separators; local derivations only
  EC-VRF (RFC 9381)                   tendermint-rs; on-chain verifiable selection
(validator, committee, relay)
  null_v derivation                   Poseidon(sk, "null_v") at setup
  SUSP_SMT                            jellyfish-merkle or rs-merkle; append-only
  DECRYPTION_SMT                      Same structure; dec_nullifier =
Poseidon(verdict_hash, null_v)
  commit_T per-epoch publication      Exempt from n_commit; every epoch
  DKG for committee threshold BLS     Per-epoch; K≈21; blst crate
  Verdict_commit transaction type     Commit phase; H(share ‖ verdict ‖ nonce)
  Verdict_reveal transaction type     Reveal phase; match against commit
  null_v_decryption transaction       Permissionless aggregation + SUSP_SMT
insertion
  Watchdog signal                     Public chain rate monitoring
  Statement 5 circuit                 Four checks + ForwardCommit + SMT non-
membership (Plonky3)
  Staggered epoch offsets             Poseidon(sk, "epoch_offset") mod
epoch_duration
  Variable chopping n_v(T)            Poseidon(sk, beacon_T, "chop_n") mod
n_jitter
  Niche announcement delay            Poseidon(sk, item_hash, beacon_T,
"niche_delay") mod max_delay
  VDF — identity chain                vdf crate (Chia); per-node sequential
```

```
  VDF — block chain                vdf crate (Chia); per-block sequential
  Pedersen commitments             curve25519-dalek / arkworks-rs
  Merkle tree                      rs-merkle; fixed max padding; Poseidon salts
  HNSW snapshot storage            Periodic local snapshots for rewind
  Shamir k-of-n                    sharks crate; extend for k > 3
  Noise Protocol sessions          snow crate
  Dandelion++ gossip               libp2p-gossipsub; broadcast fluff phase
only; timeout retry
  Loopix/Sphinx transport          katzenpost (Go); Sphinx packet format; SURB
replies; provider-based NAT
  LSH                              FALCONN
  HNSW                             hnswlib / instant-distance
  Plonky3 proofs                   plonky3 crate
  Bulletproofs                     bulletproofs (dalek-crypto)
  Laplace DP (S=2, sign-bounded)   trivial with sign preservation constraint
  BLS signatures                   blst crate
  Item-based CF                    standard linear algebra
  Dislike-aware CF                 trivial post-processing
  Uniform frame padding            trivial
  Peer attestation collection      trivial — receipts in pull responses
  Per-n-epoch commit batching      trivial epoch counter
  Rewind signal cohort correlation local HNSW provenance tracking
  provider_id field in first-obs   trivial schema extension
  Burst score aggregation          committee-side inter-arrival computation
over first-observation reports
  PSI acknowledgment message       new message type; signed; rate-limited (OQ-
61)
  PSI ack witness collection       local retention; fed into handoff ZK proof

TIER 2 — Adapt prior work
  Tendermint-style BFT             tendermint-rs; VRF proposer + VDF-chained
  BLS threshold aggregation        blst; threshold scheme
  Double-signing detection         Standard BFT; epoch_id key structure
  Light client headers + proofs    Standard Merkle SPV
  Genesis transition protocol      Bootstrap behavioral cluster relaxation
  Asymmetric PSI (Pinkas 2018)     Unbalanced; Jaccard threshold; Loopix mix-
routed with SURB
  PSI cache asymmetric decay       Jøsang & Ismail basis
```

```
  Two-tier peer selection          Cluster + bridge logic; bridge weight opt-
out
  Class 2 passive audit            Nonce-in-pull; hash-in-response
  Auditor handoff flow             Successor ZK proof + Statement 5 + PSI ack
witnesses
  Multi-auditor committee          VRF with dual cluster constraints
  DSybil with noisy ratings        Binary fallback option
  k_min impact bounding            PSI neighborhood size gate (§7.4)
  SSP-adapted simulation           Werthenbach & Pouwelse 2023
  Trust attenuation by hop         Stannat et al. AAMAS 2021
  Asymmetric penalty               Sun et al. 2008
  Smoothness detection             Srivatsa et al. 2005; recovery trajectory in
test suite
  Relay node batching              VRF selection; reputation integration
  Validator service score bonus    Trivial score component addition
  Commit-reveal verdict flow       Two-phase; ordering enforcement in block
validation
  Permissionless BLS aggregation   Standard threshold BLS
  Recursive oversight chain        Meta-committee VRF; same commit-reveal;
depth limit
  Item-velocity rewind correlation Cohort_epoch × trust_total velocity compound
signal

TIER 3 — Original design required
  Chain + arbitration committee    Single public chain + on-demand threshold-
custody committee
  ZK continuity circuit            Plonky3; Poseidon PRF relation; arbitration
committee only
  ZK consistency (Stmts 1-3)      Statement 2 requires sign preservation first
  ZK successor proof (handoff)     C_p + M_v successor + leaf count + Statement
5 + PSI ack witnesses
  ForwardCommit.Verify in circuit  BLS-based decryption verification as
constraint
  Block VDF chaining protocol      Proposer timeout; liveness guarantees
  SUSP_SMT maintenance             Append-only; validator insertion flow
  Behavioral cluster computation   On-chain derived fingerprints
  Merkle peer-attested leaves      Fixed padding; attestation assembly; partial
reveal
```

```
  Justified disclosure              Compound flag justification
  Full rate-limit enforcement       All event categories
  Cluster formation from PSI        Entry protocol; cache management
  Cover item selection              Post-n_cover local pool
  Epoch interaction checkpoints     VRF scheduling; verification
  Score band attestation flow       Committee threshold BLS → public chain
  Rewind HNSW rollback              Snapshot provenance; cohort identification
  Watchdog rate estimation          Per-node expected suspension rate
  Oversight chain termination       Hard depth limit + escalating committee
sizes
  Geographic concentration signal  Provider_id aggregation + compound flag
entry

TIER 4 — Research required before deployment
  Poseidon PRF security analysis   OQ-1 — closed (primitive split)
  EC-VRF security                   RFC 9381; DLEQ/DDH; tendermint-rs
  Domain separator enumeration      OQ-4 — closed
  ForwardCommit security analysis  OQ-2 — closed
  dec_nullifier collision resist.  OQ-5 — closed
  Statement 5 mobile benchmark      OQ-3 — open (required before mobile
deployment)
  Noise system per segment          OQ-54 — open (head / long-tail)
  Reputation weight calibration     OQ-9 — open
  Variable chopping calibration     n_base and n_jitter per sparsity tier (head
/ long-tail)
  Niche announcement delay          τ_niche and max_delay_epochs
  DSybil composition lemma          Continuous ratings; binary fallback if
unproven
  Δ_rise calibration                Path-responsiveness vs. on-off defense (OQ-
6, §8.2 T1)
  k_min calibration                 Community density profiles; OQ-24
  ZK Statement 2 mobile bench       d=128/256 Plonky3/WASM (OQ-3b)
  ZK successor proof cost           Mobile hardware; includes PSI ack witness
cost
  Bootstrap critical mass           Genesis transition; cluster relaxation (OQ-
12)
  Validator incentive analysis      Long-term shirking risk (OQ-13)
  Oversight chain termination       Depth limit + damage bound (OQ-49)
```

```
    Off-chain collusion bounds       OQ-53
    PSI ack rate limit + privacy     OQ-61 — required before deployment
    Burst score threshold            OQ-59 — Phase 5 experiment 5.48
    Sybil influence empirical data   OQ-10 — Phase 5 + live deployment
    [Remaining calibration items: see §10.1]
```

Appendix F — Configuration Reference

```
CONFIG A — Open (invite-only, maximum Sybil resistance)
  Transport:       Loopix/Sphinx + Dandelion++ (broadcast only)
  Identity:        Persistent (no epoch rotation)
  Noise:           Chopping + cover items
  ZK:              On by default
  Admission:       n=24
  Chain:           Full participation (can be validator)

CONFIG B — Balanced (recommended default)
  Transport:       Loopix/Sphinx + Dandelion++ (broadcast only)
  Identity:        Poseidon PRF rotation
  Noise:           Chopping + cover items
  ZK:              On by default
  Admission:       n=24
  Chain:           Light client

CONFIG C — High-Security (sensitive content, political contexts)
  Transport:       Loopix/Sphinx + Dandelion++ (broadcast only)
  Identity:        Poseidon PRF rotation
  Noise:           Chopping + cover items
  ZK:              On by default
  Admission:       n=168
  Chain:           Light client

CONFIG D — Mainstream DP (formal guarantees on popular content)
  Transport:       Loopix/Sphinx + Dandelion++ (broadcast only)
  Identity:        Poseidon PRF rotation
```

```
  Noise:          Laplace S=2 per segment (head / long-tail), sign-bounded — NO
chopping
  ZK:             On by default
  Admission:      n=24
  Chain:          Light client

CONFIG E — Binary Ratings (formal DSybil guarantee)
  Transport:      Loopix/Sphinx + Dandelion++ (broadcast only)
  Identity:       Poseidon PRF rotation
  Noise:          Chopping + cover items
  Ratings:        Binary (liked / not liked)
  ZK:             On by default
  Admission:      n=24
  Chain:          Light client
```

	A	B	C	D	E
Niche rec quality	5	5	4	3	4
Mainstream rec quality	4	4	4	3	3
Preference privacy	3	4	5	3	4
Identity privacy	1	4	5	4	4
Sybil resistance	5	3	2	3	5
Suspension persistence	5	5	5	5	5
Formal DSybil guarantee	1	1	2	4	5
Formal DP guarantee	1	1	2	4	1
Mobile / battery	5	4	2	4	4
Cold start speed	5	4	3	4	4

All configs: deployment must select a conformant transport profile per §5.1; self-mixing Loopix is the reference default for MVP, Tor/I2P is a specified alternate. Dandelion++ retained for epidemic broadcast (fluff) phase only across profiles. Clearnet: dev/test only. Nullifier mechanism, SUSP_SMT, DECRYPTION_SMT, ForwardCommit, and commit-reveal verdict flow apply universally. commit_T is published every epoch regardless of n_commit setting. Bridge peer weight opt-out is a per-node local setting available in all configs.

Appendix G — Node Relationship Diagram

```
                    ┌──────────┐
                    │  NODE A  │
                    └────┬─────┘
                         │
          ┌──────────────┼─────────────────┐
          │              │                 │
     ┌────┴────┐   ┌─────┴────┐    ┌───────┴────┐
     │ Interest│   │ Interest │    │ Bridge Peer│
     │ Cluster │   │ Cluster  │    │(random DHT)│
     │ Peer B  │   │ Peer C   │    │(weight     │
     │ (PSI ✓) │   │ (PSI ✓)  │    │  opt-out   │
     └─────────┘   └──────────┘    │  available)│
                                   └────────────┘

     ┌─────────────────────────────────────────────────────┐
     │ AUDITOR COMMITTEE (k members, rotating per epoch)   │
     │ VRF-selected · unpredictable before beacon publish  │
     │ Different interest AND behavioral clusters          │
     │ Reputation ≥ median · temporal depth ≥ D_min        │
     │ Run DKG at epoch start → threshold_BLS_pk           │
     │ Shamir share of encrypted state chain               │
     │ Threshold BLS signature for handoff acceptance      │
     │ Commit-reveal verdict process for suspensions       │
     │ Publish verdicts to public chain                    │
     │ Hold Shamir-shared snapshots (threshold custody)    │
     │ Aggregate burst score + provider_id distributions   │
     └─────────────────────────────────────────────────────┘

     ┌─────────────────────────────────────────────────────┐
     │ VALIDATOR SET (K_validators members, per epoch)     │
     │ VRF-selected · unpredictable before beacon publish  │
     │ Different interest AND behavioral clusters          │
     │ Reputation ≥ rep_validator · depth ≥ D_validator    │
     │ One proposer assembles block · others validate      │
     │ Threshold BLS signatures for block finality         │
     │ Verify commit_T ZK proofs and DKG completion        │
```

```
  │  Maintain DECRYPTION_SMT                                   │
  └────────────────────────────────────────────────────────────┘


  ┌────────────────────────────────────────────────────────────┐
  │  RELAY NODES (submission timing obfuscation)               │
  │  VRF-selected · different behavioral cluster               │
  │  Batch submissions · reputation model same as              │
  │  auditor committee                                         │
  └────────────────────────────────────────────────────────────┘
```

Appendix H — Identity and Privacy Relationship Diagram

```
┌──────────────────────────────────────────────────────────────────┐
│  WHAT THE ADVERSARY SEES vs. WHAT IS ACTUALLY HAPPENING          │
│                                                                  │
│  ┌──────────────┐    epoch rotation (Poseidon PRF)  ┌───────────────┐      │
│  │ epoch_id_T   │◄─── Poseidon(sk,   ───┤  secret key  │       │
│  │ (pseudonym)  │     beacon_T,null_v)  │     sk       │       │
│  └──────┬───────┘                        └──────┬───────┘        │
│         │ unlinkable across epochs              │                │
│         │                               ┌───────┴──────┐         │
│         │                               │  null_v      │         │
│         │                               │  Poseidon    │         │
│         │                               │  (sk,"null_v")         │
│         │                               │  stable,     │         │
│         │                               │  private     │         │
│         │                               └──────┬───────┘         │
│         │                                      │ private witness │
│         │                               ┌──────┴───────┐         │
│         │ non-suspension proof          │  ZK circuit  │         │
│         │ (boolean only)       ◄────────┤  null_v ∉    │         │
│         │                               │  SUSP_SMT    │         │
│         │                               │  + commit_T  │         │
│         │                               │  correctly   │         │
│         │                               │  formed      │         │
│         │                               └──────────────┘         │
```

```
│         │                                                   │
│  ┌──────┴──────┐    preference privacy                      │
│  │   C_p(T)    │◄──── Pedersen(p_v,r_p)                     │
│  └──────┬──────┘                                            │
│         │                                                   │
│  ┌──────┴──────┐    behavior privacy                        │
│  │   M_v(T)    │◄── Merkle root only                        │
│  └──────┬──────┘                                            │
│         │                                                   │
│  ┌──────┴──────┐                                            │
│  │ ZK proofs   │── auditor committee only ──► arbitration committee │
│  │ continuity  │   never on public chain                    │
│  └─────────────┘                                            │
│                                                             │
│  ┌─────────────┐    nullifier custody (opaque)              │
│  │ commit_T    │◄── ForwardCommit(null_v, epoch_id_T,       │
│  │ (public)    │    threshold_BLS_pk(committee_T); r)       │
│  └──────┬──────┘                                            │
│         │ decryptable only by valid verdict signature       │
│         │ verdict requires public commit-reveal process     │
│  ┌──────┴──────┐                                            │
│  │ null_v      │ recovered by anyone after verdict          │
│  │ (post-      │ → inserted into SUSP_SMT                   │
│  │ verdict)    │ → dec_nullifier into DECRYPTION_SMT        │
│  └─────────────┘                                            │
│                                                             │
│  ADMISSION COST                                             │
│  ┌─────────────────────────────────────────────────────┐    │
│  │ VDF chain: n sequential proofs, one per epoch       │    │
│  │ each proof depends on the previous — no parallelism │    │
│  │ interaction checkpoints require real network contact│    │
│  └─────────────────────────────────────────────────────┘    │
└─────────────────────────────────────────────────────────────┘
```

Appendix I — PSI Peer Selection Flow

```
PSI PEER SELECTION FLOW
──────────────────────────────────────────────────────────

Node A                      Node B
     │                           │
     ├── initiate PSI ──────────►│
     │   (Loopix mix-routed,     │
     │    content encrypted,     │
     │    SURB included)         │
     │                           │ issue signed psi_ack immediately
     │                           │ (regardless of handshake outcome)
     │◄── psi_ack ───────────────┤
     │   (confirms attempt       │
     │    received; no set       │
     │    contents revealed)     │
     │                           │ compute Jaccard:
     │                           │ |A∩B| / |A∪B| ≥ θ?
     │◄── PSI result ────────────┤
     │   (similarity score       │
     │    only — no set          │
     │    contents revealed)     │
     │                           │
     ├── similarity ≥ θ_cluster? │
     │       YES: add to         │
     │            interest tier  │
     │       NO:  discard        │
     │                           │
    A retains psi_ack as ZK      │
    witness for handoff proof    │
──────────────────────────────────────────────────────────
```